MUST-SCORE FACTS

Environment and Ecology

Consolidated Must-Score Facts and must-score drill cards for every chapter.

UPPCS + UKPCS Study Library · chapter order follows Subject Notes.

Chapters

1. Daily Read — High-Yield Revision Facts
2. Topic 1 — Environment Basics
3. Topic 2 — Ecology & Ecosystem
4. Topic 3 — Food Chain & Energy Flow
5. Topic 4 — Biodiversity
6. Topic 5 — Habitat, Flora & Fauna
7. Topic 6 — Protected Areas & Conservation
8. Topic 7 — Wildlife Conservation
9. Topic 8 — Forests & Forest Management
10. Topic 9 — Pollution & Waste Management
11. Topic 10 — Climate Change
12. Topic 11 — Ozone Layer
13. Topic 12 — Acid Rain
14. Topic 13 — Desertification & Land Degradation
15. Topic 14 — Environmental Impact Assessment
16. Topic 15 — Sustainable Development & Environmental Governance
17. Topic 16 — Environmental Organizations (India)
18. Topic 17 — Environmental Laws & Policies
19. Topic 18 — International Environmental Agreements & Conferences
20. Topic 19 — Climate & Environmental Institutions
21. Topic 20 — Biodiversity Conservation Methods
22. Topic 21 — Species & Ecology
23. Topic 22 — Renewable Energy
24. Topic 23 — Disaster & Environment
25. Topic 24 — Current Environmental Issues
26. Topic 25 — Global Environmental Geography
27. Topic 26 — Water Resources & Water Conservation
28. Topic 27 — Renewable & Non-Renewable Energy
29. Topic 28 — Environmental Research & Institutions
30. Topic 29 — Indian Environmental Movements
31. Topic 30 — Environmental Literature & Awareness
32. Topic 31 — Environmental Days
33. Topic 32 — National Parks & Protected Areas (Advanced)
34. Topic 33 — Biosphere Reserves
35. Topic 34 — Climate Change (Advanced)
36. Topic 35 — Atmosphere
37. Topic 36 — Ozone Layer
38. Topic 37 — Greenhouse Gases
39. Topic 38 — Pollution (Advanced)
40. Topic 39 — Acid Rain
41. Topic 40 — Desertification
42. Topic 41 — Environmental Monitoring
43. Topic 42 — International Environmental Organizations
44. Topic 43 — International Environmental Agreements
45. Topic 44 — Current Environmental Issues
46. Uttarakhand — 01 — Natural Resources & Climate Contribution
47. Uttarakhand — 02 — Biodiversity & Protected Areas
48. Uttarakhand — 03 — Climate Vulnerability & Governance

Daily Read — High-Yield Revision Facts

1. Core Environmental Legislation in India (Exact Years & Articles)

Legislation / Policy Name	Year of Enactment	Constitutional Backing / Key Mandate	Landmark Institutional Outcome
Indian Forest Act	**1927** (earlier 1865, 1878)	Consolidated forest laws; categorized into **Reserved, Protected, and Village Forests**	State Forest Departments
Wildlife (Protection) Act (WPA)	**1972** (9 Sept 1972)	Framework for protecting wild animals, birds, and plants; **6 original schedules restructured into 4 schedules in 2022 Amendment**	National Board for Wildlife (NBWL, chaired by PM); NTCA; Wildlife Crime Control Bureau (WCCB)
Water (Prevention and Control of Pollution) Act	**1974**	Prevention and abatement of water pollution; maintains wholesomeness of water	Established **Central Pollution Control Board (CPCB)** & State PCBs
Forest (Conservation) Act (FCA)	**1980** (25 Oct 1980)	Prohibits diversion of forest land for non-forest purposes without prior central approval	Forest clearance mechanism
Air (Prevention and Control of Pollution) Act	**1981**	Prevention, control, and abatement of air pollution; **Noise pollution included as an air pollutant by 1987 Amendment**	Empowered CPCB and SPCBs to set ambient air standards
Environment (Protection) Act (EPA)	**1986** (23 May 1986)	**Umbrella legislation** enacted under **Article 253** in the wake of the **Bhopal Gas Tragedy (Dec 1984)**	Empowered Union Govt to coordinate state actions, set emission norms; CRZ notifications; EIA rules
Public Liability Insurance Act	**1991**	Mandatory public insurance for providing immediate relief to victims of accidents occurring while handling hazardous substances	Environment Relief Fund
National Environment Tribunal Act	**1995**	Strict liability for damages arising out of any accident while handling hazardous substances	Precursor to NGT
Biological Diversity Act (BDA)	**2002**	Realized commitments under CBD (1992); 3-tier structure for conservation and equitable benefit sharing	**National Biodiversity Authority (NBA, Chennai)**, State Biodiversity Boards (SBB), Biodiversity Management Committees (BMC)
Forest Rights Act (FRA)	**2006**	Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act; vests forest land rights	Gram Sabha is the statutory authority to initiate rights recognition process
National Green Tribunal (NGT) Act	**2010** (18 Oct 2010)	Created dedicated environmental tribunal under **Article 21 (Right to a healthy environment)**; mandated disposal within **6 months**	**Principal Bench: New Delhi**; 4 Regional Benches: **Bhopal, Pune, Kolkata, Chennai**
CAMPA Act	**2016**	Compensatory Afforestation Fund Act; manages funds collected for diversion of forest land (90% to State, 10% to Centre)	National & State Compensatory Afforestation Funds

2. Environmental Movements in India (Pioneers & Locations)

Movement	Year & Place	Key Leaders / Activists	Core Environmental Objective / Strategy
Bishnoi Movement	**1730**, Khejarli village (Jodhpur, Rajasthan)	**Amrita Devi Bishnoi**	King Abhay Singh ordered felling of **Khejri trees** (*Prosopis cineraria*); Amrita Devi and her 3 daughters hugged trees; **363 Bishnois sacrificed their lives**; Amrita Devi Bishnoi Wildlife Protection Award instituted in memory.
Chipko Movement	**1973**, Reni village, Chamoli (**Uttarakhand**)	**Sunderlal Bahuguna**, **Chandi Prasad Bhatt** (Ramon Magsaysay awardee), **Gaura Devi** (*Chipko Mother*)	Protest against commercial logging of ash trees in Alaknanda basin; villagers hugged trees to prevent felling; coined slogan: *"Ecology is permanent economy"*; led to 15-year ban on green felling in UP/UK hills.
Appiko Movement	**1983**, Salkani, Uttara Kannada (Karnataka)	**Pandurang Hegde**	Southern version of Chipko; hugged trees in Western Ghats to save tropical forests from commercial monoculture plantations.
Silent Valley Movement	**1978**, Palakkad (Kerala)	Kerala Sastra Sahitya Parishad (**KSSP**), Romulus Whitaker	Protest against proposed **hydroelectric project on Kunthipuzha River** that would submerge the pristine moist evergreen rainforest habitat of the endangered **Lion-Tailed Macaque**; project scrapped; declared Silent Valley National Park (1984).
Narmada Bachao Andolan (NBA)	**1985**, Narmada Valley (Gujarat, MP, Maha)	**Medha Patkar**, **Baba Amte**	Mass movement against large dams on Narmada (particularly **Sardar Sarovar Dam**); demanded comprehensive rehabilitation, environmental impact assessment, and rights of displaced tribal villagers.
Jungle Bachao Andolan	**1982**, Singhbhum (Bihar / Jharkhand)	Local tribal communities	Tribals agitated against government scheme to replace indigenous, multipurpose **Sal forests** with commercially profitable **Teak (Sagwan)** plantations; known as "Greed Game Political Populism".
Tehri Dam Protest	1980s–1990s, Tehri (**Uttarakhand**)	**Sunderlal Bahuguna**	Agitation against the construction of the mega Tehri Dam on the seismic fault lines of Bhagirathi River; multiple hunger strikes.

3. Environmental Research Institutes & Organizations (Match Goldmine)

Institute / Organization	Location / Headquarters	Established Year / Key Function
Botanical Survey of India (BSI)	**Kolkata** (West Bengal)	**1890**; survey and conservation of plant resources of India
Zoological Survey of India (ZSI)	**Kolkata** (West Bengal)	**1916**; survey, exploration, and taxonomic research on animal diversity
Forest Research Institute (FRI)	**Dehradun (Uttarakhand)**	**1906** (Imperial Forest Research Institute); Greco-Roman architectural marvel
Wildlife Institute of India (WII)	**Dehradun (Uttarakhand)**	**1982**; premier autonomous institution for wildlife research and management
National Environmental Engineering Research Institute (NEERI)	**Nagpur** (Maharashtra)	**1958**; pioneer in environmental monitoring, sewage, industrial pollution solutions (CSIR)
Central Pollution Control Board (CPCB)	**New Delhi**	**1974**; statutory organization under Water Act 1974; sets national air/water norms
National Biodiversity Authority (NBA)	**Chennai** (Tamil Nadu)	**2003**; statutory body under Biological Diversity Act 2002
Sálim Ali Centre for Ornithology & Natural History (SACON)	**Coimbatore** (Tamil Nadu)	Dedicated to conservation of avifauna and wetland ecology

Institute / Organization	Location / Headquarters	Established Year / Key Function
Indian Institute of Forest Management (IIFM)	**Bhopal** (Madhya Pradesh)	**1982**; premier forestry management and planning institute
G.B. Pant National Institute of Himalayan Environment	**Kosi-Katarmal, Almora (Uttarakhand)**	**1988**; apex research body on sustainable development and ecology of Indian Himalayas
Central Arid Zone Research Institute (CAZRI)	**Jodhpur** (Rajasthan)	**1959**; research on desertification and arid land agriculture
National Disaster Management Authority (NDMA)	**New Delhi**	**2005**; apex body under Disaster Management Act 2005; **Chaired by the Prime Minister**

4. Indian Wildlife Conservation Projects (Chronological Ladder)

Project Name	Year Launched	Launch Location / Nodal Agency	Key Objective / Status / Prelims Fact
Project Tiger	**1 April 1973**	**Jim Corbett National Park (Uttarakhand)**	Central scheme; initiated with 9 reserves under Kailash Sankhala ("Tiger Man of India"); statutory backing via NTCA (2006); **58 Tiger Reserves** (Madhav TR, MP, is the 58th). Corbett has among the highest tiger densities.
Project Crocodile	**1975**	Hyderabad / Odisha (with UNDP & FAO aid)	Conservation of 3 crocodilian species: Gharial (*Gavialis gangeticus*), Mugger crocodile, and Saltwater crocodile (Bhitarkanika, Odisha)
Project Hangul	**1970**	Dachigam National Park (J&K)	Conservation of Kashmir Stag (*Cervus elaphus hanglu* - critically endangered) with IUCN/WWF aid
Project Rhino (IRV 2020)	**1987 / 2005**	Assam (Kaziranga, Manas, Orang, Pobitora)	Indian Rhino Vision 2020 aimed to attain a wild population of at least 3,000 Greater One-Horned Rhinos in Assam
Project Elephant	**February 1992**	Jharkhand (Singhbhum)	Centrally sponsored; protects elephant habitats and corridors; addresses human-elephant conflict; **Mike sites** (Monitoring of Illegal Killing of Elephants); campaign: *Haathi Mere Saathi*
Project Sea Turtle	**1999**	Odisha coast (Gahirmatha / Rushikulya)	Conservation of **Olive Ridley turtles** (*Lepidochelys olivacea*) during synchronized mass nesting (**Arribada**) with MoEFCC & UNDP
Project Snow Leopard	**2009**	High-altitude Himalayan states (Hemis, Gangotri, Nanda Devi)	Conservation of high-altitude ecosystem and the elusive Ghost of the Mountains (*Panthera uncia*)
Project Dolphin	**2020**	Ministry of Environment, Forest & Climate Change	Gangetic Dolphin (*Platanista gangetica*) declared **National Aquatic Animal on 5 October 2009** (5 Oct celebrated as **National Dolphin Day**); blind; uses echolocation; Vikramshila Sanctuary (Bihar)
Project Cheetah	**September 2022**	**Kuno National Park (Madhya Pradesh)**	Intercontinental translocation of African cheetahs from Namibia (8) and South Africa (12); Asiatic Cheetah declared extinct in India in **1952**

5. Ramsar Wetlands in Uttar Pradesh & Uttarakhand

- **Uttarakhand**:
- **Asan Conservation Reserve** (Dehradun; confluence of Asan and Yamuna; **first Ramsar of Uttarakhand, Oct 2020**).
- **Jhilmil Jheel Conservation Reserve** (Haridwar, **2005**) is the state's swamp-deer / barasingha wetland — **not** the Ramsar first (that is Asan).
- **Uttar Pradesh (12 Ramsar sites)** — do not stop at the older list of 10: 1. **Bakhira Wildlife Sanctuary** (Sant Kabir Nagar). 2. **Haiderpur Wetland** (Muzaffarnagar–Bijnor; Solani / Hastinapur WLS). 3. **Nawabganj Bird Sanctuary** (*Chandra Shekhar Azad*, Unnao). 4. **Parvati Arga Bird Sanctuary** (Gonda). 5. **Saman Bird Sanctuary** (Mainpuri). 6. **Samaspur Bird Sanctuary** (Rae Bareli). 7. **Sandi Bird Sanctuary** (Hardoi). 8. **Sarsai Nawar Jheel**

(Etawah; Sarus Crane). 9. **Sur Sarovar (Keetham Lake)** (Agra). 10. **Upper Ganga River** (Brijghat to Narora). 11. **Patna Bird Sanctuary** (Etah). 12. **Shekha Jheel** (Aligarh).

> **Traps:** Sultanpur NP = **Haryana**. Rudrasagar = **Tripura**. Samaspur and Sarsai Nawar **are** in UP.

5A. Protected areas of UP & UK — Declared / still in force

PA	State	From (year)	High-yield
Hailey → Ramganga → Jim Corbett NP / TR	**Uttarakhand**	NP **1936** (Hailey); Project Tiger launch site **1973**	India's first National Park. Nainital–Pauri belt.
Rajaji NP / TR	**Uttarakhand**	NP **1983**; tiger reserve **2015**	Elephant corridor; Haridwar–Dehradun–Pauri
Nanda Devi NP	**Uttarakhand**	**1982** (BR **1988**; UNESCO mixed serial with VoF)	Chamoli; Nanda Devi peak
Valley of Flowers NP	**Uttarakhand**	**1982** (UNESCO **1988**, expanded **2005**)	Smallest UK NP; Chamoli
Gangotri NP	**Uttarakhand**	**1989**	Largest UK NP (~2390 sq km); Uttarkashi
Govind NP / WLS	**Uttarakhand**	WLS **1955**; NP later	Snow leopard belt; Uttarkashi
Dudhwa NP / TR	**Uttar Pradesh**	NP **1977**; TR **1987**	Lakhimpur Kheri Terai; rhino from **1984**; barasingha
Pilibhit TR	**Uttar Pradesh**	**2014**	Terai; tiger + woodland swamp deer
Ranipur TR	**Uttar Pradesh**	**2022/2023**	Chitrakoot–Banda (Vindhyan); not Terai
Hastinapur WLS	**Uttar Pradesh**	**1986**	**Largest UP sanctuary**; Meerut–Bijnor belt; Haiderpur Ramsar inside
Katarniaghat WLS	**Uttar Pradesh**	**1975**	Terai Arc toward Nepal; gharial / dolphin

UP has **no** biosphere reserve and **no** marine PA. UK biosphere = **Nanda Devi (1988; UNESCO MAB 2004)**.

6. National Air Quality Index (AQI)

Launched in **October 2014** under Swachh Bharat Abhiyan by CPCB: - **Six Categories**: 1. *Good* (0–50) 2. *Satisfactory* (51–100) 3. *Moderately Polluted* (101–200) 4. *Poor* (201–300) 5. *Very Poor* (301–400) 6. *Severe* (401–500). - **Eight Monitored Pollutants (Frequently Tested in PCS!)**: 1. **Particulate Matter 10 (PM10)** 2. **Particulate Matter 2.5 (PM2.5)** 3. **Nitrogen Dioxide (NO2)** 4. **Sulphur Dioxide (SO2)** 5. **Carbon Monoxide (CO)** 6. **Ozone (O3)** 7. **Ammonia (NH3)** 8. **Lead (Pb)**. *(Notice: Carbon Dioxide (CO2) is NOT part of the AQI 8 pollutants!)*

7. The 18 Biosphere Reserves of India & 12 UNESCO MAB Reserves

No.	Biosphere Reserve	Year Set Up	State / Region	Part of UNESCO MAB (WNBR)?
1	**Nilgiri**	**1986**	Tamil Nadu, Kerala, Karnataka	**Yes (2000 — First in India)**
2	**Nanda Devi**	**1988**	**Uttarakhand** (Chamoli, Pithoragarh, Bageshwar)	**Yes (2004)**
3	**Nokrek**	**1988**	Meghalaya (Garo Hills)	**Yes (2009)**
4	**Great Nicobar**	**1989**	Andaman and Nicobar Islands	**Yes (2013)**
5	**Gulf of Mannar**	**1989**	Tamil Nadu (marine)	**Yes (2001)**
6	**Manas**	**1989**	Assam	No

No.	Biosphere Reserve	Year Set Up	State / Region	Part of UNESCO MAB (WNBR)?
7	**Sundarbans**	**1989**	West Bengal	**Yes (2001)**
8	**Similipal**	**1994**	Odisha (Mayurbhanj)	**Yes (2009)**
9	**Dibru-Saikhowa**	**1997**	Assam (*Smallest Biosphere Reserve*)	No
10	**Dehang-Debang**	**1998**	Arunachal Pradesh	No
11	**Pachmarhi**	**1999**	Madhya Pradesh (Satpura Range)	**Yes (2009)**
12	**Khangchendzonga**	**2000**	Sikkim	**Yes (2018)**
13	**Agasthyamalai**	**2001**	Kerala and Tamil Nadu	**Yes (2016)**
14	**Achanakmar-Amarkantak**	**2005**	Madhya Pradesh and Chhattisgarh	**Yes (2012)**
15	**Great Rann of Kutch**	**2008**	Gujarat (*Largest Biosphere Reserve*)	No
16	**Cold Desert**	**2009**	Himachal Pradesh (Pin Valley, Kibber, Chandratal)	No
17	**Seshachalam Hills**	**2010**	Andhra Pradesh (Tirupati)	No
18	**Panna**	**2011**	Madhya Pradesh	**Yes (2020)**

8. International Environmental Conventions & Protocols

Year	Convention / Agreement	Place / City	Primary Objective / Regulatory Scope
1971 (2 Feb)	**Ramsar Convention on Wetlands**	Ramsar (Iran)	Conservation and wise use of wetlands; entered into force 1975; **World Wetlands Day (2 Feb)**; India joined **1 Feb 1982**; Keoladeo & Chilika were first sites; **Montreux Record** (Keoladeo NP & Loktak Lake)
1972 (5–16 Jun)	**Stockholm Conference (UNCHE)**	Stockholm (Sweden)	UN Conference on the Human Environment; **Created UNEP (Nairobi, Kenya)**; celebrated as **World Environment Day (5 June)**
1973 (3 Mar)	**CITES (Washington Convention)**	Washington, D.C. (USA)	Convention on International Trade in Endangered Species of Wild Fauna and Flora; Appendix I, II, III; **World Wildlife Day (3 March)**
1979	**Bonn Convention (CMS)**	Bonn (Germany)	Convention on the Conservation of Migratory Species of Wild Animals
1982	**UNCLOS**	Montego Bay (Jamaica)	Law of the Sea: Territorial waters (12 nm), Contiguous zone (24 nm), Exclusive Economic Zone (200 nm)
1985	**Vienna Convention**	Vienna (Austria)	Framework convention for the protection of the stratospheric ozone layer
1987 (16 Sep)	**Montreal Protocol**	Montreal (Canada)	**Legally binding phase-out of Ozone Depleting Substances (ODS / CFCs)**; entered into force 1 Jan 1989; **World Ozone Day (16 Sept)**; **Kigali Amendment (2016)** phase out HFCs
1987	**Brundtland Report**	UN World Commission	Report titled ***Our Common Future***; formally defined **Sustainable Development**
1989	**Basel Convention**	Basel (Switzerland)	Control of transboundary movements of **Hazardous Wastes and their disposal**; entered into force 1992
1992 (3–14 Jun)	**Rio Earth Summit (UNCED)**	Rio de Janeiro (Brazil)	Landmark summit producing 3 Rio Conventions: 1. **UNFCCC** (Climate Change) 2. **CBD** (Biological Diversity) 3. **UNCCD** (Combating Desertification) Also produced: **Agenda 21**, Rio Declaration, and Forest Principles
1997 (11 Dec)	**Kyoto Protocol**	Kyoto (Japan)	UNFCCC COP-3; **Legally binding greenhouse gas emission reduction targets for Annex-I developed countries**; 6 greenhouse gases (CO2, CH4, N2O, HFCs, PFCs, SF6); Clean Development Mechanism (CDM); entered into force 2005; expired 2020

Year	Convention / Agreement	Place / City	Primary Objective / Regulatory Scope
1998	**Rotterdam Convention**	Rotterdam (Netherlands)	Prior Informed Consent (PIC) procedure for certain hazardous chemicals and pesticides in international trade
2000	**Cartagena Protocol on Biosafety**	Montreal / Cartagena	Protocol to CBD governing the safe transfer, handling, and use of **Living Modified Organisms (LMOs)**
2001	**Stockholm Convention on POPs**	Stockholm (Sweden)	Elimination or restriction of **Persistent Organic Pollutants (POPs)** (original "Dirty Dozen" including DDT, Dioxins, PCBs)
2010	**Nagoya Protocol**	Nagoya (Japan)	Supplementary agreement to CBD on **Access to Genetic Resources and the Fair and Equitable Sharing of Benefits (ABS)**
2013	**Minamata Convention**	Kumamoto (Japan)	Global treaty to protect human health and the environment from adverse effects of **Mercury (Hg)**
2015 (12 Dec)	**Paris Agreement**	Paris (France)	UNFCCC COP-21; keep global temperature rise **well below 2°C** above pre-industrial levels, pursue efforts to limit rise to **1.5°C**; Nationally Determined Contributions (NDCs); entered into force **4 Nov 2016**
2015	**Sustainable Development Goals (SDGs)**	UN General Assembly	**17 Goals, 169 Targets (2015–2030)**; SDG 1: No Poverty, SDG 2: Zero Hunger, SDG 3: Good Health, SDG 4: Quality Education, SDG 5: Gender Equality, SDG 6: Clean Water & Sanitation, SDG 13: Climate Action, SDG 14: Life Below Water, SDG 15: Life on Land
2022	**Kunming-Montreal GBF**	Montreal (CBD COP-15)	Global Biodiversity Framework; includes landmark **"30x30 target"** (protect at least 30% of planet's land and ocean by 2030)

9. Fundamental Ecological Principles & Laws

A. Lindeman's 10% Energy Law (1942)

- Formulated by **Raymond Lindeman**.
- Only **approximately 10% of the energy** entering any trophic level is transferred and organized into organic matter at the next higher trophic level.
- 90% of energy is lost as metabolic heat, respiration, and catabolism.
- Explains why food chains rarely exceed 4 to 5 trophic levels.

B. Ecological Pyramids (Charles Elton, 1927)

1. **Pyramid of Energy**: - **ALWAYS UPRIGHT, WITHOUT EXCEPTION!** - Energy decreases progressively at each successive trophic level according to thermodynamics.
2. **Pyramid of Biomass**: - **Terrestrial Ecosystems (Grassland, Forest)**: **UPRIGHT** (Producer biomass > Herbivore > Carnivore). - **Aquatic Ecosystems (Pond, Lake, Ocean)**: **INVERTED** (Phytoplankton standing crop biomass is much smaller than zooplankton and predatory fish).
3. **Pyramid of Numbers**: - **Grassland & Cropland**: **UPRIGHT** (Millions of grasses support fewer herbivores, and even fewer carnivores). - **Tree / Parasitic Ecosystem**: **INVERTED** (A single large tree supports hundreds of birds, which host thousands of parasites and hyper-parasites). - **Forest Ecosystem**: **Spindle-shaped** (Few large trees support many herbivores, which are eaten by fewer carnivores).

C. Bioaccumulation vs Biomagnification

- **Bioaccumulation**: Increase in the concentration of a chemical pollutant in an **individual organism over time** (rate of intake > rate of excretion).
- **Biomagnification**: Progressive increase in the concentration of persistent, non-biodegradable, fat-soluble toxins (e.g., **DDT, Mercury, PCBs**) at **successively higher trophic levels** of a food chain.

- *Classic example:* Phytoplankton (0.00005 ppm) → Zooplankton → Small Fish → Fish-eating birds / Falcons (25 ppm causing egg-shell thinning and population collapse).

D. Greenhouse Gases: Global Warming Potential (GWP, 100-Year Horizon)

Ranked from highest warming impact per molecule to lowest (relative to CO2 = 1):

SF6 (23,500) > PFCs (7,000–9,000) > HFCs (1,400–3,000) > N2O (273) > CH4 (28) > CO2 (1)

- **Water Vapour**: Most abundant natural greenhouse gas contributing greatest share to natural greenhouse effect (~60%), but has short atmospheric residence time.
- **Carbon Dioxide (CO2)**: Primary anthropogenic driver of global climate change due to massive volume and century-long residence time.
- **Methane (CH4)**: Major sources: Paddy/rice fields, ruminant livestock enteric fermentation, landfills, wetlands, coal mining.

10. Important Environmental Days (UPPCS Direct Match)

Date	Environmental Observance	Connected Milestone / Significance
2 February	**World Wetlands Day**	Signing of the Ramsar Convention on Wetlands (1971)
3 March	**World Wildlife Day**	Adoption of CITES (Washington Convention, 1973)
21 March	**International Day of Forests**	Promotes awareness of the importance of all types of forests
22 March	**World Water Day**	Focuses on the importance of freshwater
23 March	**World Meteorological Day**	Establishment of WMO (1950)
22 April	**Earth Day**	First held in 1970; global environmental movement
22 May	**International Day for Biological Diversity**	Adoption of the agreed text of the CBD (1992)
5 June	**World Environment Day**	Opening day of the historic Stockholm Conference (1972)
8 June	**World Oceans Day**	Concept originally proposed at Rio Earth Summit 1992
29 July	**International Tiger Day**	Instituted at the Saint Petersburg Tiger Summit (2010)
12 August	**World Elephant Day**	Dedicated to the preservation and protection of elephants
16 September	**World Ozone Day**	Signing of the Montreal Protocol (1987)
5 October	**National Dolphin Day (India)**	Declaration of Gangetic Dolphin as National Aquatic Animal (2009)
16 October	**World Food Day**	Founding of the FAO (1945)
5 December	**World Soil Day**	Promotes importance of healthy soil and sustainable soil management

11. Foundational Ecological Terms & Coining Scientists / Pioneers

Term / Ecological Concept	Coined By / Pioneer	Year	Core Definition / Exam Relevance
Ecology (*Oekologie*)	**Ernst Haeckel**	**1866**	Coined from Greek *oikos* (house/dwelling) + *logos* (study). Study of interactions between living organisms and their physical environment.
Ecosystem	**Arthur Tansley (A.G. Tansley)**	**1935**	The basic functional ecological unit consisting of biotic community together with its abiotic environment interacting as a system.
Biodiversity (Biological Diversity)	**Walter G. Rosen** (coined); popularized by **Edward O. Wilson (E.O. Wilson)**	**1985 / 1988**	Coined as contraction of "biological diversity". E.O. Wilson published landmark book *Biodiversity* (1988); known as "Father of Biodiversity".

Term / Ecological Concept	Coined By / Pioneer	Year	Core Definition / Exam Relevance
Biodiversity Hotspots	**Norman Myers**	**1988**	Biogeographic regions characterized by exceptional levels of plant endemism and high levels of habitat loss (≥70% loss). Adopted by Conservation International.
Ecological Niche	**Joseph Grinnell** (habitat niche); refined by **Charles Elton** & **G.E. Hutchinson**	**1917 / 1927 / 1957**	The specific functional role and physical space occupied by an organism within its ecosystem (*address + profession of species*). Hutchinson introduced "fundamental vs realized niche".
Deep Ecology	**Arne Naess**	**1973**	Radical environmental philosophy rejecting anthropocentrism; advocates that all living beings have intrinsic moral value regardless of their utility to humans.
Ecological Footprint	**William Rees & Mathis Wackernagel**	**1992**	Measure of human demand on Earth's ecosystems; the area of biologically productive land and water required to produce goods consumed and absorb waste.
Gaia Hypothesis	**James Lovelock & Lynn Margulis**	**1972**	Proposes that living organisms interact with their inorganic surroundings on Earth to form a synergistic and self-regulating complex system that helps maintain conditions for life.
Acid Rain	**Robert Angus Smith**	**1872**	Discovered connection between atmospheric pollution (SO2 & NOx from coal burning) and acid precipitation in Manchester, UK. Normal rain pH ~5.6; acid rain pH < 5.6.
Smog (Smoke + Fog)	**H.A. Des Voeux**	**1905**	Coined at the Public Health Congress in London to describe the deadly mixture of coal smoke and dense winter fog (**Classical / London Smog — reducing in nature, contains SO2**).
Photochemical Smog (Los Angeles Smog)	**Arie Haagen-Smit**	**1950s**	Sunlight-induced photochemical reaction between Nitrogen Oxides (NOx) and Volatile Organic Compounds (VOCs). **Oxidizing in nature; major components: Ozone (O3), PAN (Peroxyacetyl nitrate), and Aldehydes**.
Discovery of Antarctic Ozone Hole	**Joe Farman, Brian Gardiner, Jonathan Shanklin**	**May 1985**	British Antarctic Survey scientists discovered drastic springtime thinning of the stratospheric ozone layer over Halley Bay, Antarctica, caused by Chlorofluorocarbons (CFCs).
Biocoenosis	**Karl Möbius**	**1877**	An interacting community of different living species occupying a common biotope/habitat.
Father of Indian Ecology	**Prof. Ramdeo Misra**	1950s–1970s	Established pioneering ecological research and education in India at Banaras Hindu University (BHU), Varanasi.
Ecotone & Edge Effect	**Frederic Clements**	1905	**Ecotone**: Transition zone between two distinct biomes/communities (e.g., mangrove between marine & terrestrial; marshland between aquatic & dry land). **Edge Effect**: Phenomenon where species richness and population density are significantly greater in the ecotone than in either adjoining community (e.g., birds in forest edge).

12. Major Environmental Pollutants, Toxic Heavy Metals & Occupational Diseases

Toxic Pollutant / Metal	Source / Environmental Medium	Characteristic Disease / Syndrome	Clinical Pathology / Prelims Highlights
Methylmercury (Hg)	Industrial effluent discharged into coastal waters; biomagnified in fish	**Minamata Disease**	First discovered in **Minamata Bay (Kumamoto, Japan, 1956)**; caused by Chisso Corp chemical plant; severe neurological syndrome, ataxia, loss of motor control, sensory impairment, convulsions, congenital deformities.

Toxic Pollutant / Metal	Source / Environmental Medium	Characteristic Disease / Syndrome	Clinical Pathology / Prelims Highlights
Cadmium (Cd)	Mining runoff, zinc smelting, contaminated irrigation water for paddy fields	**Itai-Itai Disease ("Ouch-Ouch Disease")**	First discovered in **Toyama Prefecture (Japan, 1912)**; severe osteomalacia (extreme bone softening and painful skeletal fractures) and kidney failure; *Itai-Itai* means "it hurts, it hurts" in Japanese.
Nitrate (NO_3^-)	Runoff from nitrogenous chemical fertilizers in groundwater drinking water	**Blue Baby Syndrome (Methemoglobinemia)**	Nitrate converts to nitrite in infant gut; binds with infant hemoglobin to form **methemoglobin**, which cannot release oxygen to tissues; skin turns blue due to asphyxiation; fatal if untreated.
Arsenic (As)	Natural geochemical leaching into shallow groundwater aquifers	**Blackfoot Disease, Hyperkeratosis & Arsenicosis**	Severe peripheral vascular gangrene ("black foot"), skin hyper-pigmentation, keratosis, and bladder/skin cancer. Rampant in **Ganga-Brahmaputra alluvial plains: West Bengal, UP, Bihar, and Bangladesh**. Permissible limit: **0.01 mg/L** (WHO).
Fluoride (F^-)	Over-extracted groundwater leaching from fluorapatite minerals	**Dental & Skeletal Fluorosis**	Permissible limit: **1.0 to 1.5 mg/L**. Excess causes mottling of teeth, calcification of ligaments, severe crippling skeletal deformity, and **Knock-Knee Syndrome (Genu Valgum)**. Widespread in Rajasthan, Telangana, UP.
Silica (SiO_2 dust)	Inhalation of crystalline silica dust in stone quarries, agate cutting, sandblasting	**Silicosis**	Incurable occupational fibrotic lung disease causing permanent pulmonary scarring and progressive respiratory failure.
Asbestos Fibers	Asbestos cement manufacturing, building demolition, brake lining manufacture	**Asbestosis & Mesothelioma**	Asbestos fibers penetrate deep into lungs causing pulmonary fibrosis; strongly carcinogenic, inducing **malignant mesothelioma** (cancer of pleural lining).
Coal Dust	Underground coal mining and extraction	**Pneumoconiosis (Black Lung Disease / Anthracosis)**	Chronic inflammation and irreversible progressive massive fibrosis of lungs due to accumulated inhaled coal mine dust.
Cotton Dust	Cotton mills, textile spinning, and ginning factories	**Byssinosis ("Brown Lung Disease" / Monday Fever)**	Occupational asthma and chest tightness occurring on the first workday after a weekend break (*Monday disease*).
Sugarcane Bagasse	Inhalation of dry bagasse fibers during sugarcane processing & paper milling	**Bagassosis**	Hypersensitivity pneumonitis triggered by thermoactinomycetes in moldy bagasse.
Lead (Pb)	Lead-acid batteries, leaded gasoline (historic), paints, smelters, e-waste	**Plumbism / Saturnism**	Neurotoxic; causes severe cognitive and developmental impairment in children, anemia (inhibits heme synthesis), kidney damage, and characteristic **Burtonian blue line along gums**.
E-Waste Hazardous Elements	Discarded computers, mobile phones, printed circuit boards (PCBs)	Multi-organ toxicity	Lead (solder & CRTs), Cadmium (rechargeable Ni-Cd batteries & chip resistors), Mercury (switches & flat screens), Brominated Flame Retardants (BFRs - endocrine disruption).

13. Biodiversity Hotspots of India & Global Criteria

A. Global Criteria for Biodiversity Hotspot (Norman Myers, 1988)

To qualify as an international biodiversity hotspot, a biogeographic region must fulfill **TWO STRICT QUANTITATIVE CRITERIA**: 1. **High Endemism**: It must contain at least **1,500 species of vascular plants (>0.5% of the world's**

total) as endemics (species found nowhere else on Earth). 2. **High Degree of Threat**: It must have lost at least **70% of its original primary native vegetation** (i.e., has 30% or less of its original natural habitat remaining). - Globally, there are **36 recognized Biodiversity Hotspots** covering only 2.5% of Earth's land surface, but supporting over 50% of world's endemic plant species and 43% of endemic bird, mammal, reptile, and amphibian species.

B. The 4 Biodiversity Hotspots Covering India

Hotspot Name	Geographical Extent in India	Adjoining International Regions	Landmark Endemic Species / Key Features
1. Himalaya Hotspot	Covers the **entire Indian Himalayan region**: Jammu & Kashmir, Ladakh, Himachal Pradesh, **Uttarakhand**, Sikkim, and Arunachal Pradesh.	Pakistan, Nepal, Bhutan, Tibet / Southern China	Snow Leopard, Himalayan Musk Deer, Himalayan Tahr, Blue Sheep (*Bharal*), Red Panda; vast alpine and temperate rhododendron-oak ecosystems.
2. Indo-Burma Hotspot	Covers all of **North-East India south of the Brahmaputra River** (Assam plains/hills, Meghalaya, Nagaland, Manipur, Mizoram, Tripura) and the **Andaman Islands**.	Myanmar, Thailand, Laos, Vietnam, Cambodia, and Southern China	Hoolock Gibbon (India's only ape), Golden Langur, Stump-tailed Macaque; extraordinary diversity of orchids, ginger species, and freshwater turtles.
3. Western Ghats & Sri Lanka Hotspot	Covers the continuous mountain chain along the western coast across **Gujarat, Maharashtra, Goa, Karnataka, Kerala, and Tamil Nadu**.	Island nation of Sri Lanka	**Lion-Tailed Macaque** (Silent Valley), **Nilgiri Tahr** (Eravikulam), Nilgiri Langur, Malabar Gliding Frog; exceptional concentration of endemic amphibians (~80% endemic), reptiles, and flowering plants.
4. Sundaland Hotspot	Covers the **Nicobar Islands** of India (Great Nicobar, Little Nicobar).	Indonesia (Sumatra, Java, Borneo), Malaysia, Singapore, Brunei, and Southern Thailand	Nicobar Megapode (mound-building bird), Nicobar Tree Shrew, Saltwater Crocodile; rich coastal coral reefs and dense tropical rainforests.

> **Crucial Exam Trap:** Andaman Islands fall under **Indo-Burma Hotspot**, while Nicobar Islands fall under **Sundaland Hotspot**!

Topic 1 — Environment Basics

Consolidated — 31 Must-Score Facts

1. Environment means the sum of **biotic and abiotic** factors that affect living organisms. Notes do not treat "only physical" or "only living" as the full definition.
2. The four (चातुर्याम) spheres are **lithosphere, hydrosphere, atmosphere, and biosphere**. Biosphere roughly spans from about **200 m below soil** to about **6 km** above sea level.
3. **Ecology** (पारिस्थितिकी) is the **study** of organism–environment links. An **ecosystem** (पारिस्थितिकी तंत्र) is the **functional unit**. A **habitat** (वास स्थान) is the **place**. A **niche** is the **role**.
4. Abiotic factors include light, temperature, water, soil, gases, pH, and salinity. **Edaphic** means soil; **climatic** means weather; **topographic** means slope and altitude.
5. Biotic structure runs **producers → consumers → decomposers**. **Rhizobium** fixes nitrogen in legumes.
6. **Liebig's law** says growth is limited by the **scarcest** essential resource. **Shelford's law** says a species survives only inside a **min–max tolerance** range.
7. Land is about **29%** of Earth's surface and water about **71%**. **Lentic** means standing water; **lotic** means flowing water. Mangroves and wetlands are **transitional**.
8. Human–environment thought moves **environmental determinism (Ratzel) → possibilism (Vidal de la Blache) → neo-determinism (Griffith Taylor)**.
9. **IPAT** is Impact = Population (जनसंख्या) × Affluence × Technology. Hardin's **Tragedy of the Commons** dates to **1968**.
10. Sustainable development is defined in the **Brundtland Report (1987)** (ब्रंटलैंड रिपोर्ट). **Rio 1992** produced **Agenda 21**, **UNFCCC** (यूएनएफसीसीसी), and **CBD**.
11. India's **LiFE** Mission launched in **2022**; the idea was floated at **COP26 in 2021**, not COP25.
12. **Chipko** (चिपको) began in **1973** at **Reni, Uttarakhand (उत्तराखंड)**. **JFM** guidance is **1990**. **FRA** is **2006**.
13. **Article 48A** is the State DPSP (राज्य के नीति निदेशक तत्व) on environment. **Article 51A(g)** is the citizen's Fundamental Duty. **Article 48** is about agriculture and cows — not environment.
14. **Article 21** (अनुच्छेद 21) has been read to include a **healthy environment**. The **42nd (42वां) Amendment (1976)** added 48A and 51A(g).
15. The **Environment (Protection) Act, 1986** is India's post-**Bhopal (1984)** umbrella law.
16. **MoEF** began in **1985** and became **MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) in 2014**. Headquarters is New Delhi.
17. **Stockholm 1972** (स्टॉकहोम सम्मेलन) was the first UN human-environment conference and led to **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) at **Nairobi**.
18. World Environment Day (विश्व पर्यावरण दिवस) is **5 June**. Earth Day (पृथ्वी दिवस) is **22 April**. World Wildlife Day (विश्व वन्यजीव दिवस) is **3 March**, not 22 March.
19. World Wetlands Day (विश्व आर्द्रभूमि दिवस) is **2 February**. International Ozone Day is **16 September**. Desertification (मरुस्थलीकरण) and Drought (सूखा) Day is **17 June**. International Day for Biological Diversity is **22 May**.
20. UNEP headquarters is **Nairobi, Kenya**. CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड) and MoEFCC sit in **New Delhi**.
21. The **Brundtland Commission (WCED)** worked **1983–1987** and defined sustainable development in *Our Common Future*.
22. Uttar Pradesh (उत्तर प्रदेश) mainly falls in the **Gangetic Plain** biogeographic zone (**Zone 7**).
23. Classic UP river–city facts include Ganga (गंगा) at Varanasi (वाराणसी) and Kanpur (कानपुर), Yamuna (यमुना) at Agra, and Gomti at Lucknow (लखनऊ).
24. Pollution hotspots often tested for UP include **Kanpur tanneries**, NCR air (Ghaziabad/Noida), and Ganga ghats at **Varanasi** (वाराणसी).
25. **Ganga Action Plan** (गंगा एक्शन प्लान) began in **1985**. **Namami Gange** (नमामि गंगे) launched in **2014**.

26. A standard triad classifies environment as **physical**, **biological**, and **cultural**. **Operational environment** is **not** part of that triad.
27. Natural versus anthropogenic, terrestrial versus aquatic, and internal versus external are further common classification axes.
28. Micro-, meso-, and macro-scales describe environment size from a pond edge to a region or the globe. Competitive papers usually default to the **natural/physical** sense of environment.
29. Agenda 21 belongs to **Rio 1992**, not 1995. Mixing Brundtland 1987 with Agenda 21's year is a common trap.
30. **Carrying capacity** is the maximum load an environment can sustain indefinitely. Resource extraction must stay **below** regeneration, and waste must stay **within** absorption capacity — not the reverse.
31. **Ecological footprint** (पारिस्थितिक पदचिह्न) measures the land/water area needed to support consumption, in **global hectares (gha)**.

📋 Must-score facts — spheres, laws, articles, days

Core tags

Item	Lock
Ecology / Ecosystem / Habitat / Niche	Study / functional unit / place / role
Four spheres	Lithosphere · hydrosphere · atmosphere · biosphere
Biosphere span	~200 m below soil → ~6 km above sea level
Edaphic / Climatic / Topographic	Soil / weather / slope–altitude
Liebig / Shelford	Scarcest resource / min–max tolerance
Lentic / Lotic	Standing / flowing water
Thought chain	Ratzel determinism → Vidal possibilism → Griffith Taylor neo-determinism
IPAT / Tragedy of Commons	P×A×T / Hardin **1968**

Milestones & law

Item	Lock
Brundtland / Rio	**1987** SD definition / **1992** Agenda 21 + UNFCCC + CBD
LiFE	Idea COP26 **2021**; launch **2022**
Chipko / JFM / FRA	**1973** Reni / **1990** / **2006**
Art. 48A / 51A(g) / 48	State DPSP env / citizen duty / agri–cows (**not** env)
42nd Amd / EPA	**1976** adds 48A+51A(g) / EPA **1986** post-Bhopal
MoEF → MoEFCC	**1985** → **2014**; HQ New Delhi
Stockholm / UNEP	**1972** → UNEP **Nairobi**
GAP / Namami Gange	**1985** / **2014**

Days (quick)

Item	Lock
WED / Earth / Wildlife	5 Jun / 22 Apr / **3 Mar** (≠ 22 Mar)
Wetlands / Ozone / Desertif. / Biodiversity	2 Feb / 16 Sep / 17 Jun / 22 May

EARTH'S FOUR SPHERES & ATMOSPHERIC STRATIFICATION

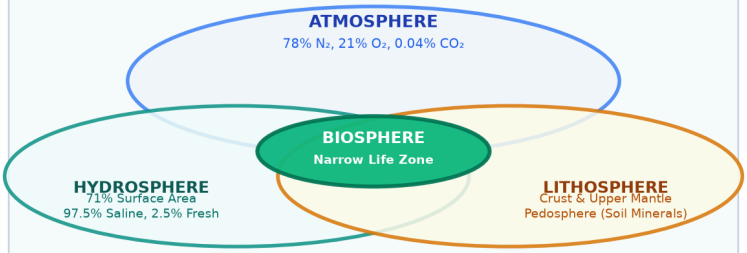
Visual Conceptual Guide: The Three Abiotic Realms, The Biosphere Life Zone & Troposphere/Stratosphere Boundary

VISUAL REVISION AID

1. THE FOUR INTERACTING SPHERES: BIOSPHERE AT THE OVERLAPPING TRI-JUNCTION

THERMODYNAMICS

- ENERGY: OPEN SYSTEM
  Constant solar influx and heat dissipation into space.
- MATTER: CLOSED SYSTEM
  Nutrients (C, N, P, H_2O) recirculate endlessly; cannot escape Earth!
- FOUNDERS:
  Suess (1875): Term
  Vernadsky (1926): Cycle



VERTICAL LIMITS

- UPPER LIMIT:
  ~6,000 m (Alpine)
  Extreme low pressure, hypoxia & sub-zero.
- DEEP LIMIT:
  ~11,000 m (Trench)
  Mariana Trench vents; benthic barophiles.
- EXAM TRAP:
  Biosphere is a razor-thin planetary film!

2. ATMOSPHERIC VERTICAL STRATIFICATION & THE STRATOSPHERIC OZONE SHIELD

STRATOSPHERE (~12 to 50 km)

- Calm, dry, horizontal airflow; ideal cruising altitude for commercial jet aircraft
- **Thermal Inversion: Temperature RISES with altitude due to UV absorption by Ozone!**

OZONOSPHERE / OZONE LAYER (15 to 35 km)

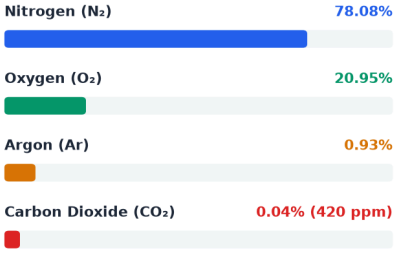
Absorbs 99% of lethal solar UV-C and harmful UV-B radiation!

TROPOPAUSE (Thermal Boundary ~12km)

TROPOSPHERE (0 to ~12 km)

- Holds 75% of total atmospheric mass & virtually all water vapour / clouds
- **ALL weather phenomena (clouds, rain, storms, convection) occur here!**
- Normal Lapse Rate: Temperature drops ~6.5°C per 1,000m rise (cooling with height)

ATMOSPHERIC GAS COMPOSITION



Topic 2 — Ecology & Ecosystem

Consolidated — 39 Must-Score Facts

1. **Ecology** is the **study** of organism–environment (पर्यावरण) interactions. **Haeckel (1866)** coined the term. **Ramdeo Misra** is called the father of Indian ecology. **Bionomics** (बायोनॉमिक्स) is synonymous with ecology; it does **not** mean "management of life."
2. An **ecosystem** (पारिस्थितिकी तंत्र) is a **functional unit** of biotic and abiotic interaction with energy flow and nutrient cycling. **Tansley (1935)** coined the word.
3. Ecosystems are normally **open systems**. Calling an ecosystem a closed system is wrong.
4. Four (चातुर्याम) basic components often listed together are **abiotic materials, producers, consumers, and decomposers**.
5. The seven ecological levels run **organism → population (जनसंख्या) → community → ecosystem → landscape → biome (जीवोम) → biosphere**. A frequent shorter order asked is **Population → Community → Ecosystem → Landscape**.
6. The **biosphere** is the largest ecosystem of Earth. Among ecosystem *types*, the **marine/ocean** ecosystem covers the largest area and is cited as the **most stable**.
7. Ecosystems may be natural or anthropogenic, terrestrial or aquatic, and **lentic** (standing) or **lotic** (flowing).
8. Artificial / anthropogenic examples include **farmland / rice field**, orchard, garden, aquarium, and human-made reservoir.
9. A pond's vertical/horizontal zones include **littoral → limnetic → profundal**. Marine zones include littoral, neritic, pelagic, benthic, and abyssal.
10. Structure runs **producers → primary/secondary/tertiary consumers → decomposers and detritivores**. Producers generally outnumber primary consumers.
11. Core ecosystem functions are **productivity, decomposition, nutrient cycling, and unidirectional energy flow**.
12. A **niche** is a species' **role**, not its place. A **habitat** (वास स्थान) is the place. **Joseph Grinnell (1917)** coined the ecological-niche concept.
13. A **fundamental niche** (मूल (मौलिक) निकेत) is the full potential role without competitors. A **realized niche** (वास्तविक निकेत) is the narrower role after competition.
14. **Gause's competitive exclusion** says two species with identical niches cannot stably coexist when resources are limited.
15. **Liebig** = limiting factor; **Shelford** = tolerance range; **Lindeman (1942)** ≈ **10%** energy transfer between trophic levels.
16. **NPP = GPP − plant respiration**. Plants capture only about **1%** of incident sunlight on leaves.
17. Decreasing productivity order often tested: **Mangroves > grasslands > lakes > oceans**.
18. **Primary succession** (प्राथमिक अनुक्रमण) starts on a lifeless substrate. **Secondary succession** (द्वितीयक अनुक्रमण) starts where **soil already exists** after disturbance.
19. Full succession phases: **Nudation → Migration → Ecesis → Reaction → Stabilisation**. The full community sequence is a **sere** (क्रमक). UPPCS 2024 often starts from Migration when Nudation is omitted.
20. Pioneers such as lichen and moss lead through seral stages to a **climax community** (चरम समुदाय). **Monoclimax** theory is linked to **F.E. Clements**. **Hydrosere** is wet; **xerosere** is dry.
21. In grasslands, trees often fail to replace grasses because of **water limits and fire**.
22. An **r-strategist** produces many offspring with fast growth and short life. A **K-strategist** produces few offspring with slow growth and long life.
23. Population growth models contrast the **J-curve** (exponential) with the **S-curve** (logistic) that levels near **carrying capacity**.

24. MEA (**2005**) ecosystem services are usually **provisioning, regulating, supporting, and cultural**. Some WDR/UPSC (संघ लोक सेवा आयोग) framings add **preserving**. Supporting examples include nutrient cycling and (in that framing) crop pollination.
25. Ecosystem balance is a **dynamic equilibrium**. **Resilience** is the capacity to recover after disturbance. **Deforestation** is a main cause of ecological imbalance in India.
26. **Eutrophication** (सुपोषण) is nutrient overload that drives algal bloom and a dissolved-oxygen crash.
27. Energy flow is **one-way**. Nutrients **cycle**. Saying energy cycles is a classic wrong option.
28. A **community** is all populations of different species in an area. A **population** is one interbreeding species in one area.
29. An **ecotone** (इकोटोन) is the transition zone between two communities or ecosystems.
30. Aquatic clines: **pycnocline** = density gradient; **halocline** = salinity gradient; **thermocline** = temperature-with-depth gradient.
31. In the **phosphorus cycle**, weathering of rocks is the main source releasing the nutrient into the cycle.
32. **Upwelling** brings cold, nutrient-rich water to the surface and raises marine productivity.
33. **Eucalyptus** is often treated as an environmental hazard because it draws heavy groundwater and can lower the water table.
34. UP lotic examples include the **Ganga (गंगा), Yamuna (यमुना), and Gomti**. Terai (तराई) forest and Vindhyan scrub are terrestrial examples.
35. UP wetland examples include **Sur Sarovar (Agra)** and **Nawabganj (Unnao)**. Western UP sugarcane belts are classic agro-ecosystems.
36. Odum is linked to modern ecosystem ecology; keep him separate from Haeckel (term ecology) and Tansley (term ecosystem). **Deep ecology** was coined by **Arne Næss (1973)**. **Ecological Transition** was used by **John W. Bennett**.
37. **Allen's Rule:** colder climates → smaller appendages (less heat loss). **Bergmann's Rule:** colder climates → larger body size (heat conservation). Do not swap the pair.
38. **Gloger's Rule:** darker pigmentation is more common in warm, humid regions.
39. **Limiting similarity** means coexisting species differ enough in niche use that they do not compete for exactly the same resource at the same time.

📋 Must-score facts — terms, succession, productivity

Founders / levels

Item	Lock
Ecology term / Indian ecology	Haeckel **1866** / Ramdeo Misra
Ecosystem term	Tansley **1935**; normally **open** system
Levels (short)	Population → Community → Ecosystem → Landscape
Niche coined	Grinnell **1917** (role ≠ place)
Lindeman	**10%** energy transfer (**1942**)
NPP	GPP − plant respiration; plants ~**1%** of sunlight

Succession / services

Item	Lock
Primary vs secondary	Lifeless substrate / soil already present
Sere phases	Nudation → Migration → Ecesis → Reaction → Stabilisation
Monoclimax	F.E. Clements

Item	Lock
r vs K	Many offspring–short life / few–long life
MEA 2005 services	Provisioning · regulating · supporting · cultural
Eutrophication	Nutrient overload → bloom → DO crash
Energy vs nutrients	Energy **one-way**; nutrients **cycle**
Ecotone / Deep ecology	Transition zone / Arne Næss **1973**

Topic 3 — Food Chain & Energy Flow

Consolidated — 30 Must-Score Facts

1. A **food chain** is a single linear who-eats-whom sequence. A **food web** is a network of interconnected chains and is more stable than one chain alone.
2. A **grazing food chain** starts from **living green plants**. A **detritus food chain** starts from **dead organic matter**.
3. Classic terrestrial chain: **Grass → Grasshopper → Frog → Snake**. Classic human-linked chain: **Grass → Goat → Human**.
4. Classic marine chain: **Diatoms → Crustaceans → Herrings**.
5. **Trophic levels** are feeding ranks counted from producers: **T1 producers, T2 herbivores, T3–T5 carnivores**.
6. The **Sun is the energy source**, not a trophic level. Never count sunlight as T1. **Solar energy** is the driving force of the ecosystem.
7. Energy flow is **unidirectional**: Sun → producers → consumers → heat. Nutrients cycle; **energy does not**.
8. About **10%** of energy passes to the next trophic level (**Lindeman, 1942**). About **90%** is lost as heat, respiration, and movement. Energy **decreases** upward.
9. Food chains stay short, usually about **3–5 levels**, because usable energy runs out. Food chains are **not** found within a single species.
10. **Humans** are **primary as well as secondary** consumers (plants and meat).
11. Main marine primary producers are **phytoplankton**.
12. Primary consumers are mainly herbivores such as **deer**; **ants** may also act as primary consumers (and scavengers/decomposers in broader roles). Fox and tiger are not primary consumers.
13. Decomposers are **fungi and bacteria**, not viruses.
14. The **energy pyramid** (kcal/m^2/yr) is **always upright**. An inverted energy pyramid is never correct.
15. A **biomass pyramid** can invert in **pond / marine** systems. A **numbers pyramid** can invert in a parasitic chain (one tree, many insects).
16. **Biomagnification** raises toxin concentration up the chain. For DDT, the **snake** (tertiary consumer) shows higher concentration than grasshopper or toad.
17. Butterfly decline harms **pollination** and can cut populations of predators such as wasps, spiders, and birds. It does not by itself force a drastic rise in fungal infections of crops.
18. Grazing does **not** start from dead matter. Detritus does **not** start from living green plants.
19. More links in a food web raise stability, but energy still flows **one way**.
20. Decomposers and detritivores close the detritus path and recycle nutrients back to producers.
21. Apex predators sit at the top trophic levels and are often scarce because little energy remains.
22. Dudhwa (दुधवा)'s grass → deer → tiger chain is a classic UP **grazing** example with tiger as apex predator.
23. A Ganga (गंगा) algae → fish → dolphin web is a standard UP **lotic aquatic** food-web example.
24. Terai (तराई) sal leaf litter → earthworm is a UP **detritus** start. Do not call it grazing.
25. Western UP sugarcane → pest → predator is a short **cropland** chain of about three levels.
26. Vulture decline after veterinary **diclofenac** (डाइक्लोफेनाक) disrupts the **detritus / scavenging** pathway.
27. Ecological pyramids may show energy, biomass, or numbers — always note which quantity is plotted.
28. Saying "energy is cyclic like nutrients" is a frequent false statement.
29. Match pyramid type to exception: energy never inverts; biomass may invert at sea/pond; numbers may invert with parasites.
30. Keep Lindeman's **10% rule** as an approximate key fact, not an exact laboratory constant for every ecosystem.

📋 Must-score facts — chains, pyramids, biomagnification

Chain types

Item	Lock
Food chain vs web	Linear sequence / interconnected network (more stable)
Grazing vs detritus	Living green plants / dead organic matter
Trophic ranks	T1 producers → T2 herbivores → T3–T5 carnivores
Sun	Energy **source** — **not** a trophic level
Lindeman 10%	~10% up; ~90% lost; energy ↓ upward
Humans	Primary **and** secondary consumers
Decomposers	Fungi + bacteria (≠ viruses)

Pyramids / toxins

Item	Lock
Energy pyramid	**Always upright**
Biomass / numbers invert	Pond–marine / parasitic chain
Biomagnification	Toxin ↑ up chain (DDT highest in apex)
Diclofenac	Vulture crash → scavenging pathway break
UP grazing example	Dudhwa grass → deer → tiger

Topic 4 — Biodiversity

Consolidated — 36 Must-Score Facts

1. Biodiversity is variability among living organisms at **genetic, species, and ecosystem** levels — the range of plants and animals in an area.
2. **Walter G. Rosen (1985)** coined the word *biodiversity*. **E.O. Wilson** popularised it. **Whittaker (1960)** advanced **alpha, beta, and gamma** diversity.
3. **Genetic diversity** is variation **within** a species. **Species diversity** is variety **among** species. **Ecosystem diversity** is variety of habitats and ecosystems.
4. **Alpha** = mean species diversity at a local habitat (वास स्थान). **Beta** = change between ecosystems. **Gamma** = overall regional diversity.
5. The most significant aspect of biodiversity is **maintenance of the ecosystem**, not food or drugs alone.
6. Species richness is generally highest in the **tropics / tropical rainforest** (optimum biome (जीवोम)) and **increases toward the equator**; it decreases toward the poles and with altitude.
7. On land, tropical rainforest is richest; in the sea, **coral reefs** are among the richest. UPPCS has accepted both rainforest and coral reefs for "relatively higher diversity."
8. India holds about **2.4%** of world land but about **7–8%** of global species (rough shares often cited: mammals ~7.6%, birds ~12.6%, reptiles ~6.2%, amphibians ~4.4%).
9. India's four (चातुर्याम) biodiversity hotspots are **Himalaya (हिमालय), Indo-Burma, Western Ghats–Sri Lanka, and Sundaland**. **Andamans** fall under **Indo-Burma**; **Nicobar** (निकोबार) under **Sundaland**. Eastern Ghats (पूर्वी घाट) / Vindhyas / Naro Hills are **not** hotspots.
10. A hotspot needs about **≥1500 endemic vascular plants** and **≥70%** original habitat lost. Criteria also stress **species richness, endemism, and threat**. There are **36** global hotspots (~2.5% of land; >50% endemic plants).
11. **Silent Valley (साइलेंट वैली) (Kerala)** is among India's richest rainforest biodiversity sites (NP **1984**; Silent Valley Project = Kerala). **Valley of Flowers** (फूलों की घाटी) is in **Chamoli (चमोली), Uttarakhand (उत्तराखंड)**. Himalayan richness reflects confluence of biogeographic zones.
12. The top cause of biodiversity loss is **habitat destruction**. HIPPO = Habitat, Invasive species (आक्रामक विदेशी प्रजाति), Pollution, Population (जनसंख्या) pressure, Over-exploitation. Vegetarianism is **not** a threat. Fewer trophic levels reduce diversity.
13. India aims for **Land Degradation (भू-क्षरण) Neutrality by 2030**.
14. **In-situ** (स्थान पर) = natural habitat (NP, sanctuary, BR, sacred grove). **Ex-situ** (स्थानांतरित) = outside habitat (zoo, botanical garden (वनस्पति उद्यान), seed bank (बीज बैंक)). Cryo-banks use **liquid nitrogen**. Biosphere reserves (जैवमंडल आरक्षित क्षेत्र) best join biodiversity with traditional human life / cultural diversity.
15. The **CBD (1992, Rio)** rests on three pillars: conserve, sustainable use, fair benefit-sharing. **Cartagena Protocol** = biosafety / LMOs. **Nagoya (2010)** = ABS.
16. India's **Biological Diversity Act, 2002** creates **NBA (Chennai, 2003)**, SBBs, BMCs, and **PBRs**. BMCs are central to Nagoya ABS, including collection fees.
17. On the IUCN Red List, **Threatened** = **VU + EN + CR**. **Endangered** = **EN** only. Red Data Book (रेड डेटा बुक) lists threatened species. A species not seen for **50 years** may be treated as extinct in older definitional stems.
18. An **endemic** species occurs only in one area. An **invasive alien** is non-native and harmful (**Prosopis juliflora** reduces native biodiversity).
19. Keystone / flagship / umbrella / indicator roles stay distinct.
20. International Day for Biological Diversity is **22 May** (earlier linked to **29 December**). UN **Decade on Biodiversity = 2011–2020**.
21. India's first **National Centre for Marine Biodiversity (NCMB)** is at **Jamnagar (Gujarat)**.
22. The **Kunming–Montreal GBF (2022)** includes the **30×30** target. CBD COP-10 = **Nagoya 2010**; COP-11 = **Hyderabad (हैदराबाद) 2012**.

23. Hotspot concept = **Norman Myers** (not BirdLife). **BirdLife International** identifies Important Bird and Biodiversity Areas.
24. **TRAFFIC** (WWF (डब्ल्यूडब्ल्यूएफ)–IUCN alliance, 1976) monitors wildlife trade (पण्याध्यक्ष) — not a UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) bureau.
25. India coral reefs: **Lakshadweep (लक्षद्वीप) (atolls), Gulf of Mannar, Gulf of Kachchh, Andaman (अंडमान) & Nicobar** — **not** Sundarbans (सुंदरबन) (mangroves). Coral bleaching (प्रवाल विरंजन) mainly from **warming seawater / global warming (भूमंडलीय तापन)**.
26. **Diclofenac** (डाइक्लोफेनाक) veterinary painkiller caused India's vulture crash. **Dugong** = herbivorous marine mammal (Schedule I); Gulf of Mannar / Palk. **Indian wild ass** = Rann of Kachchh. **Lion-tailed macaque** = Western Ghats (Kerala, Karnataka, Tamil Nadu (तमिलनाडु)).
27. India has **80 Ramsar sites**. **Tamil Nadu 16** (highest), **Uttar Pradesh (उत्तर प्रदेश) 10**, **Punjab 6**. **Montreux Record** lists Ramsar sites with adverse ecological change from human interference.
28. UP Terai (Dudhwa (दुधवा), Pilibhit) sits on the **Himalaya hotspot** fringe. Key UP species: tiger, rhino, barasingha, Gangetic dolphin; state bird **Sarus crane**.
29. Common UP invasives: water hyacinth, Parthenium, Lantana.
30. NBA HQ = **Chennai**. CBD ≠ Nagoya year. Threatened ≠ Endangered.
31. **Sea buckthorn** (Ladakh (लद्दाख) / cold Himalaya) checks erosion and is nutritious — not primarily a biodiesel (बायो-डीजल) timber crop.
32. **TEEB** makes nature's economic values visible for decision-makers. **Wetlands International** is an independent NGO (HQ Netherlands), not a Ramsar intergovernmental organ.
33. **CEBPOL** (Centre for Biodiversity Policy and Law) sits with NBA, **Chennai** (India–Norway).
34. Maximum butterflies in tropics because they **cannot tolerate low temperatures**.
35. **Gulf of Mannar Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र)** is often cited as India's richest marine biodiversity region (21 islands mosaic).
36. Afforestation does **not** cause biodiversity loss; road expansion, urbanisation, and agriculture expansion do (UKPCS-style stems).

📋 Must-score facts — levels, hotspots, CBD, IUCN

Concepts

Item	Lock
Term / populariser	Rosen **1985** / E.O. Wilson
Alpha / Beta / Gamma	Local / between ecosystems / regional
Genetic / Species / Ecosystem	Within species / among species / habitats
Richest land / sea	Tropical rainforest / coral reefs
India land vs species	~**2.4%** land; ~**7–8%** species
4 Indian hotspots	Himalaya · Indo-Burma · WG–SL · Sundaland
Andaman / Nicobar	Indo-Burma / Sundaland
Hotspot criteria	≥1500 endemic plants + ≥70% habitat lost; **36** global

Law / lists

Item	Lock
In-situ / Ex-situ	Natural habitat / outside (zoo, seed bank)
CBD / Cartagena / Nagoya	1992 Rio / biosafety LMOs / ABS **2010**
BDA / NBA	**2002** / NBA **Chennai 2003**

Item	Lock
Threatened (IUCN)	**VU + EN + CR** (Endangered ≠ all Threatened)
Biodiversity Day	**22 May**
Kunming–Montreal	**2022**; 30×30
India Ramsar	**80** sites; TN highest; UP **10** class fact
Hotspot concept	Norman Myers (≠ BirdLife)

Topic 5 — Habitat, Flora & Fauna

Consolidated — 24 Must-Score Facts

1. A **habitat** (वास स्थान) is the physical **place** where an organism lives. A **niche** is the **role**. An **ecosystem** (पारिस्थितिकी तंत्र) is the functional unit. **Environment** (पर्यावरण) is the total surroundings.
2. Habitats are commonly classed as **terrestrial, aquatic, or transitional** (mangrove/wetland). Transitional habitats are not purely terrestrial.
3. Adaptation labels include **arboreal** (trees), **fossorial** (burrows), **cursorial** (open plains), and **xericole** (desert).
4. **Habitat loss** is the number-one biodiversity threat. **Fragmentation** splits patches; **corridors** reconnect them.
5. **MISHTI** is the Mangrove Initiative for Shoreline Habitats & Tangible Income from Budget **2023–24** under MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय). It is **not** a 2019 scheme.
6. **NUHHP** is the National Urban (नगरीय) Housing and Habitat Policy of **2007** (MoHUA).
7. Habitat–scheme chronology: **JNNURM 2005 → NUHHP 2007 → AMRUT 2015 → Jal Jeevan Mission 2019**.
8. **Flora** means plant life of a region. **Fauna** means animal life. Do not swap the two.
9. India has roughly **45,000** plant species and **91,000+** animal species on about **2.4%** of world land.
10. **BSI (1890, Kolkata)** surveys plants. **ZSI (1916, Kolkata)** surveys animals. Both sit in Kolkata; kingdoms differ.
11. Engler is linked to world floral regions. **C.O. Sauer** is linked to planting hearths in cultural geography.
12. A **microhabitat** (सूक्ष्म आवास) is a small subdivision inside a larger habitat.
13. In-situ (स्थान पर) conservation (स्व-स्थाने संरक्षण) protects species in natural habitat; ex-situ (स्थानांतरित) is outside habitat — same fact as biodiversity sheets.
14. Uttar Pradesh (उत्तर प्रदेश)'s richest wildlife habitat belt is the **Terai** (तराई) (Dudhwa (दुधवा), Pilibhit, Kishanpur).
15. UP state animal is **barasingha**. UP state bird is **Sarus crane**.
16. The **Terai Arc Landscape** corridor links UP Terai protected areas toward Nepal.
17. Aquatic fauna facts for UP include the **Gangetic dolphin** in the Ganga (गंगा)–Yamuna (यमुना) system.
18. **Water hyacinth** is a common invasive threat in UP village ponds.
19. Uttar Pradesh has **no mangrove habitat** purposes — it is landlocked for that category.
20. Habitat is an address; niche is a job description. Mixing the two loses marks on definition stems.
21. Wetlands and mangroves are **transitional**, not "purely terrestrial forest" options.
22. MISHTI is mangrove-focused coastal income and protection — not a Terai tiger scheme.
23. Jal Jeevan Mission (**2019**) is the latest fact in the urban/habitat chronology set above; do not date NUHHP as 2019.
24. Corridor questions for UP should start from Terai Arc links, not from Western Ghat mangrove corridors.

📋 Must-score facts — habitat, flora–fauna, schemes

Tags

Item	Lock
Habitat / Niche / Ecosystem	Place / role / functional unit
Habitat classes	Terrestrial · aquatic · transitional
Arboreal / Fossorial / Cursorial / Xericole	Trees / burrows / open plains / desert
#1 threat	Habitat loss (+ fragmentation; corridors reconnect)
Flora / Fauna	Plants / animals — do not swap
India species share	~45k plants; ~91k+ animals on ~2.4% land

Schemes

Item	Lock
MISHTI	Mangrove initiative; Budget **2023–24** (≠ 2019)
NUHHP	Urban Housing & Habitat Policy **2007**
Chronology	JNNURM 2005 → NUHHP 2007 → AMRUT 2015 → JJM 2019

Topic 6 — Protected Areas & Conservation

Consolidated — 32 Must-Score Facts

1. The **Wildlife (Protection) Act, 1972** is India's main protected-area statute.
2. Terrestrial WPA strictness runs **National Park > Wildlife Sanctuary > Conservation/Community Reserve**.
3. A **Wildlife Sanctuary** (Section **18**) allows limited human use with **Chief Wildlife Warden** permission.
4. A **National Park** (Section **35**) is the strictest WPA protected area. Grazing and forestry are largely banned. Boundary change needs a resolution of the **State Legislature** (not Parliament).
5. A **Biosphere Reserve** (जैवमंडल आरक्षित क्षेत्र) follows the UNESCO MAB three-zone model: **core** (no exploitative use) → **buffer** (research, education, regulated tourism/grazing) → **transition** (settlements, cultivation, livelihoods). It has **no WPA statutory basis**.
6. India has **18** biosphere reserves (जैवमंडल आरक्षित क्षेत्र); about **12–13** are in the UNESCO network (count drifts with new WNBR listings). The first was **Nilgiri (नीलगिरि) (1986)**. **Nokrek** is in **Meghalaya** (मेघालय), not Manipur.
7. A **Conservation Reserve** (Section **36A**, 2002) is a government/community forest buffer near a protected area.
8. A **Community Reserve** (Section **36B**, 2002) sits on private or community land with voluntary consent.
9. A **Tiger Reserve** rests on WPA Section **38V** (2006 path) with **core + buffer** under **NTCA**. India has **58** tiger reserves. **Corbett** was first in **1973**.
10. An **Elephant Reserve** comes from **Project Elephant (प्रोजेक्ट एलीफेंट) (1992)** (प्रोजेक्ट एलिफेंट) by executive notification. It has **no WPA statutory basis**. India has about **33** elephant reserves.
11. India's first Marine National Park is **Gulf of Kutch, Gujarat (1980)**. **Gulf of Mannar (TN, 1980)** is also a Marine NP. Uttar Pradesh (उत्तर प्रदेश) has **no marine PA**.
12. **Ramsar (1971)** marks wetlands of international importance. India joined in **1982**. Uttar Pradesh has **12** Ramsar sites.
13. **Rudrasagar** is in **Tripura**. **Sultanpur** is in **Haryana**. Neither is an Uttar Pradesh Ramsar site.
14. An **Eco-Sensitive Zone (ESZ)** (पर्यावरण-संवेदनशील क्षेत्र) is an **EPA 1986** (पर्यावरण संरक्षण 1986) buffer around protected areas that restricts mining and polluting industry. ESZ is **not** a WPA category.
15. Protected-area stock (ENVIS-style ~2023): about **106** National Parks, **573** Wildlife Sanctuaries, **123** Conservation Reserves, **220** Community Reserves. **Madhya Pradesh** leads NP count; **Andaman (अंडमान) & Nicobar (निकोबार)** leads sanctuary count.
16. In a National Park, local people are **not** allowed to collect/use biomass the way limited use is permitted in a sanctuary. BR is for landscape conservation of **all forms of life**, not "a few species only."
17. WPA basis yes: National Park, Sanctuary, Tiger Reserve, Conservation Reserve, Community Reserve. WPA basis no: Biosphere Reserve, Elephant Reserve, Ramsar, ESZ.
18. Uttar Pradesh's only classic National Park fact is **Dudhwa** (दुधवा) (Lakhimpur Kheri; rhino rehabilitation from **1984**, later phases including ~**2018**).
19. UP tiger reserves are **Dudhwa, Pilibhit, and Ranipur**.
20. Important UP sanctuaries include **Katarniaghat (1975), Kishanpur, Chandra Prabha (Chandauli), Nawabganj, and Sur Sarovar**.
21. UP elephant reserves include **UP ER (2009)** and **Terai (तराई) ER (2022)**.
22. UP Ramsar sites include Upper Ganga (गंगा) (Brijghat–Narora), Nawabganj, Parvati Arga, Saman, Samaspur, Sandi, Sarsai Nawar, Sur Sarovar, Haiderpur, Bakhira, Patna (पटना), and Shekha Jheel.
23. River–park traps: **Corbett = Ramganga (रामगंगा) + Kosi (कोसी)** (not Ganga); **Kaziranga (काजीरंगा) = Brahmaputra (ब्रह्मपुत्र) / Diphlu system** (not Manas); **Silent Valley (साइलेंट वैली) = Kunthipuzha** (not Kaveri (कावेरी)).
24. The **Terai Arc** corridor links Dudhwa–Kishanpur–Katarniaghat toward Nepal.
25. Tiger Reserves are statutory under WPA; Elephant Reserves are not — that swap is a high-yield trap.

26. Biosphere Reserve zoning is landscape management under MAB; it is not the same as a National Park notification under Section 35.
27. Community Reserve is private/community land; Conservation Reserve is mainly government forest buffer — do not reverse Sections **36A/36B**.
28. ESZ regulates harmful activity around a PA; it does not itself create a new WPA Schedule animal list.
29. Sanctuary permits regulated use; National Park does not allow the same degree of use — strictness order must stay NP > WLS.
30. Match "firsts" carefully: first BR = **Nilgiri 1986**; first TR = **Corbett 1973**; first Marine NP (प्रथम समुद्री उद्यान) = **Gulf of Kutch 1980**; world first NP = **Yellowstone 1872**.
31. **International Tiger Day = 29 July** (St Petersburg 2010). **M-STrIPES** monitors tiger reserves (patrol + ecology (पारिस्थितिकी) GIS tool).
32. **Sathyamangalam TR (TN)** is a classic ecological link between Eastern and Western Ghats (पश्चिमी घाट).

📋 Must-score facts — WPA categories, tiger, Ramsar

Strictness / categories

Item	Lock
WPA year	**1972**
Strictness (terrestrial)	NP > Sanctuary > Conservation/Community Reserve
NP / Sanctuary	Sec. **35** strictest / Sec. **18** limited use
Biosphere Reserve	UNESCO MAB; core → buffer → transition (**not** WPA)
India BRs	**18** national; first **Nilgiri 1986**
Conservation / Community Res.	Sec. 36A / 36B (**2002**)
Tiger Reserve	Sec. **38V**; NTCA; **58** TRs; Corbett first **1973**
Elephant Reserve	Project Elephant **1992**; ~**33**; **no** WPA statute

Marine / wetland / ESZ

Item	Lock
First Marine NP	Gulf of Kutch, Gujarat **1980**
Ramsar	**1971**; India joined **1982**
Rudrasagar / Sultanpur	Tripura / Haryana (**not** UP)
ESZ	EPA **1986** buffer (≠ WPA category)

Topic 7 — Wildlife Conservation

Consolidated — 36 Must-Score Facts

1. Wildlife conservation protects species and habitats, with **in-situ** (स्थान पर) primary and **ex-situ** (स्थानांतरित) as backup.
2. Major threats include habitat (वास स्थान) loss, poaching, human–wildlife conflict, invasive species (आक्रामक विदेशी प्रजाति), and veterinary **diclofenac** (डाइक्लोफेनाक) for vultures.
3. The **Wildlife (Protection) Act, 1972** followed the Stockholm moment and uses **six Schedules**. **Schedule I** is maximum protection.
4. Schedule I facts include tiger, elephant, rhino, gharial, and vultures among frequently asked species. A **Schedule I tortoise** enjoys the **same protection level as the tiger**.
5. **Schedule VI** plants cannot be cultivated without a **licence** from the Chief Wildlife Warden (or authorised officer) — not "banned under all circumstances."
6. The **Chief Wildlife Warden** is the State implementing officer. **NBWL** is the PM-chaired national advisory body.
7. **NTCA** became statutory through the **2005/2006** WPA amendments and governs **Tiger Reserves only**.
8. **Project Tiger** (प्रोजेक्ट टाइगर) began in **1973**. Corbett was the first tiger reserve. India has **58** tiger reserves. The **2022** census figure often cited is **3,682** tigers.
9. **Project Elephant** (प्रोजेक्ट एलीफेंट) began in **1992** with about **33** elephant reserves and **no WPA statutory basis** — it covers **wild and captive** elephants.
10. The **Crocodile Project (1975)** covers gharial, mugger, and saltwater crocodile. Best natural gharial view: **Chambal** (चंबल) (also Girwa); IUCN (आईयूसीएन) status **Critically Endangered**.
11. **Project Lion (2020)** focuses on Asiatic lions at **Gir (गीर), Gujarat**. Uttar Pradesh (उत्तर प्रदेश) has **no** wild Asiatic lion population (जनसंख्या).
12. **IRV 2020** planned rhino recovery across seven PAs, including **Dudhwa** (दुधवा) in Uttar Pradesh.
13. **Project Snow Leopard (2009)** and **GSLEP (2013)** cover high Himalaya (हिमालय). UP has **no** snow leopard habitat.
14. The **Vulture Action Plan (2006)** and veterinary **diclofenac ban (2006)** address renal failure in Indian vultures. **SAVE** (Nepal–India) targets **vulture** recovery.
15. An **umbrella** species (tiger, elephant) protects wide ecosystems. A **flagship** species mobilises public support. A **keystone** species' loss collapses function (vulture example).
16. World Wildlife Day (विश्व वन्यजीव दिवस) is **3 March**. Wildlife Week in India is **2–8 October**. WWF (डब्ल्यूडब्ल्यूएफ) logo animal is the **Giant Panda**.
17. **Operation Olivia** is an Indian Coast Guard (तटरक्षक बल) nesting patrol on the **Odisha** (ओडिशा) coast.
18. UP tiger reserves are **Dudhwa, Pilibhit, and Ranipur**.
19. UP rhino fact is **Dudhwa National Park** (दुधवा राष्ट्रीय उद्यान) (reintroduced population).
20. UP gharial sites include **Katarniaghat Wildlife Sanctuary** and the **Chambal** sanctuary landscape.
21. UP elephant reserves are **UP ER (2009)** and **Terai (तराई) ER (2022)**.
22. Programme-year chain: WPA **1972** → Project Tiger **1973** → Crocodile Project **1975** → Project Elephant **1992** → Biodiversity Act **2002** → NTCA statutory **2005/2006** → Vulture ban **2006** → Project Snow Leopard **2009** → Project Lion **2020**.
23. **Whale shark** is India's largest fish; seasonal aggregations off **Gujarat** (Mar–May); protection programme from **2004**.
24. **Kharai camel** (Kutch, Gujarat) can swim in seawater and graze mangroves; can be domesticated — **not** "wild only."
25. **Pugmark** technique estimates (प्राकलन समिति) wild animal populations from footprints.

26. **Spider monkey** is Central/South American — **not** naturally found in India (Star tortoise, Monitor lizard, Pygmy hog are).
27. Scientific names: Asiatic wild ass (**Equus hemionus**), Barasingha (**Rucervus/Cervus duvaucelii**), Chinkara (**Gazella bennettii**), Nilgai (**Boselaphus tragocamelus**).
28. **Community Reserve** (सामुदायिक आरक्षित क्षेत्र) (WPA): CWLW authority applies; hunting banned; NTFP collection allowed; **traditional agriculture / jhum not** allowed after notification.
29. **Animal Welfare Board of India** (1962) is under the **Prevention of Cruelty to Animals Act, 1960** — **not** EPA 1986 (पर्यावरण संरक्षण 1986). NTCA is statutory; NGRBA was PM-chaired.
30. Gharial, Indian wild ass, and wild buffalo cannot be hunted except under limited WPA provisions.
31. India holds about **2.4%** of world land area but about **8%** of known biodiversity (~**7.28%** of known animal species in Alfred-type figures).
32. First International Tiger Forum / World Tiger Summit **2010** was at **Saint Petersburg**, Russia.
33. **New York Declaration on Forests (2014)** is voluntary/non-binding; India was **not** an original signatory.
34. **UNCCD (1994)** (मरुस्थलीकरण रोकथाम संधि) is the sole legally binding desertification (मरुस्थलीकरण) convention; bottom-up local participation; no exclusive South Asia finance monopoly claim.
35. Conservation is ecological protection; **Wildlife Protection** language often means the **WPA enforcement** track.
36. NTCA does not run Elephant Reserves; confusing NTCA with Project Elephant administration is a trap.

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India's Wildlife Conservation Projects Chronology
                                  |
1972: Wildlife (Protection) Act, 1972 (Six Schedules; Schedule I = Apex)
                                  |
1973: Project Tiger (Corbett NP; launched 1 April 1973 by Indira Gandhi)
                                  |
1975: Crocodile Conservation Project (UNDP / FAO; Gharial in Chambal)
                                  |
1986: Environment (Protection) Act, 1986 (Post-Bhopal Gas Tragedy)
                                  |
1987: Project Rhino / Rhino Vision (One-horned Rhino in Kaziranga & Dudhwa)
                                  |
1992: Project Elephant (Launched at Singhbhum, Jharkhand; non-statutory)
                                  |
2002: Biological Diversity Act, 2002 (National Biodiversity Authority, Chennai)
                                  |
2005–06: NTCA Established (Statutory status to Tiger Conservation under WPA)
                                  |
2006: National Vulture Conservation Plan & Veterinary Diclofenac Ban
                                  |
2009: Project Snow Leopard (High-altitude Himalaya & Central Asia)
                                  |
2010: National Green Tribunal Act, 2010 (Article 21 environmental justice)
                                  |
2020: Project Lion & Project Dolphin Announced (74th Independence Day)
                                  |
2022: Project Cheetah (Kuno National Park, MP - African Cheetah reintroduction)
```

📋 Must-score facts — projects, schedules, NTCA

Law / schedules

Item	Lock
WPA Schedules	Six; **Schedule I** = maximum protection
Schedule VI	Plants; licence from CWLW (not absolute (निरपेक्ष) ban)
NBWL / CWLW	PM-chaired advisory / State implementing officer
NTCA	Statutory via 2005–06 path; Tiger Reserves only

Projects

Item	Lock
Project Tiger	**1973**; Corbett first; **58** TRs; ~3682 (2022)
Project Elephant	**1992**; ~33 reserves; wild + captive
Crocodile Project	**1975**; gharial–mugger–saltwater; Chambal
Project Lion	**2020**; Gir only (no wild UP lions)
IRV 2020	Rhino recovery; includes **Dudhwa**
Project Snow Leopard / GSLEP	**2009** / **2013** (no UP habitat)
Vulture plan / diclofenac ban	**2006** / **2006**

Topic 8 — Forests & Forest Management

Consolidated — 36 Must-Score Facts

1. Forest cover (वन आवरण) in ISFR means a patch of at least **1 ha** with canopy **≥10%**, on any land ownership.
2. Canopy classes: **VDF ≥70%**, **MDF 40–70%**, **OF 10–40%**.
3. ISFR **2023** India figures: forest cover about **21.76%**, tree cover (वृक्ष आवरण) about **3.41%**, total green cover about **25.17%**. ISFR **2021** held forest cover at about **21.71%** and forest+tree cover about **24.62%** (still used in older stems).
4. **Madhya Pradesh** has the largest forest **area**. **Mizoram** has the highest **percentage** cover among States. **Arunachal Pradesh** often leads **very dense forest** area.
5. **Forest cover ≠ Recorded Forest Area ≠ tree cover**. Recorded forest area is a legal-notification figure near about **23%**.
6. Under the **Indian Forest Act, 1927**, **Reserved Forest** is strictest (acts banned unless permitted). **Protected Forest** is State-regulated with some customary rights possible.
7. **Champion and Seth (1968)** classify Indian forests into **16 types in 6 groups**. The largest share is **Tropical Dry Deciduous** (about **38%**).
8. **Tropical wet evergreen** (>**200 cm** rain) holds maximum plant diversity / biomass among common options — Western Ghats (पश्चिमी घाट), NE hills, A&N (not Himachal (हिमाचल) as wet evergreen).
9. **Teak** dominates **tropical moist deciduous** forests; **Madhya Pradesh** has the maximum teak area among common options.
10. **Red Sanders (*Pterocarpus santalinus*)** is a dry-deciduous South India endemic (Palkonda–Seshachalam / southern Eastern Ghats (पूर्वी घाट)) — **not** a tropical rainforest tree.
11. **Taxus** (Himalaya (हिमालय); Red Data) yields **taxol** (anti-cancer; also cited vs Parkinson's in older stems).
12. **Amazon rainforest** is often called the **"lungs of the Earth"** (~**20%** of terrestrial photosynthetic oxygen share in coaching figures).
13. UP type facts: Terai (तराई) = moist deciduous; Vindhyan = dry deciduous; Bundelkhand = thorn.
14. Law chronology: **IFA 1927 → FCA 1980 → NFP 1988 → JFM 1990 → FRA 2006 → CAMPA Act 2016**.
15. The **Forest (Conservation) Act, 1980** requires **Central** approval before diverting forest land. Compensatory afforestation is mandatory.
16. **National Forest Policy 1988** targets about **33%** national forest/tree cover and **66%** in hills. **NFP 1952** classes: Protected/Conserved, National, Village Forests, Tree Lands — **not** "National Park" as a forest class.
17. NFP 1988 includes afforestation/wasteland, reforestation, and wood substitutes — **not** wide promotion of insecticides/pesticides.
18. **Chipko (1973, Reni, Uttarakhand (उत्तराखंड))** — Bahuguna, Bhatt, **Gaura Devi**. The **Godavarman (1996)** line broadened "forest."
19. Afforestation tools include **NAP (2000)**, **Green India Mission (2014)**, **Miyawaki** mini-forests, and **CAMPA**.
20. **Social forestry** grows trees on **non-forest** land. **Agroforestry** integrates trees with crops/livestock (**CAFRI Jhansi (झांसी)**).
21. **JFM (1990)** is State–village co-management of **degraded forest**. **FRA 2006** recognises dweller rights (**IFR ~4 ha**).
22. **FCA** controls **diversion**. **FRA** recognises **rights**. **JFM** is executive co-management — three different tools.
23. UP ISFR **2023** forest cover is about **6.24%** of State area.
24. **Sonbhadra** (सोनभद्र) has UP's highest district forest cover. Districts above **20%** include **Chandauli** and **Shravasti**.
25. **Urbanisation is a cause of deforestation**, not an "impact/effect" of deforestation (classic trap).
26. Forests are a **renewable** resource; they also enhance environment (पर्यावरण) quality — but the second fact does **not** alone explain renewability.

27. **Aluminium** is called a **"green metal"** for recyclability/eco profile — not primarily "to save forests by replacing wood."
28. **Living root bridges** are characteristic of **Meghalaya** (मेघालय).
29. ISFR 2021 top absolute (निरपेक्ष) gains: **Andhra Pradesh, Telangana, Odisha**. NE states showed large losses.
30. Mangrove area rank (ISFR 2021 style): **WB > Gujarat > A&N (~616–617 km^2) > AP**. **Coringa** = Godavari (गोदावरी) delta (डेल्टा) (AP).
31. **Andaman (अंडमान) & Nicobar (निकोबार)** uniquely combine mangrove + evergreen + deciduous in one option set.
32. International Day of Forests = **21 March** (UN **2012**). Wildlife Week = **2–8 October**. WWF (डब्ल्यूडब्ल्यूएफ) logo = **Giant Panda**.
33. Bhutan constitutionally aims for **≥60%** forest cover (actual cover much higher).
34. Bamboo on **non-forest** land was omitted from "tree" under the **Indian Forest (Amendment) Act, 2017**.
35. **FRI Dehradun (देहरादून) (1906)**; **ICFRE HQ Dehradun**; **IIFM Bhopal**; Survey of India ≠ environment agency.
36. CAMPA manages compensatory afforestation funds after diversion — it is not FRA Gram Sabha (ग्राम सभा) rights recognition (मान्यता).

📋 Must-score facts — ISFR, types, law chain

Cover definitions

Item	Lock
Forest cover (ISFR)	≥1 ha; canopy ≥**10%**; any ownership
VDF / MDF / OF	≥70% / 40–70% / 10–40%
ISFR 2023 class	Forest ~**21.76%**; tree ~3.41%; total green ~25.17%
Largest area / highest %	MP / Mizoram
Cover ≠ RFA ≠ tree cover	Legal RFA ~23% class figure

Types / law

Item	Lock
Champion & Seth	**1968**; 16 types / 6 groups; Tropical Dry Deciduous largest
Wet evergreen rain	>**200 cm**; max diversity/biomass
Teak / Red Sanders	Moist deciduous / dry-deciduous S. India endemic
UP belts	Terai moist deciduous · Vindhyan dry · Bundelkhand thorn
Law chain	IFA 1927 → FCA 1980 → NFP 1988 → JFM 1990 → FRA 2006 → CAMPA 2016
Reserved vs Protected Forest	Strictest IFA / State-regulated lighter

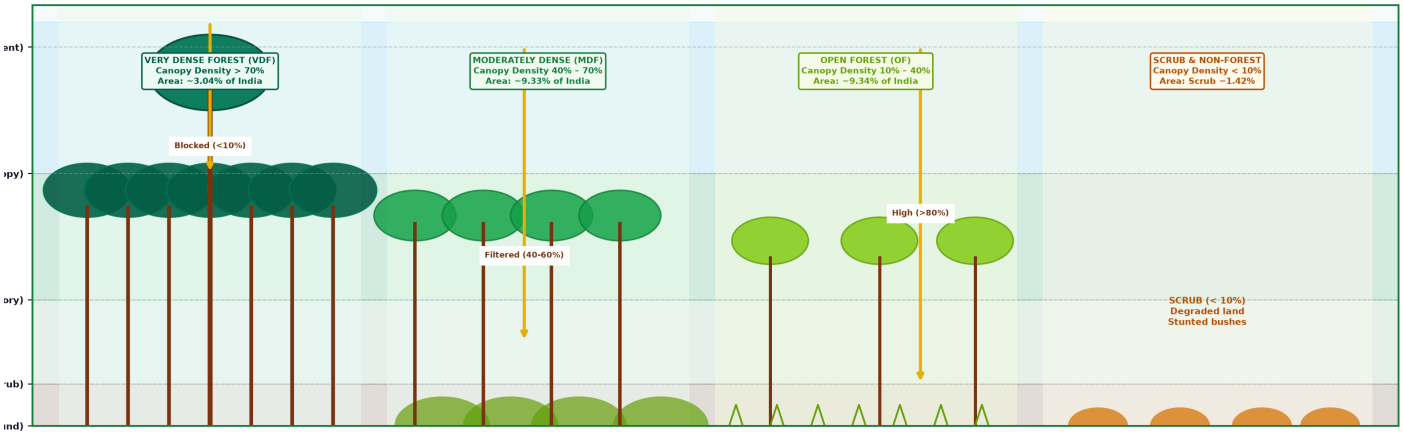
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FORESTS OF INDIA: CANOPY STRATIFICATION ARCHITECTURE & ISFR EXAM GEOGRAPHY

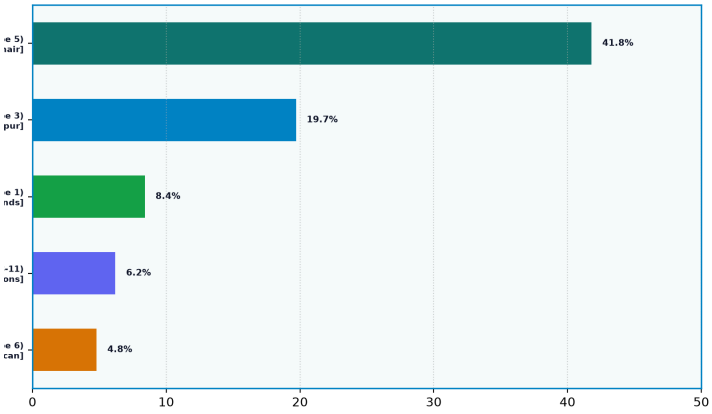
Physical Forest Profile (0 to 45m) | FSI Canopy Density Tiers (VDF >70%, MDF 40-70%, OF 10-40%) | Champion-Seth 16 Classes & UP Profile

FORESTRY MODEL

ISFR VERTICAL CANOPY ARCHITECTURE & CROWN CLOSURE TIERS (FSI / MoEFCC CRITERIA)



CHAMPION & SETH 16 TYPES — TOP 5 DOMINANT CLASSES IN INDIA



ISFR ALL-INDIA & UTTAR PRADESH EXAM HIGHLIGHTS

ALL-INDIA ISFR BENCHMARKS

- Total Forest Cover: 21.76%
- Total Tree Cover: 2.91%
- Total Green Cover: 24.67% (Target: 33.3% - NFP 1988)

★ LARGEST FOREST AREA:
1. Madhya Pradesh (77,493 km²)
2. Arunachal Pradesh
3. Chhattisgarh

★ HIGHEST FOREST %:
1. Mizoram (84.53%)
2. Arunachal Pradesh (79.33%)
3. Meghalaya (76.00%)

★ MAX FOREST GAIN: AP > Telangana

UTTAR PRADESH EXAM PROFILE

- UP Forest Cover: 6.15% (14,818 km²)
- UP Tree Cover: 3.08% (7,421 km²)
- UP Total Green Cover: 9.23%

★ TOP DISTRICT BY AREA & %:
1. SONBHADRA: 34.87% (2,544 km²) [Highest in both Area & %!]
2. Chandauli (21.78%)
3. Chitrakoot (19.64%)

★ LOWEST DISTRICT BY AREA & %:
1. BHADOHI (Sant Ravidas Nagar) Cover: 0.37% (3.71 km² - Lowest!)
2. Mau | 3. Mainpuri

Topic 9 — Pollution & Waste Management

Consolidated — 39 Must-Score Facts

1. **Pollution** is harmful addition beyond the environment (पर्यावरण)'s absorption capacity. **Degradation** is the broader fall in environmental quality and productivity.
2. Human-caused pollution / human-modified environment = **anthropogenic**. Anthropogenic pollution often raises persistent **non-biodegradable** loads.
3. **Biodegradable** pollutants (sewage, faecal matter, urine, domestic waste, rubber in some stems) break down by microbes. **Non-biodegradable** include pesticides, plastics, mercury, asbestos.
4. **Point sources** are chimneys and drains. **Non-point sources** include farm runoff and roads.
5. **Primary pollutants** are emitted directly (SO_2, PM, CO, NOx, lead). **Secondary pollutants** form in air (O_3, PAN, photochemical smog, acid rain (अम्ल वर्षा)).
6. Law years: **Water Act 1974** → **Air Act 1981** → **EPA 1986 (पर्यावरण संरक्षण 1986)**. Noise Rules date to **2000**. Air Act amended **1987** to include noise.
7. **CPCB** (केंद्रीय प्रदूषण नियंत्रण बोर्ड) sets ambient air and surface-water standards. **SPCB** enforces in States. **CGWB** handles **groundwater** assessment — not CPCB.
8. India's AQI uses **eight** pollutants: **PM10, PM2.5, NO_2, CO, O_3, SO_2, NH_3, Pb** — **not CO_2 or CH_4** (those are climate gases). Bands: **0–50 Good** … **401–500 Severe**.
9. London smog is **SO_2 + smoke**. Los Angeles / photochemical smog is **NOx + VOCs + sunlight → O_3 / PAN**. **PAN** is a powerful eye irritant.
10. **CO** binds haemoglobin **~200–300×** more strongly than O_2 (carboxyhaemoglobin). Lead antiknock in petrol harms IQ/brain development.
11. **Fly ash** from coal TPPs: usable in bricks / Portland cement replacement; contains SiO_2, Al_2O_3, CaO **plus toxic metals** — not "oxides only."
12. **Lichens** are SO_2 bio-indicators and die in polluted air. **Radon** is the classic major **indoor** air pollutant (lung cancer risk).
13. **NCAP (2019)** covers non-attainment cities with a **20–30%** PM reduction target. **BS-VI** fuel arrived in **2020**. **WAYU** air-purifying units were installed in **Delhi**.
14. **CCAC** (2012, with UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम)) focuses on **methane, black carbon, HFCs** — not a G20-only club.
15. Water facts: high **BOD** means worse organic pollution; **COD ≥ BOD**; clean DO is often **>6 mg/L**, while **<5 mg/L** / **<4 mg/L** stresses aquatic life.
16. **Eutrophication** (सुपोषण) is nutrient overload (N/P) → algal bloom → dissolved-oxygen crash — a **water** phenomenon, not air pollution.
17. Disease locks: **Minamata–Hg**, **Itai-itai–Cd**, **Knock-knee / fluorosis–F**, **Black Foot–As**, blue-baby–nitrate.
18. **Damodar** (दामोदर) is classically called a **biological desert** from mining/industry pollution. Ganga (गंगा) BOD often highest **Kanpur (कानपुर)–Allahabad (इलाहाबाद)** stretch.
19. Arsenic groundwater hotspots: Ganga–Brahmaputra (ब्रह्मपुत्र) / Bangladesh Padma (पद्मा)–Meghna; Indian states include WB, Bihar, UP, Jharkhand, Assam, Manipur, Chhattisgarh.
20. About **80%** of marine pollution comes from **land**. Ocean acidification (महासागरीय अम्लीकरण) from CO_2 harms calcareous plankton and corals. **IMO** HQ = **London**.
21. Noise is measured in **decibels**. Under **Noise Pollution (Regulation and Control) Rules, 2000**, a person may complain when noise exceeds the ambient standard by **10 dB(A) or more**. A **10 dB** rise also means **10×** intensity on the logarithmic scale. **Green muffler** = roadside tree belts for noise (and dust).
22. Plastic: **PWM Rules 2016**, SUP ban push **2022**, microplastics **<5 mm**; **microbeads** harm marine ecosystems. Polythene = ethylene polymer, non-biodegradable.

23. **Bioremediation** uses microbes to clean pollution; **cannot** readily/completely treat **Cd/Pb** heavy metals. **Oilzapper** (TERI) is for oily sludge/spills. Genetic engineering can create specialised strains (e.g. *Pseudomonas putida*).
24. Waste rules cluster in **2016**: SWM, BMW, Hazardous Waste, and Plastic Waste. **E-waste Rules** updated in **2022**. **EPR** first featured strongly in plastic/e-waste rules **2011+**.
25. **3R** priority is **Reduce > Reuse > Recycle**. **Swachh Bharat** began in **2014**. NAQI launched **17 Oct 2014**.
26. BMW colour codes: **Yellow** incineration; **Red** contaminated recyclables (IV tubes/sets, catheters, urine bags, syringes **without** needles) for disinfection then recycle; **White** sharps (needles); **Blue** glass vials / metal.
27. **Euro / BS** emission norms limit CO, HC, NOx, PM. Euro-II ultra-low sulphur diesel ≈ **≤0.05% S**. India skipped BS-V to **BS-VI (1 Apr 2020)**.
28. Catalytic converters use **Pt / Pd / Rh** to convert CO → CO_2, NOx → N_2, HC → CO_2 + H_2O.
29. UP air NCAP cities include Lucknow (लखनऊ), Kanpur, Ghaziabad, Agra, Varanasi (वाराणसी), Meerut (मेरठ), Noida, Prayagraj (प्रयागराज), Moradabad, and Bareilly.
30. UP issue facts: Kanpur Ganga tanneries; Ghaziabad/Noida air; **Moradabad** noise; **Taj Trapezium** at Agra; Varanasi ghat/plastic pressure.
31. FSI publishes forest cover (वन आवरण), not pollution standards. CWC (केंद्रीय जल आयोग) is surface-water resources, not the groundwater quality report.
32. EPA **1986** is the umbrella for many later rules; it does not replace Water and Air Acts.
33. Secondary ozone at ground level is a pollutant; do not confuse it with stratospheric protective ozone.
34. Hazardous waste follows HW Rules **2016** plus Basel controls — not SWM municipal rules alone.
35. **Asian Brown Cloud** / atmospheric brown cloud is especially noted over **South Asia** (winter aerosols, black carbon, sulphate).
36. Bag filters trap finer particulates (**<~50 µm**); cyclone separators/collectors handle coarser (**>~50 µm**). **Cyclone divider** is not a standard control device name.
37. **Radioactive pollution** is ionizing radiation beyond safe dose. **Becquerel (Bq)** counts disintegrations per second; **Sievert (Sv)** is biological dose; **half-life** is time for **50%** decay — not complete disappearance.
38. **AERB** (Atomic Energy Regulatory Board) regulates nuclear safety in India. **Chernobyl (26 April 1986)** is the classic nuclear-accident teaching case; **Bhopal (1984)** is MIC industrial gas, not radioactive pollution.
39. **Thermal pollution** raises water temperature (mainly power-plant cooling water). Warm water holds **less DO**; even a **2–3°C** rise can stress aquatic life. **Cooling towers/ponds** mitigate discharge under Water Act consent conditions.

📋 Must-score facts — pollutants, AQI, smog, laws

Basics

Item	Lock
Primary / Secondary	Emitted direct / formed in air (O_3, PAN, acid rain)
Point / Non-point	Chimney–drain / farm runoff–roads
Law years	Water **1974** → Air **1981** → EPA **1986**; Noise Rules **2000**
AQI 8 pollutants	PM10, PM2.5, NO_2, CO, O_3, SO_2, NH_3, Pb (**not** CO_2/CH_4)
London / LA smog	SO_2+smoke / NOx+VOCs+sun → O_3+PAN
CO / Lead	Hb affinity ~200–300× / IQ–brain harm
Lichens / Radon	SO_2 bio-indicator / top indoor pollutant
NCAP / BS-VI	**2019** / **2020**

Topic 10 — Climate Change

Consolidated — 36 Must-Score Facts

1. **Global warming** is the long-term rise in average surface temperature. It is a subset of broader **climate change**, which also includes rainfall, extremes, and sea-level shifts.
2. IPCC (आईपीसीसी) AR6 facts about **1.1°C** above 1850–1900 globally. India has warmed about **0.7°C** since 1901 (IMD class figure). Past-century surface rise is about **1°C ≈ 1.8°F**.
3. The natural greenhouse effect (प्राकृतिक ग्रीनहाउस प्रभाव) keeps Earth near about **+15°C**. Without it, Earth would be near about **−18°C**.
4. **Joseph Fourier (1820s)** postulated the greenhouse-effect idea. Sun sends **shortwave** (UV + visible + IR); Earth re-emits **longwave IR**. Ozone absorbs harmful UV; GHGs trap outgoing IR.
5. Enhanced greenhouse effect (मानवजनित संवर्धित प्रभाव) means excess anthropogenic GHGs trap outgoing infrared radiation. Main cause of recent warming is **CO_2** (highest concentration among long-lived GHGs).
6. Kyoto-basket majors: **CO_2** (GWP **1**), **CH_4** (GWP about **28–36**), **N_2O** (GWP about **265**), plus HFCs, PFCs, **SF_6** (and **NF_3** in later inventories).
7. Rough **WMO-style relative (सापेक्ष) shares** often used in Indian papers: **CO_2 ~64%, CH_4 ~19%, N_2O ~6%, CFCs + others ~11%**. Decreasing order: **CO_2 > CH_4 > CFCs > N_2O**.
8. **Water vapour** is Earth's most abundant natural GHG (NASA: ~half of the natural greenhouse effect) but mainly a **feedback**, not the primary policy driver.
9. **Not GHGs: argon, nitrogen (N_2), oxygen, hydrogen, propane**. **SO_2 / NOx** do **not** contribute to warming **directly** (indirect / other roles).
10. **Direct** GHGs include CO_2, CH_4, N_2O, CFCs, SF_6, NF_3. **Indirect** agents include **NOx, CO, NMVOCs, SO_2**.
11. India is about the **third**-largest total emitter (**China** first); **Bhutan** is often cited as **carbon-negative**.
12. A **carbon footprint** totals GHGs in tCO_2e across Scope **1** (direct), **2** (energy), and **3** (supply chain).
13. A **carbon sink** absorbs more CO_2 than it releases (forests, oceans, soils, mangroves / **blue carbon** (नीला कार्बन)). **Sequestration** is the storage process.
14. **Carbon fertilization** = more plant growth from higher atmospheric CO_2 (not the same as ocean acidification (महासागरीय अम्लीकरण) or warming).
15. Pre-industrial CO_2 ~**278–280 ppm (~0.03%)**; mid-20th century ~**315 ppm**; now **415–420+ ppm**.
16. **Methane ("marsh gas")** sources: paddies, cattle, coal mines, landfills, termites; natural wetlands dominate natural CH_4 (~**75–76%** of natural). **Rice** is a key anthropogenic source of **both CH_4 and N_2O**.
17. **Methane hydrates** under Arctic tundra (टुंड्रा) / seafloor can release CH_4 if warmed; atmospheric CH_4 oxidises to CO_2 in roughly a decade or two.
18. Impacts include heat waves, glacier melt, sea-level rise, coral bleaching (प्रवाल विरंजन), disease spread, stronger storms. **Arctic / Greenland ice** is among the most fragile first-hit systems.
19. Rough IPCC-style risk: beyond about **+2°C** pre-industrial → widespread coral mortality; beyond about **+3°C** → terrestrial biosphere tends toward a **net carbon source**.
20. **Adaptation** = coping with impacts. **Mitigation** = cutting emissions and enhancing sinks.
21. **NAPCC (2008)** has **eight** national missions — **not** nuclear power as a listed mission.
22. India's updated **NDC** path includes about **45%** emission-intensity cut and about **50%** non-fossil electricity capacity by **2030** (not "60% energy from renewables" as a false paraphrase).
23. India's **Net Zero** pledge is **2070**. Net zero means emissions balanced by removals — **not** literally zero every emission.
24. **LiFE Mission** launched **June 2022**; idea at **COP26 (2021)**, not COP25.
25. Climate treaty chain: **UNFCCC (यूएनएफसीसीसी) 1992** → **Kyoto 1997 (force 2005)** → **Paris 2015**. **Montreal 1987** = ozone, not climate.

26. **CDM** (Kyoto Art. 12) lets Annex-I parties fund projects in developing countries for **CERs** (1 CER ≈ 1 tCO_2). **Carbon credit** (कार्बन क्रेडिट) concept originates from Kyoto.
27. **Green Climate Fund** was set up at **Cancun COP-16 (2010)**, not Durban. **Earth Hour** (अर्थ आवर) = **WWF** (डब्ल्यूडब्ल्यूएफ), last Saturday of March (lights off ~8:30–9:30 pm).
28. **GHG Protocol** = **WRI + WBCSD** accounting tool. **Blue carbon** = ocean/coastal carbon (mangroves, seagrass, salt (लवणाध्यक्ष) marsh).
29. UP heat vulnerability often cites (साइट्स) Lucknow (लखनऊ), Agra, Prayagraj (प्रयागराज), and NCR.
30. UP drought (सूखा) fact is **Bundelkhand's seven districts**. Flood facts include Ballia, Ghazipur, and Varanasi (वाराणसी) belts.
31. Emissions hotspot language for UP–MP often points to the **Singrauli** coal belt.
32. Terai (तराई) forests at Dudhwa (दुधवा) and Pilibhit are UP **carbon-sink** examples.
33. **Milankovitch** orbital theory (eccentricity, obliquity, precession) — **solar irradiance** is **not** part of that astronomical set.
34. **Ice cores** are the classic **cryogenic** climate archive (Greenland / Antarctica).
35. **Solar flares** are **not** a primary cause of recent climate change in standard NASA/IPCC framing.
36. Farm practices that help soil carbon storage include **contour bunding, relay cropping, and zero tillage**.

📋 Must-score facts — warming, GHGs, sinks

Frame

Item	Lock
Warming vs climate change	Temp rise subset / broader incl. rainfall & extremes
Natural GHG effect	~+15°C with; ~−18°C without
Fourier	1820s greenhouse-effect idea
Main recent driver	Anthropogenic **CO_2**
Kyoto basket	CO_2, CH_4, N_2O, HFCs, PFCs, SF_6 (+NF_3 later)
Relative shares (WMO-style)	CO_2~64% > CH_4~19% > CFCs+~11% > N_2O~6%
Not GHGs	Ar, N_2, O_2, H_2, propane
India rank / Bhutan	~3rd absolute (निरपेक्ष) / often carbon-negative
Sink / sequestration	Net CO_2 absorb / storage process
Blue carbon	Ocean–coastal sinks (mangroves etc.)

Topic 11 — Ozone Layer

Consolidated — 30 Must-Score Facts

1. Ozone (O_3) is triatomic oxygen. **Stratospheric ozone** (समतापमंडलीय ओजोन) (about **15–30/35 km**) is protective "good" ozone. **Tropospheric ozone** (क्षोभमंडलीय ओजोन) is a ground-level pollutant.
2. About **90%** of ozone is stratospheric; about **10%** is tropospheric. Stratospheric ozone is regulated in part by **NO_2**.
3. The **Chapman cycle** describes natural formation and destruction balance of stratospheric ozone (UV acting on O_2).
4. **UV-A (315–400 nm)** reaches the surface; **UV-B (280–315 nm)** is mostly blocked by ozone; **UV-C (100–280 nm)** is fully blocked. Ozone blocks ~**93–99%** of the most damaging UV.
5. Ozone-depleting substances release chlorine and bromine radicals that catalytically destroy ozone. One chlorine atom can destroy on the order of **100,000** ozone molecules.
6. Main ODS include **CFCs, Halon-1211/1301, carbon tetrachloride, methyl chloroform, methyl bromide, and transitional HCFCs**. Fridge coolants often sold as **Mafron**; large plants may use **ammonia**.
7. An **ozone hole** is not a physical hole. It means column ozone below **220 Dobson Units**. **1 DU ≈ 0.01 mm** pure ozone at 0°C, 1 atm.
8. The Antarctic ozone hole peaks in Southern Hemisphere spring (**September–October**) under a polar vortex with polar stratospheric clouds + CFCs.
9. The Antarctic ozone hole was discovered in **1985** by **Farman (फ़रमान), Gardiner, Shanklin** (British Antarctic Survey). **Kent Moore** reported Tibetan Plateau ozone feature in **2005**.
10. Treaty chain: **Vienna Convention 1985 → Montreal Protocol 1987 (force 1 Jan 1989) → Kigali Amendment 2016 (force 1 Jan 2019)**.
11. Montreal is the binding ODS phase-out treaty. International Ozone Day is **16 September** (UN observance from **1994**).
12. **Kigali (2016)** phases down **HFCs**. HFCs are not classic ODS, but they have high global-warming potential.
13. Montreal achieved near-universal ratification and cut ODS emissions by about **99%** from peak levels.
14. Treaty trap: **Montreal = ozone**; **Kyoto/Paris = climate**. Never swap the issue tags. Carbon trading ≠ Montreal.
15. Global average ozone column is often near about **300 DU**; the hole threshold is **<220 DU**.
16. India ratified Montreal in **1992**. The nodal unit is the **Ozone Cell, MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय)**.
17. India CFC phase-out fact is **2010**. HCFC freeze is **2020**; complete HCFC phase-out aims at **2030**.
18. India's Kigali HFC phase-down commitment begins from **2028** in standard coaching facts.
19. Stratospheric ozone protects life; tropospheric ozone harms lungs and crops — same molecule, opposite verdict by layer.
20. Vienna is the framework convention; Montreal is the binding control protocol — parallel to UNFCCC (यूएनएफसीसीसी)/Kyoto structure, but for ozone.
21. Dobson Unit measures total ozone column, not ground-level ppm of pollution ozone.
22. Polar stratospheric clouds enable chlorine activation — that is why the deep hole is polar-spring specific.
23. HCFCs were transitional substitutes on the way out of CFCs; they are still ODS, just weaker/longer phase-out.
24. Kigali sits under the Montreal family, not under UNFCCC as a climate protocol.
25. Ozone Day is **16 September**; do not confuse it with World Environment (पर्यावरण) Day (विश्व पर्यावरण दिवस) (**5 June**) or Wildlife Day (**3 March**).
26. CFCs are both ODS and GHGs; CFC source is not primarily automobiles; CFC use is not tubeless tyres.
27. Extra UV mainly raises **skin cancer** risk (classic stem).
28. Homosphere (सममंडल) ≈ troposphere (क्षोभमंडल) + stratosphere (समतापमंडल) + mesosphere (मध्यमंडल); **ionosphere not** in that set.

29. Column thickness often highest in **spring**, lowest in **rainy season** (general seasonal pattern).
30. Full teaching depth for this chapter also lives in Topic **36**.

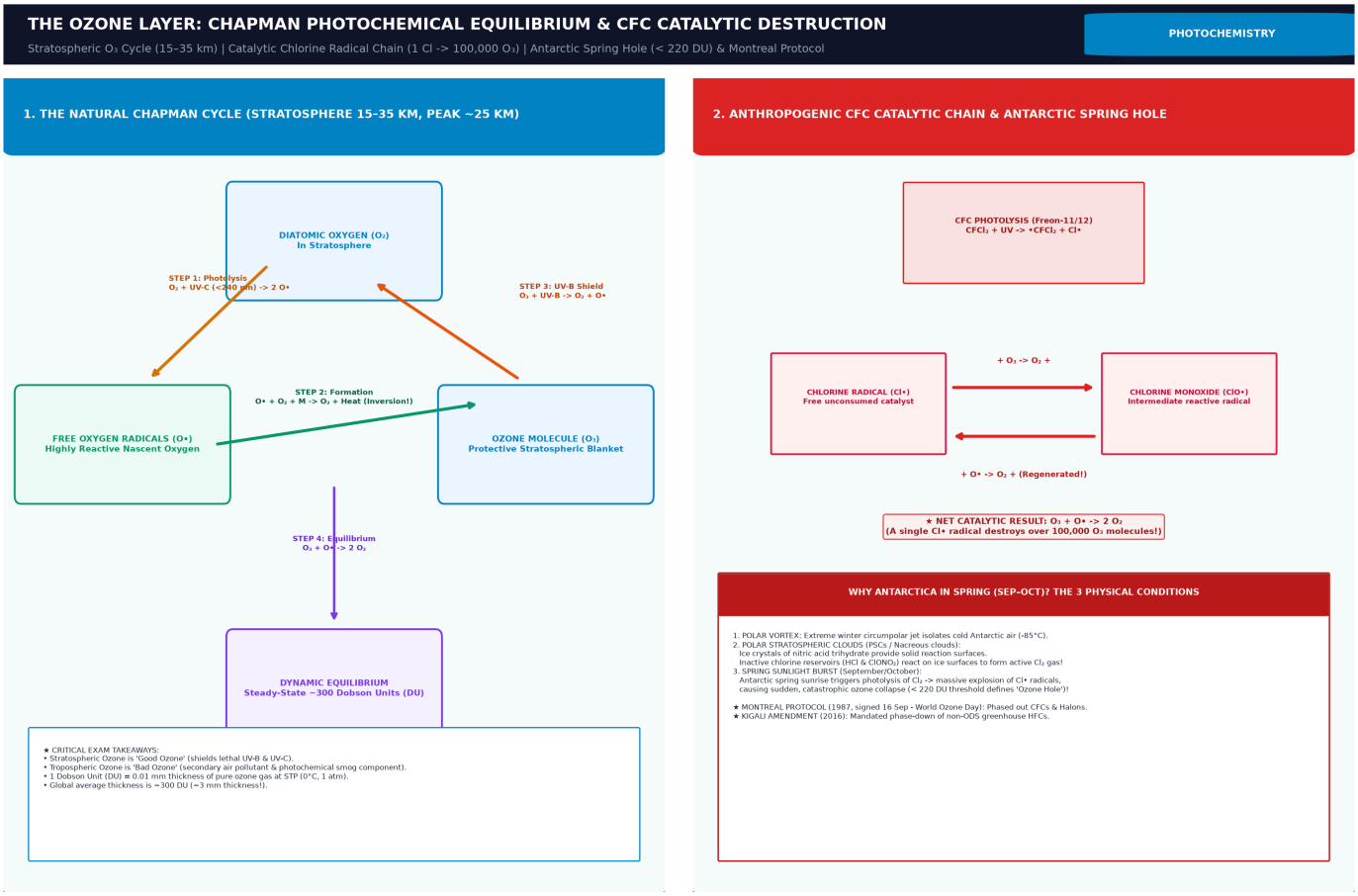
📋 Must-score facts — good/bad ozone (बुरा ओजोन), Montreal, Kigali

Science

Item	Lock
Good / bad ozone	Stratosphere (~15–35 km) / tropospheric pollutant
Share	~90% stratospheric; ~10% tropospheric
UV-A / B / C	Reaches surface / mostly blocked / fully blocked
Ozone hole	Column <**220 DU** (not a physical hole)
Antarctic peak	SH spring Sep–Oct; PSCs + vortex + CFCs
Discovery	**1985** Farman–Gardiner–Shanklin (BAS)

Treaties

Item	Lock
Chain	Vienna **1985** → Montreal **1987** → Kigali **2016**
Montreal vs Kyoto/Paris	Ozone ODS / climate GHGs — never swap
Kigali	Phases down **HFCs** (high GWP; not classic ODS)
Ozone Day	**16 September**



Topic 12 — Acid Rain

Consolidated — 24 Must-Score Facts

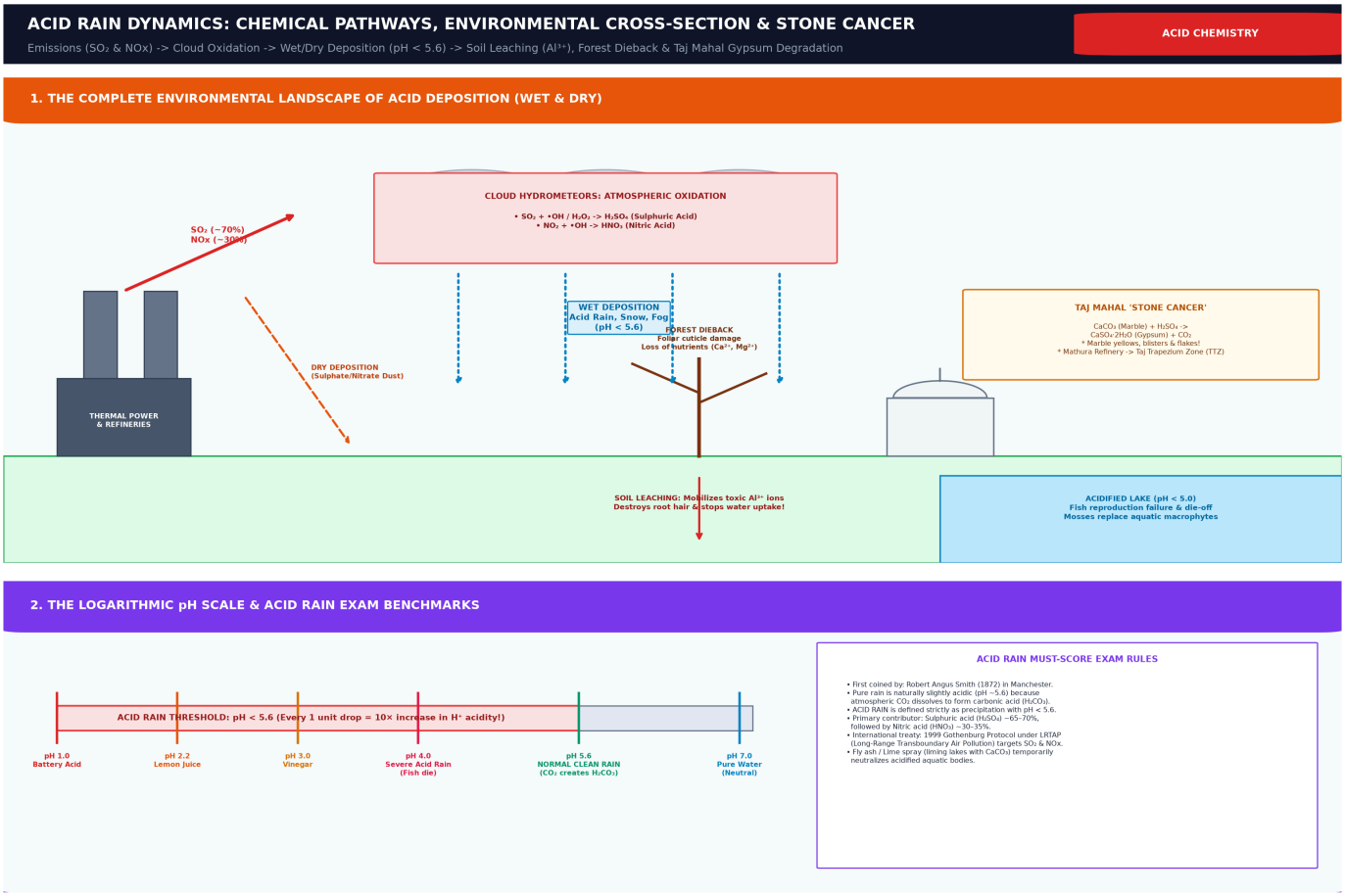
1. **Acid rain** means wet or dry acidic deposition with pH below **5.6**.
2. Normal unpolluted rain is already about **pH 5.6** because dissolved CO_2 forms weak carbonic acid — it is **not** pH 7.
3. **Wet deposition** is acids in rain, snow, or fog. **Dry deposition** is acidic gases and particles settling without rain.
4. **SO_2** oxidises to **H_2SO_4** (sulphuric acid), mainly from coal TPPs, smelters, and refineries.
5. **NOx** oxidises to **HNO_3** (nitric acid), mainly from vehicles and high-temperature combustion.
6. CO_2 forms only weak H_2CO_3 and sets the **5.6 baseline**; it is **not** the main culprit of severe acid rain.
7. On the pH scale, acidic is **<7**, basic is **>7**, neutral is **7**. Each step down multiplies H^+ by **10**.
8. Freshwater below about **pH 5** stresses fish and can release toxic **Al^{3+}**.
9. Acid deposition leaches soil **Ca^{2+} and Mg^{2+}** and can free toxic aluminium.
10. Conifers are especially sensitive forest receptors. Marble and limestone monuments form a **gypsum crust** ($CaSO_4$) under sulphuric attack — the Taj story.
11. **Lichens** die in high SO_2 and act as pollution bio-indicators.
12. Prevention tools include **FGD scrubbers** (lime/limestone), catalytic converters, low-sulphur coal, CNG/LPG switches, and **BS-VI** fuel (about **10 ppm** sulphur).
13. The **Gothenburg Protocol (1999)** (गोथेनबर्ग प्रोटोकॉल) under **CLRTAP (1979)** cuts transboundary acidifying emissions.
14. Treaty trap: Gothenburg/CLRTAP = acidifying air pollution; Montreal = ozone; Kyoto (क्योटो) = climate GHGs.
15. The **Taj Trapezium Zone** is about **10,400 km^2** around the Taj Mahal (ताज महल), strengthened by Supreme Court orders from **1996**.
16. **Mathura (मथुरा) refinery** is a classic SO_2 source in the Taj Trapezium narrative.
17. The **Singrauli–Sonbhadra (सोनभद्र)** coal TPP belt is a major UP–MP SO_2 source region.
18. UP NCAP (राष्ट्रीय स्वच्छ वायु कार्यक्रम) cities linked to acidifying/urban (नगरीय) air pressure include Kanpur (कानपुर), Ghaziabad, Lucknow (लखनऊ), and Noida.
19. SO_2 is denser than air — a physical property sometimes used in distractors.
20. Acid rain is **not** "rain only"; dry deposition counts too.
21. Secondary pollutants H_2SO_4 and HNO_3 form in the atmosphere from primary SO_2 and NOx.
22. FGD is flue-gas desulphurisation at power-plant chimneys — a mitigation hardware fact, not a treaty name.
23. Calling CO_2 the chief severe acid-rain driver is wrong; SO_2 and NOx are the pair.
24. Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल) does **not** control acid rain. Keep ozone chemistry out of acid-rain treaty options.

📋 Must-score facts — pH, acids, Taj, Gothenburg

Chemistry

Item	Lock
Acid rain pH	Below **5.6**
Normal rain (सामान्य वर्षा)	~pH **5.6** from CO_2 → H_2CO_3 (≠ 7)
Wet / Dry	Rain–snow–fog / gases–particles without rain
SO_2 / NOx paths	→ H_2SO_4 (TPPs) / → HNO_3 (vehicles)
CO_2 role	Sets 5.6 baseline — **not** main severe-acid culprit

Item	Lock
Receptors	Fish <~pH 5; Al^{3+} release; conifers; marble → gypsum
Lichens	SO_2 bio-indicators
Gothenburg / CLRTAP	**1999** / **1979** acidifying emissions



Topic 13 — Desertification & Land Degradation

Consolidated — 24 Must-Score Facts

1. **Desertification** is land degradation in **arid, semi-arid, and dry sub-humid (ASDM)** areas from climate variation plus human pressure.
2. Desertification is **not** only "desert expanding." Any dryland losing productivity can count.
3. **Land degradation** is broader: decline in productive capacity on any land through erosion, salinization, waterlogging, mining, or pollution.
4. India has about **96.4 million ha** degraded land — roughly **29–30%** (ISRO Atlas **2016** class figure).
5. **Normal geological erosion** is slow and roughly balanced by soil formation. **Accelerated erosion** is human-driven and faster than formation — the focus.
6. Water-erosion sequence: **sheet** → **rill** → **gully** → **ravine**. Chambal (चंबल) badlands are the ravine end-member.
7. Wind erosion works by **deflation** (lifting) and **abrasion** (sandblasting).
8. Leading causes include overgrazing in drylands, deforestation, unsustainable farming, excessive irrigation (salinization/waterlogging), and mining.
9. India hotspots: **Thar** (थार) wind erosion; **Chambal** (चंबल) ravines (MP–UP–Rajasthan); **Bundelkhand** drought (सूखा)–degradation cycle; Punjab–Haryana salinization belts.
10. Conservation tools include contour bunding, terraces, shelter belts, afforestation, gully plugging, check dams, and watershed management (वाटरशेड (जलग्रहण) प्रबंधन).
11. Earthworms are soil-fertility allies in restoration narratives.
12. **UNCCD** was adopted (अंगीकृत) in **1994**. Desertification and Drought Day is **17 June**.
13. **LDN** (Land Degradation Neutrality) under **SDG 15.3** targets no net loss of healthy land by **2030**.
14. **DPAP** launched in **1973**. **DDP** is **1977–78**. **IWMP** is **2009**. **MGNREGA (2005)** funds bunds and check dams.
15. Treaty trap: **UNCCD ≠ UNFCCC (यूएनएफसीसीसी)**. Desertification is not the climate convention.
16. Afforestation helps **prevent** degradation; it is not a cause of desertification.
17. Gully erosion (अवनलिका अपरदन) is not the same as sheet erosion (परत (चादर) अपरदन) — sheet is the first thin uniform stage.
18. Chambal ravine area is about **3 million ha** in standard coaching numbers.
19. UP Chambal ravine districts include **Etawah (इटावा) and Agra** (with Morena–Bhind on the MP side).
20. Bundelkhand's **seven UP districts** are the drought-plus-degradation fact.
21. Thar-margin UP questions emphasise wind erosion and dune encroachment.
22. **Sonbhadra** (सोनभद्र) is a UP mining-related land-degradation hotspot.
23. Wasteland means degraded land poorly suited to cultivation — related to, but not identical with, desertification's ASDM definition.
24. LDN by 2030 is an SDG land target; do not date UNCCD itself as 2030.

📋 Must-score facts — drylands, erosion, UNCCD, schemes

Concepts

Item	Lock
Desertification	Degradation in arid–semi-arid–dry sub-humid (ASDM)
≠ expanding desert only	Any dryland productivity loss can count
India degraded land	~**96.4 mha**; ~**29–30%** (ISRO Atlas 2016)
Water erosion sequence	Sheet → rill → gully → ravine (Chambal)

Item	Lock
Wind tools	Deflation + abrasion
UNCCD / Day	**1994** / **17 June**
LDN	SDG **15.3**; no net loss by **2030**
DPAP / DDP / IWMP	**1973** / **1977–78** / **2009**

Topic 14 — Environmental Impact Assessment

Consolidated — 24 Must-Score Facts

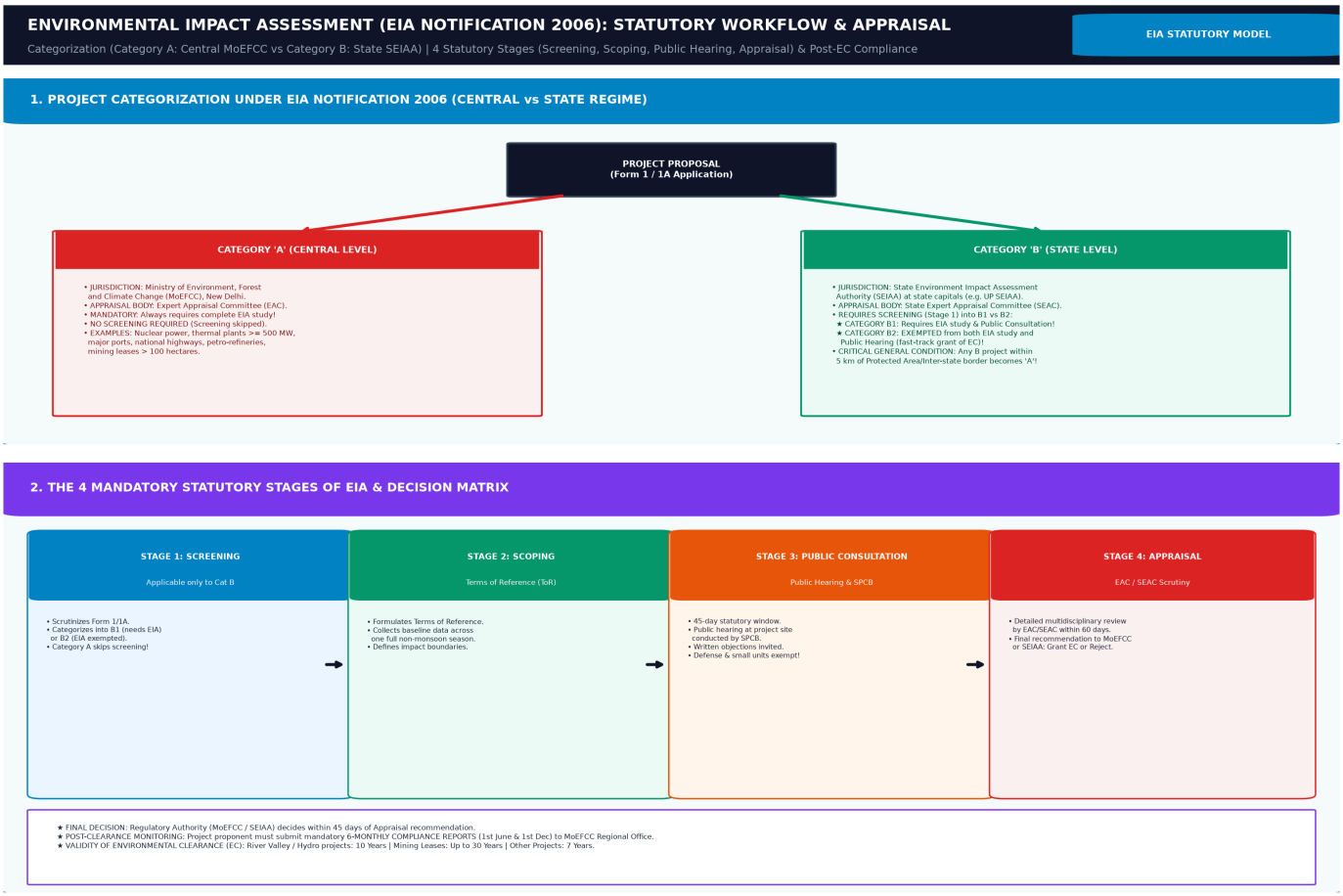
1. **EIA** studies a project's likely environmental effects **before** approval. The aim is anticipatory prevention, not reactive cleanup.
2. Legal base is the **Environment (पर्यावरण) (Protection) Act, 1986** (Section **3**). First EIA Notification was **1994**; the current framework is **EIA Notification 2006 (14 September)**.
3. **Environmental Clearance** is mandatory prior clearance for listed projects.
4. **Category A** is appraised centrally by **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) through the **EAC**.
5. **Category B** is appraised by **SEIAA** through **SEAC**. **B1** needs full EIA (usually with public hearing); **B2** is simplified.
6. Under a **General Condition**, a Category B project in a sensitive area is treated as Category A.
7. The four (चातुर्याम) classic stages are **screening** → **scoping** → **public consultation** → **appraisal**. Screening applies to Category B only.
8. **Scoping** fixes the **Terms of Reference (ToR)** (संदर्भ की शर्तें) for the EIA study.
9. An EIA report covers baseline, impact prediction, mitigation, and an **Environmental Management Plan (EMP)** (पर्यावरण प्रबंधन योजना).
10. **Rapid EIA** uses one season; **comprehensive EIA** covers all seasons.
11. MoEFCC/SEIAA **grant** EC. EAC/SEAC only **recommend** (सिफारिश). CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड)/SPCB set standards and monitor — they do **not** grant EC.
12. **NGT (Act 2010)** hears EC and pollution appeals. **PARIVESH** is MoEFCC's online EC portal.
13. **SEA** assesses policies, plans, and programmes upstream of projects. India has project-level EIA law but **no standalone SEA statute**.
14. SEA references include the Kiev SEA Protocol **2003** (Espoo family) and the EU SEA Directive.
15. Rio Principle **17** supports EIA for projects with significant impacts.
16. EC must come **before** construction. Ex-post facto clearance is illegal in the standard line.
17. Timeline: Bhopal **1984** → EPA **1986** → EIA Notification **1994** → EIA Notification **2006** → NGT Act **2010** → draft EIA amendment debates **2020** (not a full replacement of 2006).
18. UP examples: Ganga (गंगा) Expressway as Category A; Singrauli TPP expansions; Upper Ganga ESZ plan-level restrictions before project EC.
19. EIA is project-level; SEA is plan/policy-level — do not merge the acronyms.
20. Water Act **1974**/Air Act **1981** create consent regimes; they are **not** the EIA enabling Act (**EPA 1986** (पर्यावरण संरक्षण 1986) is).
21. DEIAA appears in the minor-minerals / B2 district track — not as the Category A central authority.
22. Public consultation includes hearing plus written comments; skipping it on B1 where required is a process defect.
23. EMP is mitigation and monitoring inside the EIA package — not a substitute for the EC grant itself.
24. PARIVESH tracks applications; it does not replace NGT's appellate role.

📋 Must-score facts — EIA stages, categories, SEA

Framework

Item	Lock
Legal base	EPA **1986**; Notification **1994** → **2006** frame
Category A / B	MoEFCC–EAC central / SEIAA–SEAC state
B1 / B2	Full EIA (+ hearing usual) / simplified

Item	Lock
General Condition	Sensitive-area B treated as A
Four stages	Screening → scoping → public consultation → appraisal
Screening	Category **B** only
Rapid vs comprehensive	One season / all seasons
EC granter	MoEFCC/SEIAA (EAC/SEAC recommend only)
NGT / PARIVESH	Act **2010** appeals / MoEFCC online portal
SEA	Policies–plans–programmes; India has no standalone SEA statute



Topic 15 — Sustainable Development & Environmental Governance

Consolidated — 32 Must-Score Facts

1. **Sustainable development** (Brundtland **1987**, *Our Common Future*) means meeting present needs without compromising future generations' ability to meet theirs.
2. Brundtland is **not** *Limits to Growth* (Club of Rome, **1972**).
3. SD rests on **economic, social, and environmental** pillars (plus institutional capacity). It is not environment-only.
4. Core objectives include basic needs, intra- and inter-generational equity, ecological limits, participation, and integrated planning.
5. Priority objectives also include **poverty reduction**, **biodiversity conservation**, and **pollution control**.
6. Indicators include **HDI (UNDP (यूएनडीपी))**, ecological footprint (पारिस्थितिक पदचिह्न), NITI (नीति) Aayog (नीति आयोग) SDG Index, global SDG indicator sets, and EPI.
7. **Natural capital** means renewable and non-renewable environmental resources (air, water, land, minerals, forests). **Roads** are not natural capital.
8. **SDGs** were adopted (अंगीकृत) in **September 2015**, run to **2030**, and include **17 goals** and **169 targets**.
9. SDGs replaced **MDGs** (8 goals, **2000–2015**) and apply universally to all countries.
10. Goal facts worth ratta include SDG **6** water, **7** energy, **13** climate, **14** life below water, and **15** life on land.
11. Global SDG Index ranks often place **Finland** first. India was about **116th (2017)** and about **112th (2023)**.
12. NITI Aayog **SDG India Index** has repeatedly placed **Kerala** first; Himachal (हिमाचल) Pradesh and Tamil Nadu (तमिलनाडु) sit near the top. **Gujarat** was **not** in the 2019–20 top-five set often tested.
13. Conference chain: **Stockholm 1972 (स्टॉकहोम सम्मेलन) → Brundtland 1987 → Rio and Agenda 21 in 1992 → SDGs 2015**.
14. **Agenda 21** is Rio **1992**, not 1995.
15. **LiFE Mission** launched in **June 2022**; the idea was floated at **COP26 Glasgow 2021**, not COP25.
16. MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) is the nodal environment (पर्यावरण) ministry. **NGT Act 2010** creates the green tribunal.
17. Rio principles frequently tested are **Precautionary Principle, Polluter Pays, and Public Trust**.
18. Constitutional hooks are **Article 48A**, **Article 51A(g)**, and **Article 21** (अनुच्छेद 21) read with a healthy environment.
19. UP SDG reporting typically sits with the **Planning Department**.
20. Environmental justice means fair distribution of environmental benefits and burdens plus fair participation in decisions.
21. Bhopal **1984** led to EPA **1986**. UP justice/pollution examples include Kanpur (कानपुर) tanneries and the Singrauli coal belt.
22. Ethics labels: **anthropocentrism** (human-centred), **biocentrism** (all life), **ecocentrism** (ecosystems). Leopold's Land Ethic is **1949**; deep ecology (पारिस्थितिकी) links to **Arne Næss**.
23. Gandhi (गांधी)'s **trusteeship** is a frequent Indian environmental-ethics fact.
24. **Green economy** (UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) **2011** line) means low-carbon, resource-efficient, socially inclusive growth.
25. **Circular economy** redesigns take-make-dispose into reduce-reuse-recycle loops, often with **EPR** rules. Circular sits inside, but is not identical to, green economy.
26. **Sustainable agriculture** means using land so its **quality remains intact**, with mixed cropping, organic manures, nitrogen-fixing plants, and pest-resistant varieties as the environment-friendly strategy set.
27. **NMSA** (National Mission for Sustainable Agriculture) is a **NAPCC** mission on productivity, water-use efficiency, nutrient management, and livelihood diversification.

28. Example scheme–SDG matches: Jal Jeevan / Swachh Bharat → SDG **6**; SAUBHAGYA / UJALA → SDG **7**; NAPCC / Net Zero 2070 → SDG **13**.
29. MDGs had 8 goals; SDGs have 17 — do not reverse the counts.
30. HDI is an UNDP development indicator; it is not itself an MoEFCC clearance tool.
31. Polluter Pays allocates cleanup cost to the polluter; Precautionary Principle acts under scientific uncertainty — different tools.
32. Public Trust treats certain natural resources as held for public benefit, not absolute (निरपेक्ष) private exhaustion.

📋 Must-score facts — Brundtland, SDGs, NITI Index

Spine

Item	Lock
Brundtland	**1987** *Our Common Future* (≠ Limits to Growth 1972)
Three pillars	Economic · social · environmental
SDGs	**Sep 2015** → **2030**; **17** goals / **169** targets
MDGs → SDGs	8 goals 2000–15 / universal to all countries
Key goals	6 water · 7 energy · 13 climate · 14 ocean · 15 land
Conference chain	Stockholm 72 → Brundtland 87 → Rio 92 → SDGs 15
Agenda 21	Rio **1992** (≠ 1995)
NITI SDG Index top	Kerala repeatedly first; HP/TN near top

Topic 16 — Environmental Organizations (India)

Consolidated — 28 Must-Score Facts

1. **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) is India's nodal ministry for environment (पर्यावरण), forests, wildlife, and climate change (जलवायु परिवर्तन). It began as MoEF in **1985** and was renamed MoEFCC in **2014**. Headquarters is **New Delhi**.
2. MoEFCC grants **Category A Environmental Clearance**, runs the **PARIVESH** portal, and administers EPA, Wildlife Act, Forest Conservation Act, and Biodiversity Act. It does **not** set NAAQS pollution standards.
3. **CPCB** (केंद्रीय प्रदूषण नियंत्रण बोर्ड) was created under the **Water Act, 1974**, sits in **New Delhi**, sets **NAAQS**, and coordinates State Pollution Control Boards. The Air Act **1981** widened its air-pollution role.
4. CPCB does **not** grant Environmental Clearance, does **not** publish ISFR, and does **not** prepare the Annual Groundwater Quality Report — that report belongs to **CGWB**.
5. Every state has an **SPCB**. SPCBs issue **Consent to Establish** and **Consent to Operate** under the Water and Air Acts. They do **not** grant Category B EC — that is **SEIAA**.
6. **UPPCB** is Uttar Pradesh (उत्तर प्रदेश)'s board with headquarters at **Lucknow** (लखनऊ). It enforces norms on Kanpur (कानपुर) tanneries, Noida–Ghaziabad industry, and Ganga (गंगा)–Yamuna (यमुना) discharges.
7. **NGT** was created by the **NGT Act, 2010** (operational **2011**). Headquarters is New Delhi, with regional benches at Bhopal, Pune (पुणे), Kolkata, and Chennai.
8. NGT hears environmental appeals, including EC appeals, and can award compensation under Polluter Pays. The Act aims for disposal within **six months**. NGT does **not** itself grant EC.
9. The NGT Chairperson must be a retired **Supreme Court** judge or a High Court Chief Justice.
10. **WII** (**1982**, Dehradun (देहरादून)) does wildlife research and trains forest officers. It helped frame India's **ten** biogeographic zones and tiger-census methods. It is **not** the plant or animal survey body.
11. **BSI** (**1890**, Kolkata) surveys **plants** and publishes *Flora (वनस्पति) of India* and the plant Red Data Book (रेड डेटा बुक). **ZSI** (**1916**, Kolkata) surveys **animals** and publishes *Fauna (प्राणीजात) of India* and the animal Red Data Book.
12. **FSI** (**1981**, Dehradun) maps forest cover (वन आवरण) and publishes the **India State of Forest Report** every **two years**. ISFR is **not** an annual CPCB product.
13. **NEERI** (**1958**, **Nagpur**) began as **CPHERI** and is a CSIR laboratory for environmental science and engineering under the Ministry of Science and Technology.
14. Headquarters fact: **Delhi** — MoEFCC, CPCB, NGT; **Dehradun** (देहरादून) — WII, FSI; **Kolkata** — BSI, ZSI; **Nagpur** — NEERI; **Lucknow** — UPPCB.
15. Category A EC → **MoEFCC**. Category B EC → **SEIAA**. Industry consent → **SPCB**. EC appeal → **NGT**.
16. Pollution standards → **CPCB**. Forest cover report → **FSI**. Plant taxonomy → **BSI**. Animal taxonomy → **ZSI**. Wildlife training → **WII**. Environmental engineering research → **NEERI**.
17. MoEFCC was **not** created in **1972**. That year marks the Stockholm Conference and the Wildlife (Protection) Act.
18. Consent under Water/Air Acts is **not** the same as Environmental Clearance under EPA/EIA rules. Confusing CTE/CTO with EC is a classic trap.
19. Forest clearance for diversion of forest land sits with **MoEFCC** under the Forest (Conservation) Act — not with CPCB or NGT.
20. BSI and ZSI both sit in Kolkata, but BSI is plants only and ZSI is animals only. Swapping kingdoms is a frequent wrong match.
21. WII and FSI both sit in Dehradun, but WII is wildlife science/training and FSI is forest-cover assessment. Do not give ISFR to WII.
22. NGT adjudicates; it does **not** replace CPCB on standards or MoEFCC on clearances. The older **NEAA (1997)** was dissolved after NGT was set up.

23. SPCBs work under CPCB technical guidance and state administrative control. They enforce standards; they do not write national NAAQS.
24. PARIVESH is MoEFCC's online EC application system — not a CPCB monitoring portal.
25. Match report to publisher: ISFR → **FSI**; Groundwater Quality Report → **CGWB**; NAAQS bulletins → **CPCB**; *Flora* → **BSI**; *Fauna* → **ZSI**.
26. NEERI is **Nagpur 1958**, not New Delhi 1970 and not a pollution-standards board.
27. **CAZRI = Jodhpur**; **IIFM = Bhopal**; operational Antarctica (भारती) stations = **Maitri + Bharati** (मैत्री).
28. **INCOIS = Hyderabad (हैदराबाद)**; ENVIS population (जनसंख्या)–environment centre = **IIPS Mumbai**.

📋 Must-score facts — MoEFCC, CPCB, NGT, surveys

HQ map

Item	Lock
MoEF → MoEFCC	**1985** → **2014**; Delhi
CPCB	Water Act **1974**; Delhi; NAAQS; coordinates SPCBs
UPPCB	Lucknow
NGT	Act **2010** (ops 2011); Delhi + Bhopal/Pune/Kolkata/Chennai
NGT Chair	Retired SC judge **or** HC Chief Justice
WII / FSI	**1982** / **1981**; both **Dehradun**
BSI / ZSI	**1890** / **1916**; both **Kolkata**
NEERI	**1958** Nagpur; CSIR (≠ MoEFCC)

Topic 17 — Environmental Laws & Policies

Consolidated — 28 Must-Score Facts

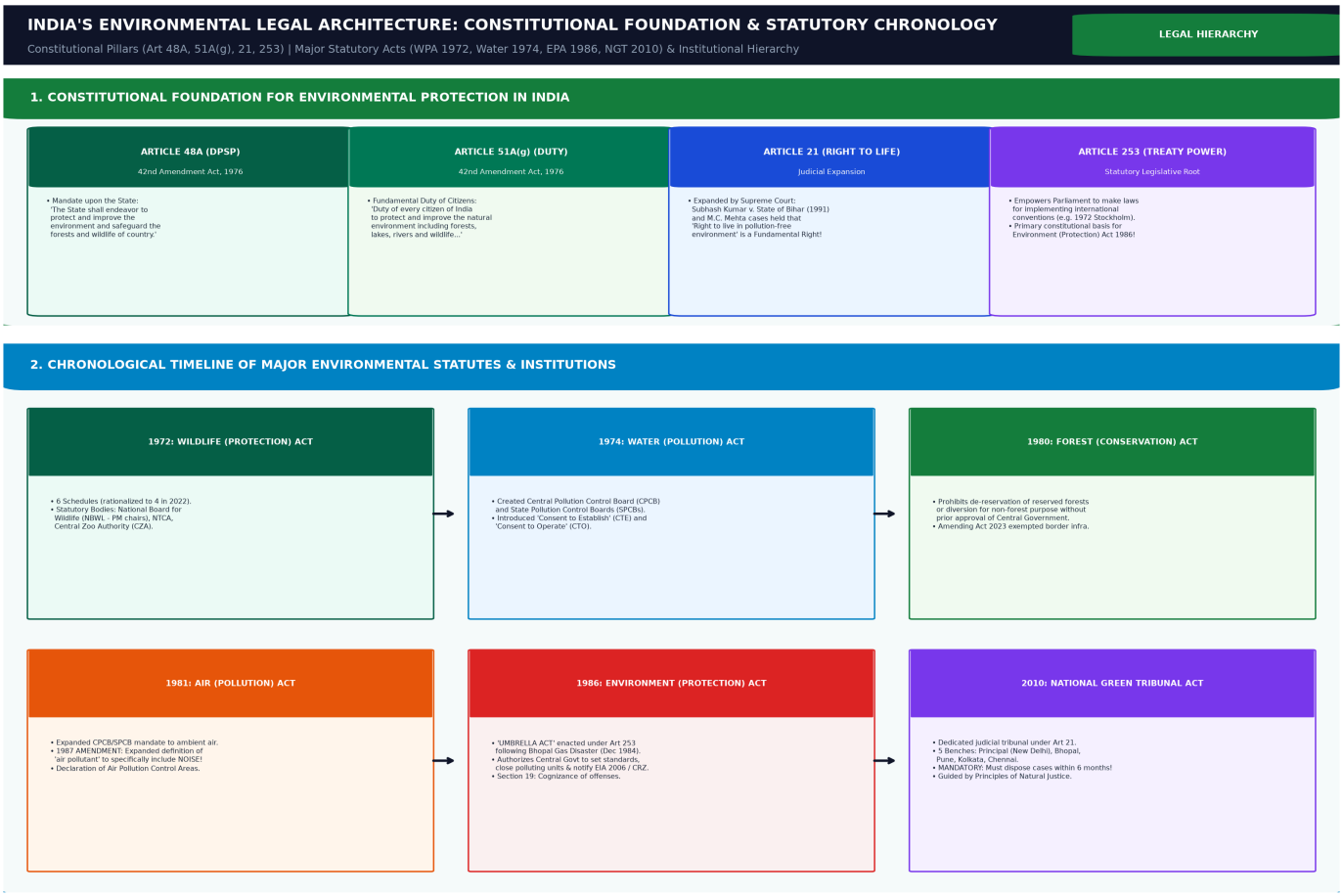
1. Memorise the act-year chain: **WPA 1972** → **Water 1974** → **FCA 1980** → **Air 1981** → **EPA 1986** → **BDA 2002** → **FRA 2006** → **NGT Act 2010**.
2. **Article 48A** (DPSP (राज्य के नीति निदेशक तत्व)) directs the State to protect the environment (पर्यावरण). **Article 51A(g)** is the citizen's Fundamental Duty to protect nature. Do not swap them.
3. **Article 21** (अनुच्छेद 21) has been read to include a healthy environment. **Article 48** is about agriculture and cattle — it is **not** the environment article (**48A** is).
4. The **Environment (Protection) Act, 1986** is the post-Bhopal umbrella law under **Article 253**. Section **3** powers support EIA, CRZ, ESZ, **GEAC**, and **CGWA**. It **supplements** Water and Air Acts; it does not repeal them.
5. EIA Notification and related clearances flow from **EPA**, not from the Water Act.
6. The **Wildlife (Protection) Act, 1972** uses **six** Schedules. Schedules **I–IV** protect animals; Schedule **V** is vermin; Schedule **VI** covers plants. **Schedule I** is the highest protection; Schedule **V** animals may be hunted as vermin.
7. Under WPA, a **National Park** (Section **35**) is stricter than a **Sanctuary** (Section **18**). Tiger Reserves rest on Section **38V** (2006 amendment path).
8. The **Forest (Conservation) Act, 1980** requires prior **Central** approval before diverting forest land. CAMPA and NPV sit in this clearance world.
9. The **Forest Rights Act, 2006** recognises rights of Scheduled Tribes and Other Traditional Forest Dwellers. Individual forest rights are capped around **4 hectares**, with a cutoff of **13 December 2005**, and recognition (मान्यता) begins at the **Gram Sabha** (ग्राम सभा).
10. **FCA** controls **diversion** of forest land. **FRA** recognises **rights** of forest dwellers. Do not treat them as one Act.
11. The **Biological Diversity Act, 2002** implements the CBD through **NBA** (national), **SBB** (state), and **BMC** (local) with People's Biodiversity Registers.
12. The **Water Act, 1974** created **CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड)/SPCB** and the consent regime. The **Air Act, 1981** covers air pollution and NAAQS. Water is **1974**; Air is **1981** — do not reverse the years.
13. Industry **consent** (CTE/CTO) is under Water/Air Acts via **SPCB**. **Environmental Clearance** is under EPA/EIA via **MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय)/SEIAA**. Forest clearance is under **FCA** via MoEFCC.
14. The **NGT Act, 2010** creates the green tribunal for environmental appeals and compensation, with a six-month disposal target.
15. National Forest Policy **1952** was revenue-leaning. Later practice follows National Forest Policy **1988** targets of about **33%** forest cover (वन आवरण) nationally and **66%** in hills.
16. EPA is **1986**, not 1972. WPA is **1972**, not 1986. Mixing those years is a standard trap.
17. Schedule **V** under WPA is **not** the highest protection tier — Schedule **I** is.
18. Consent is not EC, and EC is not forest clearance. Keep the three tools on separate legal tracks.
19. FRA's nodal line is tribal (आदिवासी) forest rights, not MoEFCC forest diversion under FCA.
20. BDA is about biodiversity access and benefit-sharing institutions — not the same as WPA schedules or NGT appeals.
21. Article **48A** binds the State as a Directive Principle; **51A(g)** binds the citizen as a duty. Both environment hooks sit beside Article **21** case law.
22. Water Act creates the pollution-board architecture; EPA is the later umbrella for broader environmental regulation including EIA.
23. Air Act year fact is **1981**. Pairing Air with **1974** or Water with **1981** is wrong.
24. NGT hears appeals; it does not rewrite the year of EPA or replace CPCB's standards role.
25. CAMPA/NPV issues arise after forest diversion under FCA — they are not FRA Gram Sabha rights tools.
26. Match clearance labels carefully: **SPCB consent**, **MoEFCC/SEIAA EC**, **MoEFCC forest clearance**.

27. Wildlife and forests sit on the **Concurrent List** (समवर्ती सूची) — Centre and states both legislate.
28. Indian courts have adopted Rio principles — **Precautionary**, **Polluter Pays**, and **Public Trust** (*Vellore Citizens' Welfare Forum*).

📋 Must-score facts — act years, articles, clearances

Act chain

Item	Lock
Spine	WPA 72 → Water 74 → FCA 80 → Air 81 → EPA 86 → BDA 02 → FRA 06 → NGT 10
48A / 51A(g) / 48	State DPSP env / duty / agri–cattle (**not** env)
EPA umbrella	Post-Bhopal; Art. 253; EIA/CRZ/ESZ/GEAC power
WPA Schedules	I–IV animals; V vermin; VI plants
NP vs Sanctuary	Sec. 35 stricter / Sec. 18
FCA vs FRA	Central diversion approval / forest-dweller rights
BDA ladder	NBA → SBB → BMC + PBR
Consent vs EC vs forest	SPCB Water/Air / MoEFCC–SEIAA EPA / FCA Central



Topic 18 — International Environmental Agreements & Conferences

Consolidated — 30 Must-Score Facts

1. Conference spine: **Stockholm 1972** → **Brundtland 1987** → **Rio 1992** → **Johannesburg 2002** → **Rio+20 2012** → **SDGs and Paris 2015**.
2. **Stockholm 1972** (स्टॉकहोम सम्मेलन) was the first UN Conference on the Human Environment (पर्यावरण). It led to **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) (Nairobi). World Environment Day (विश्व पर्यावरण दिवस) is **5 June**. The slogan fact is "Only One Earth."
3. The **Brundtland Report** (**1987**, *Our Common Future*) popularised **sustainable development**. It is **not** *Limits to Growth* (**1972**, Club of Rome).
4. **Rio 1992** (UNCED) produced the Rio Declaration, **Agenda 21**, Forest Principles, and opened **UNFCCC** (यूएनएफसीसीसी) and **CBD** for signature. **CITES** is **not** a Rio product — CITES is **1973**.
5. **Agenda 21** is Rio's non-binding sustainable-development action plan for the twenty-first century. It was **not** signed in 1995.
6. Climate chain: **UNFCCC 1992** → **Kyoto Protocol 1997** → **Paris Agreement (पेरिस समझौता) 2015**. Paris uses universal **NDCs** and the **1.5°C** ambition language.
7. Ozone chain: **Vienna Convention 1985** → **Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल) 1987** → **Kigali Amendment 2016** (HFCs). Montreal is **ozone**, not climate.
8. Treaty–issue fact: Montreal/Vienna = ozone; Kyoto/Paris/UNFCCC = climate; CBD = biodiversity; Ramsar = wetlands; CITES = wildlife **trade** (पण्याध्यक्ष).
9. **CMS (Bonn)** covers migratory species. **Basel** covers hazardous **waste**. **Rotterdam** covers hazardous **chemicals** (PIC). **Stockholm POPs** covers persistent organic pollutants. **Minamata** covers **mercury**.
10. **CBD** has three pillars: conserve biodiversity, sustainable use, and fair benefit-sharing. **Nagoya Protocol (2010)** details access and benefit-sharing.
11. **Ramsar** (रामसर) (**1971**) is the wetlands treaty under the Wise Use idea. World Wetlands Day (विश्व आर्द्रभूमि दिवस) is **2 February** (रामसर).
12. **CITES** (**1973**, Washington) regulates trade through Appendices **I–III**. IUCN (आईयूसीएन)'s Red List assesses extinction risk; CITES regulates trade — different jobs.
13. **GEF** was established in **1991** on the eve of Rio as a finance mechanism for major environmental conventions.
14. **UNCCD (1994)** (मरुस्थलीकरण रोकथाम संधि) is the sole legally binding convention linking environment, development, and sustainable land management; it stresses bottom-up local participation against desertification (मरुस्थलीकरण).
15. The **New York Declaration on Forests (2014)** is a voluntary, non-legally binding political declaration (halve forest loss by 2020 / end by 2030 ambition language); India was **not** an original signatory.
16. Kyoto set binding targets mainly for developed parties. Paris is a universal NDC architecture. Do not treat them as identical instruments.
17. Kigali amends Montreal to phase down **HFCs**. It is an ozone-protocol amendment path, not a UNFCCC protocol.
18. Basel is waste shipment control. Rotterdam is prior informed consent for chemicals. Stockholm POPs is about persistent toxics such as listed DDT/PCB classes. Minamata is mercury.
19. CMS is migratory species; CITES is trade; Ramsar is wetlands. Mixing those three is a classic match trap.
20. UNEP was born from Stockholm, not from Rio. Rio opened UNFCCC and CBD.
21. Paris is **2015**; Kyoto is **1997**; UNFCCC framework is **1992**. Keep the climate chronology straight.
22. Montreal successful ODS phase-out is often contrasted with slower climate politics — but stems still demand the correct issue tag, not slogans.
23. Rudrasagar is a **Tripura** Ramsar site. Sultanpur is **Haryana**. Neither is an Uttar Pradesh (उत्तर प्रदेश) site.

24. Uttar Pradesh Ramsar examples include sites such as Nawabganj, Samaspur, Sarsai Nawar, Upper Ganga (गंगा), Bakhira, and Haiderpur — verify the live list before the paper, but never park Rudrasagar or Sultanpur in UP.
25. CBD opened at Rio **1992**; Nagoya ABS is **2010**. Do not date CBD as 2010.
26. Forest Principles at Rio are non-binding soft law — not the same as CITES appendices.
27. Johannesburg **2002** and Rio+20 **2012** continue the sustainable-development conference line after Rio; they do not reopen CITES as a 1992 invention.
28. "Only One Earth" and UNEP creation fact to Stockholm **1972**, not to Paris **2015**.
29. Wise Use is the Ramsar philosophy. It is not the CBD's three objectives word-for-word.
30. Master year table worth ratta: Ramsar 1971, CITES 1973, Vienna 1985, Montreal 1987, Basel 1989, UNFCCC/CBD 1992, UNCCD 1994, Kyoto 1997, Rotterdam 1998, Stockholm POPs 2001, Nagoya 2010, Minamata 2013, New York Forests Declaration 2014, Paris 2015, Kigali 2016.

📋 Must-score facts — conferences, treaty–issue map

Conferences

Item	Lock
Spine	Stockholm 72 → Brundtland 87 → Rio 92 → Joburg 02 → Rio+20 12 → SDGs/Paris 15
Stockholm → UNEP	**1972**; UNEP Nairobi; WED **5 June**
Rio products	Agenda 21 · UNFCCC · CBD (CITES **not** Rio product)
Climate ladder	UNFCCC 92 → Kyoto 97 → Paris 15
Ozone ladder	Vienna 85 → Montreal 87 → Kigali 16

Issue tags

Item	Lock
Montreal / Kyoto–Paris	Ozone / climate
CBD / Ramsar / CITES	Biodiversity / wetlands / wildlife trade
Basel / Rotterdam / Stockholm POPs	Haz. waste / PIC chemicals / POPs
CMS (Bonn)	Migratory species
GEF / UNCCD	**1991** finance / **1994** desertification

GLOBAL ENVIRONMENTAL GOVERNANCE: HISTORICAL SUMMITS ROADMAP & TREATY CLUSTERS

Evolutionary Timeline (Stockholm 1972 -> Brundtland 1987 -> Rio 1992 -> Paris 2015) | Specialized Clusters: Climate, Ozone, Biodiversity & Hazardous Chemicals (BRS+M)

GLOBAL TREATIES

1. THE HISTORICAL EVOLUTIONARY SUMMITS ROADMAP (1972 TO 2030 SDGs)

1972: STOCKHOLM

UNCHE ('Only One Earth')

- Watershed global conference.
- Created UNEP (Nairobi).
- 5th June declared World Environment Day.
- Prompted India's WPA 1972!

1987: BRUNDTLAND

'Our Common Future'

- WCED Report formally defined SUSTAINABLE DEVELOPMENT: 'Meeting present needs without compromising ability of future generations to meet theirs.'

1992: RIO EARTH SUMMIT

UNCED (The Landmark)

- Rio Declaration (27 principles).
- Agenda 21 action plan.
- 3 RIO CONVENTIONS:
  1. UNFCCC (Climate)
  2. UNCBD (Biodiversity)
  3. UNCCD (Desertification)

2015: PARIS & SDGs

COP21 & 2030 Agenda

- PARIS AGREEMENT: Binding goal to limit warming well below 2°C (pursue 1.5°C) via NDCs.
- 17 SUSTAINABLE DEVELOPMENT GOALS (SDGs) & 169 targets.

2. THE FOUR SPECIALIZED MULTILATERAL ENVIRONMENTAL CONVENTIONS CLUSTERS

CLIMATE CHANGE TRACK

- UNFCCC (1992, Rio): Parent treaty.
- KYOTO PROTOCOL (1997): Annex I binding emissions targets; Clean Dev Mechanism (CDM).
- PARIS AGREEMENT (2015): Nationally Determined Contributions; Common But Differentiated Responsibilities (CBDR-RC).

OZONE PROTECTION TRACK

- VIENNA CONVENTION (1985): Framework convention for research.
- MONTREAL PROTOCOL (1987): Legally binding phase-out of CFCs, Halons & ODS (Universal ratification!).
- KIGALI AMENDMENT (2016): Mandates phase-down of HFCs (GHGs).

BIODIVERSITY & SPECIES

- RAMSAR CONVENTION (1971): Wetlands.
- CITES (1973): Trade in endangered species.
- CMS / BONN (1979): Migratory wildlife.
- UNCBD (1992, Rio):
  - * Cartagena Protocol (2000): Biosafety
  - * Nagoya Protocol (2010): ABS
  - * Kunming-Montreal (2022): 30x30 Target!

CHEMICALS CLUSTER (BRS+M)

- BASEL (1989): Control of transboundary movements of hazardous wastes.
- ROTTERDAM (1998): Prior Informed Consent (PIC) for hazardous pesticides.
- STOCKHOLM (2001): Persistent Organic Pollutants (POPs - 'Dirty Dozen').
- MINAMATA (2013): Mercury phase-out.

Topic 19 — Climate & Environmental Institutions

Consolidated — 24 Must-Score Facts

1. **IPCC** sits in **Geneva**, issues Assessment Reports, and synthesises climate science. It does **not** set NDCs and is **not** the UNFCCC treaty body.
2. IPCC was established in **1988** by **WMO and UNEP** and shared the Nobel Peace Prize in **2007**.
3. **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) headquarters is **Nairobi**. Flagship products include the **Emissions Gap Report** (उत्सर्जन अंतराल रिपोर्ट) and GEO. UNEP was created after Stockholm **1972**.
4. UNEP hosts several MEA secretariats (including Montreal, Basel, and Minamata lines) and works with GEF-linked convention finance.
5. **UNDP** headquarters is **New York**. It publishes the **Human Development Report** (from **1990**) and the **HDI**. HDR is UNDP, not UNEP.
6. HDI is linked to **Mahbub ul Haq** and **Amartya Sen**. Do not invent a wrong founding-name trap for HDR.
7. **IUCN** (आईयूसीएन) headquarters is **Gland**. It publishes the **Red List** of extinction risk. IUCN assesses risk; it does **not** regulate wildlife trade (पण्याध्यक्ष) — that is **CITES** (साइट्स).
8. On the Red List, **Threatened** means **Vulnerable + Endangered + Critically Endangered** (VU + EN + CR). "Endangered alone" is not the full threatened set.
9. **WWF** is a conservation NGO also linked to Gland, known for the panda logo and the **Living Planet Report / Living Planet Index**.
10. **FAO** headquarters is **Rome**. Environment-facing outputs include the **State of the World's Forests (SOFO)** and agriculture–forest links. FAO partners UNEP on some chemical-convention work such as Rotterdam.
11. HQ memory: Nairobi = UNEP; New York = UNDP; Geneva = IPCC; Gland = IUCN and WWF; Rome = FAO.
12. IPCC ≠ UNFCCC. Science assessment is not the same as the climate treaty and COP process.
13. UNEP ≠ UNDP. Environment (पर्यावरण) programme versus development agency — swap reports and you lose marks.
14. Emissions Gap Report → **UNEP**. Human Development Report → **UNDP**. Red List → **IUCN**. Living Planet → **WWF**. Assessment Reports → **IPCC**.
15. IUCN ≠ CITES. Risk categories versus trade appendices.
16. WWF ≠ IUCN. NGO Living Planet work versus Red List authority.
17. FAO is food and agriculture first; it is not the Red List body and not the Emissions Gap publisher.
18. GEF (**1991**) finances global environmental conventions; do not confuse GEF with IPCC authorship of Assessment Reports.
19. Red List codes to know: CR, EN, VU as the threatened band; also EX, EW, NT, LC, DD, NE in the wider ladder.
20. UNEP's Green Economy (हरित अर्थव्यवस्था) work and GEO sit in the environment-programme lane — not in UNDP's HDI lane.
21. IPCC reports inform negotiators; parties still adopt NDCs under the UNFCCC/Paris track.
22. Living Planet Index tracks vertebrate abundance trends; it is not a legal trade permit system.
23. SOFO is FAO's forest flagship — not FSI's India State of Forest Report.
24. Match org → city → product before matching slogans. Geography of headquarters is a frequent prelims filter.

📋 Must-score facts — IPCC, UNEP, IUCN HQs & reports

HQ / product

Item	Lock
IPCC	Geneva; WMO+UNEP **1988**; Nobel Peace **2007**; ARs

Item	Lock
UNEP	Nairobi; Emissions Gap + GEO
UNDP	New York; HDR/HDI from **1990**
IUCN	Gland; Red List (risk ≠ CITES trade)
WWF	Gland; Living Planet Report/LPI
FAO	Rome; SOFO
Threatened	VU + EN + CR
IPCC ≠ UNFCCC	Science assessment ≠ climate treaty/COP

Topic 20 — Biodiversity Conservation Methods

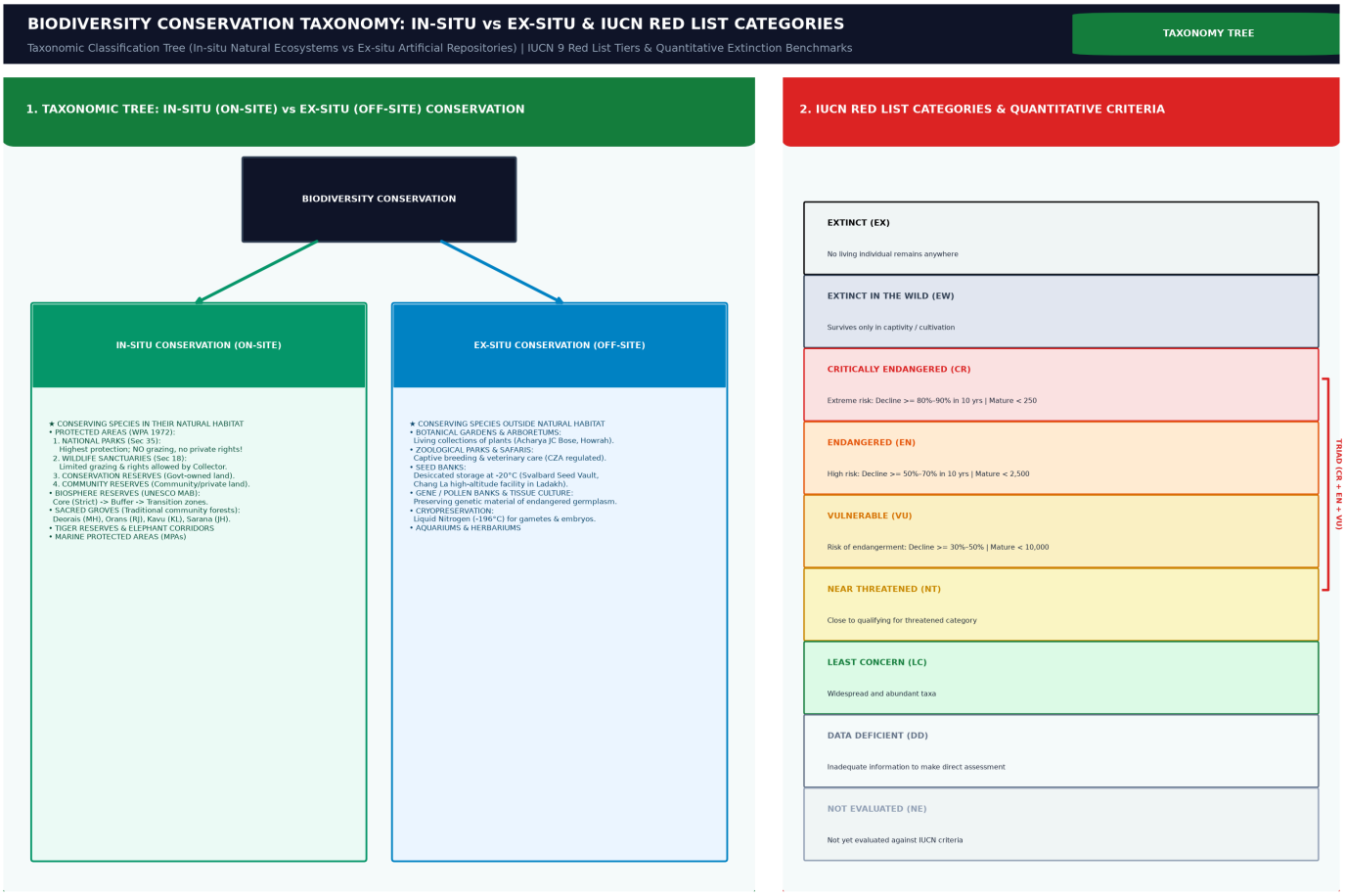
Consolidated — 24 Must-Score Facts

1. **In-situ** (स्थान पर) conservation protects biodiversity **in the natural habitat (वास स्थान)**. It is the preferred primary strategy because the whole ecosystem (पारिस्थितिकी तंत्र) and natural selection stay intact.
2. **Ex-situ** (स्थानांतरित) conservation protects biodiversity **outside** the natural habitat — zoos, botanic gardens, seed banks, gene banks, cryobanks. It is a backup, not a replacement for in-situ.
3. In-situ examples: National Park, Wildlife Sanctuary, Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र), Conservation/Community Reserve (सामुदायिक आरक्षित क्षेत्र), sacred grove, Ramsar (रामसर) wetland, wildlife corridor.
4. Ex-situ examples: zoo, botanical garden, seed bank, gene bank, captive breeding centre, tissue culture, cryopreservation (क्रायो-संरक्षण).
5. Four-way pattern: National Park and sacred grove are **in-situ**; zoo, seed bank, and botanic garden are **ex-situ**.
6. A **gene bank** stores genetic material broadly — seeds, DNA, tissue, pollen, sperm, embryos. A **seed bank** stores **seeds only**.
7. **Orthodox seeds** (सामान्य (रूढ़िवादी) बीज) (wheat, rice) tolerate drying and cold storage around **−18°C**. **Recalcitrant seeds** (कठिन (असहनीय) बीज) (mango, coconut) cannot take simple dry seed-bank storage and need living collections or cryo routes.
8. **Cryopreservation** stores material at about **−196°C** in liquid nitrogen.
9. India's plant gene/seed flagship is **NBPGR**, New Delhi (**1976/1978** line). India's animal gene-bank flagship is **NBAGR**, Karnal (**1984**).
10. The **Indian Botanic Garden**, Howrah (**1787**), is India's classic large botanic garden, linked with BSI administration. The **National Zoological Park**, New Delhi (**1959**), is the national zoo flagship.
11. **CZA** is the statutory zoo regulator under **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय).
12. The **Svalbard Global Seed Vault** in Norway is a duplicate backup vault — not India's NBPGR.
13. NBPGR is plants in Delhi; NBAGR is animals in Karnal. BSI in Kolkata surveys plants; it is **not** the national gene bank.
14. Dudhwa (दुधवा) and Katerniaghat-style protected areas are **in-situ**. Kukrail crocodile centre and garden collections are **ex-situ**.
15. Sacred groves (पवित्र उपवन) are community in-situ pockets — do not park them with zoos.
16. Wildlife corridors keep in-situ populations linked; they are not ex-situ cages.
17. Captive breeding is ex-situ recovery support; releasing only works if habitat (in-situ) remains.
18. Seed banks fail for recalcitrant species if the stem assumes every crop seed stores like wheat.
19. Gene bank ⊃ seed bank. Calling every gene bank a "seeds-only room" is wrong.
20. Biosphere Reserves (जैवमंडल आरक्षित क्षेत्र) and National Parks are in-situ frameworks; botanic gardens are living ex-situ plant collections.
21. UP colour: Dudhwa/Katerniaghat = in-situ; NBRI Lucknow (लखनऊ) garden collections = ex-situ; NBPGR in Delhi still serves national plant genetic resources including UP crops.
22. Community and Conservation Reserves are newer in-situ tools beside classic parks and sanctuaries.
23. Cryobanking is an ex-situ technology for germplasm that ordinary refrigeration cannot hold.
24. Priority line for prelims: **in-situ first, ex-situ supplementary** — never reverse the preference.

📋 Must-score facts — in-situ, ex-situ, gene banks

Methods

Item	Lock
In-situ	Natural habitat — preferred primary
Ex-situ	Zoo, botanic garden, seed/gene/cryobank — backup
Gene bank vs seed bank	Broad genetic material / seeds only
Orthodox / Recalcitrant	Dry+cold OK (wheat/rice) / cannot (mango/coconut)
Cryopreservation	~−196°C liquid nitrogen
NBPGR / NBAGR	Plants Delhi / animals Karnal
Indian Botanic Garden	Howrah **1787**; BSI link
Svalbard	Norway backup vault (≠ NBPGR)
CZA	Statutory zoo regulator, MoEFCC



Topic 21 — Species & Ecology

Consolidated — 26 Must-Score Facts

1. India's **Red Data Book** is a **national** catalogue — **BSI** for plants, **ZSI** for animals (both Kolkata). The **IUCN Red List** is the **global** extinction-risk system from IUCN (Gland).
2. IUCN has nine categories: EX, EW, CR, EN, VU, NT, LC, DD, NE. Risk order puts CR above EN above VU among living threatened tiers.
3. **Threatened** on IUCN means **VU + EN + CR** only — not "Endangered" alone.
4. IUCN assesses extinction risk. **CITES** (साइट्स) regulates international wildlife **trade** (पण्याध्यक्ष). Different institutions, different jobs.
5. **Ramsar (रामसर) (1971)** is the wetlands treaty under Wise Use. World Wetlands Day (विश्व आर्द्रभूमि दिवस) is **2 February** (रामसर).
6. Uttar Pradesh (उत्तर प्रदेश) Ramsar examples include Sarsai Nawar, Samaspur, Nawabganj, Upper Ganga (गंगा), Bakhira, and Haiderpur. **Rudrasagar is Tripura**; **Sultanpur is Haryana** — not UP.
7. **Mangroves** are salt-tolerant coastal forests with **pneumatophores** (वात-मूल) and often viviparous seeds. **Sundarbans** (सुंदरबन) is the classic largest mangrove belt; Rhizophora and Avicennia are common genera.
8. **MISHTI** is the Union Budget **2023–24** mangrove restoration push. Do not date it as 2019.
9. **Coral reefs** are built from calcium-carbonate skeletons and live with symbiotic **zooxanthellae** (ज़ूज़ैंथेली). **Bleaching** happens when heat stress expels the algae.
10. India's coral belts include Gulf of Mannar, Lakshadweep (लक्षद्वीप), Andaman (अंडमान) & Nicobar (निकोबार), and Gulf of Kutch.
11. **Grasslands** are grass-dominated systems maintained by fire and grazing. UP's Terai (तराई) is a foothill grassland–forest mosaic; **Dudhwa** (दुधवा) holds barasingha (swamp deer). Kaziranga (काजीरंगा) is the floodplain grassland rhino fact.
12. Himalayan vegetation follows altitude belts: tropical → subtropical → temperate → alpine → nival. Snow leopard is a high Himalaya (हिमालय) fact; Dudhwa is Terai foothill fringe, not alpine.
13. Marine **neritic** shelf waters are the highest fisheries-productivity zone; pelagic means open ocean. India's coast is about **7516 km**.
14. The **Gangetic dolphin** is a **freshwater river** dolphin — not a marine species trap.
15. Wetland ≠ mangrove ≠ coral. Inland marsh, saline (लवणीय) coastal forest, and reef builders are three different ecosystems.
16. Pneumatophores are mangrove breathing roots for gas exchange in waterlogged saline soil.
17. Amrit Dharohar (**2023**) is India's wetland stewardship framing beside Ramsar listing.
18. Coral bleaching (प्रवाल विरंजन) is a stress response, not a synonym for mangrove dieback.
19. Terai grasslands of UP are not Himalayan alpine meadows.
20. Red Data Book (national BSI/ZSI) is not identical to IUCN Red List categories, even when species appear on both tracks.
21. World Wetlands Day is February **2**, not Environment (पर्यावरण) Day (5 June).
22. Mangrove restoration schemes do not make landlocked UP a mangrove state — UP links are scheme awareness and river–wetland ecology.
23. Zooxanthellae live inside coral polyps; losing them bleaches the coral's colour and energy supply.
24. Threatened = three IUCN tiers combined. Memorise VU, EN, CR as the band.
25. CMS covers migratory species; Ramsar covers wetlands; do not swap those treaty jobs when ecosystems are asked.
26. Dudhwa = Terai grassland–forest and barasingha; Sundarbans = mangrove; Lakshadweep/Mannar = coral — keep the flagship map clean.

📋 Must-score facts — Red List, Ramsar, biomes

Lists / wetlands

Item	Lock
India Red Data	BSI plants + ZSI animals (Kolkata)
IUCN 9 categories	EX EW CR EN VU NT LC DD NE
Threatened	VU+EN+CR only
IUCN vs CITES	Extinction risk / trade regulation
Ramsar / Wetlands Day	**1971** / **2 February**
Rudrasagar / Sultanpur	Tripura / Haryana
MISHTI	Budget **2023–24** mangroves
India corals	Mannar · Lakshadweep · A&N · Kutch (**not** Sundarbans)
Gangetic dolphin	Freshwater river (≠ marine trap)

Topic 22 — Renewable Energy

Consolidated — 34 Must-Score Facts

1. **Renewable** energy replenishes on a human timescale — sun, wind, water, biomass, geothermal. **Nuclear** is low-carbon but usually **not** counted as renewable MCQs.
2. **Solar PV** turns sunlight directly into electricity. **CSP** (सीएसपी) uses mirrors to make heat, then steam and turbines.
3. The **National Solar Mission** began in **2010** under NAPCC (JNNSM from Jan 2010). India chased about **100 GW** solar by 2022 and **500 GW** non-fossil capacity by **2030**.
4. Solar has the **greatest RE potential** in India among common options and is the most eco-friendly among solar cell vs fossils/nuclear fission options.
5. **Bhadla** in Rajasthan is India's mega solar-park fact (not Gujarat as "largest"). **PM-KUSUM** supports farmer solar pumps. **OSOWOG** is One Sun One World One Grid.
6. **Domestic Content Requirement (DCR)** in solar news means mandating domestically manufactured PV cells/modules.
7. The **International Solar Alliance** began in **2015** (India–France, COP21). Headquarters is **Gurugram**. Not all UN members — originally sun-belt countries between Tropics. First Summit **New Delhi, March 2018**. Members target mobilising about **USD 1 trillion** for solar by **2030**.
8. **Cochin International Airport** = world's first fully solar-powered airport. **Diu** = first Indian UT / smart city running **100% solar** (daytime / UT stems). **Surat** = first district with **100% solar-powered PHCs**.
9. Wind power converts kinetic energy of air; power scales roughly with the **cube** of wind speed. Leading states include **Tamil Nadu (नाडु), Gujarat, and Karnataka**.
10. **Biomass** is energy from organic matter. **Biogas** (बायोगैस) from anaerobic digestion is mainly **methane (CH_4) and carbon dioxide (CO_2) (कार्बन डाइऑक्साइड)** — not a butane/carbon-monoxide pair.
11. Bagasse cogeneration, **GOBAR-Dhan**, and **SATAT** (bio-CNG) are biomass-policy neighbours.
12. **Geothermal** uses Earth's internal heat. Indian provinces include **Himalaya (हिमालय), Cambay, West Coast, SONATA (Son–Narmada (नर्मदा)–Tapi (तापी)), Godavari (गोदावरी), Mahanadi (महानदी)** — **not Ganga (गंगा) Delta (डेल्टा)**. Sites: **Puga, Manikaran, Tattapani**.
13. **Tidal** energy comes from Moon-driven sea-level change. India's best potential is often locked to the **Gulf of Khambhat**, then Gulf of Kutch.
14. **Small hydro** under MNRE is **≤25 MW**. Above that is treated as large hydro. Himalayan streams hold large SHP potential.
15. The **Ramganga (रामगंगा) Project** in Garhwal (गढ़वाल), Uttarakhand (उत्तराखंड), is about **198 MW** — a hydro fact that also matters for irrigation links toward UP.
16. **Green hydrogen** is H_2 from water electrolysis powered by **renewable** electricity. Grey hydrogen comes from natural gas; blue adds CCS (सुरक्षा कैबिनेट समिति) to gas routes.
17. The **National Green Hydrogen Mission** (January **2023**) targets about **5 MMT** per year by **2030**. National Hydrogen Energy Board dates to **3 Oct 2003**.
18. Hydrogen can power **IC engines**, be **blended with natural gas**, and run **fuel-cell vehicles** — all three are correct in recent stems.
19. Fuel cells with pure H_2 emit **heat and water**; they produce **DC**; they can power buildings and portable devices.
20. Solar ≠ wind ≠ tidal ≠ geothermal. Source and physics differ even when all are "clean."
21. ISA HQ is Gurugram, not Paris and not New Delhi as a casual guess.
22. Biogas major pair is CH_4 + CO_2. Wrong gas pairs are frequent traps.
23. Geothermal and nuclear are not "derived from the Sun" in the renewable-sun family sense used in many stems — though fossil-fuel energy ultimately traces to the Sun.
24. India still lacks a large commercial tidal plant even where potential maps look strong.

25. Green / grey / blue hydrogen colours mark production route and carbon intensity — not the colour of the gas.
26. UP colour: Mirzapur (मिर्ज़ापुर) solar collaboration history (इतिहास), western UP bagasse cogeneration, limited wind, Ramganga water–power links.
27. CSP is heat-first; PV is photon-to-electron. Do not describe CSP as a simple rooftop panel.
28. Wind leadership is southern and western coastal/plateau states — not a default UP fact.
29. Small hydro ceiling **25 MW** is the MNRE number; do not invent a different cutoff.
30. OSOWOG is a grid-connectivity vision around solar; it is not a biomass scheme.
31. Non-fossil **500 GW by 2030** is a capacity ambition fact beside older Solar Mission milestones.
32. Tariffs for solar are set by **CERC / SERCs**, not by SECI alone; India is **not** a top silicon-wafer manufacturer (panels often assembled from imported cells).
33. Microbial fuel cells and bioasphalt appear as advanced sustainable-energy neighbours (wastewater electricity; non-petroleum road binder).
34. Algae biofuel limits for developing countries: high expertise + large facilities with ecological/social concerns — **not** "seas only."

📋 Must-score facts — solar, wind, biomass, tidal

Solar / wind

Item	Lock
NSM / JNNSM	**2010** under NAPCC
Targets class	~100 GW solar by 2022; **500 GW** non-fossil by 2030
Bhadla	Rajasthan mega solar park
ISA	**2015** India–France COP21; HQ **Gurugram**
PM-KUSUM / OSOWOG	Farmer solar pumps / One Sun One World One Grid
Wind leaders	TN, Gujarat (power $\propto$ speed^3)
Biogas majors	**CH_4 + CO_2** ($\neq$ butane/CO)
Tidal best India	Gulf of Khambhat → Gulf of Kutch
SHP (MNRE)	$\leq$**25 MW**

Topic 23 — Disaster & Environment

Consolidated — 24 Must-Score Facts

1. Environmental degradation (पर्यावरणीय क्षरण) raises disaster risk (deforestation worsens floods and landslides). Disasters also damage ecosystems through erosion, pollution, and forest loss.
2. Flood types include riverine floods on the Ganga (गंगा) plains, flash floods/cloudbursts in Himalaya–Terai (तराई), and urban (नगरीय) waterlogging. UP river facts include Ganga, Yamuna (यमुना), Ghaghara (घाघरा), Rapti (राप्ती), and Gandak (गंडक).
3. **NDMA** (आपदा प्राधिकरण) is under the Disaster Management Act **2005**. The **Prime Minister** chairs NDMA. **NDRF** is the response force. Sendai Framework for disaster risk reduction is **2015**.
4. Drought progresses from **meteorological** (low rain) to **agricultural** (soil moisture) to **hydrological** (rivers and wells). Bundelkhand's seven UP districts are a chronic drought hotspot.
5. **DPAP** (Drought Prone Areas Programme) is **1973**. Do not date DPAP as **1952**. A rough programme spine is CDP (सीडीपी) 1952 (सामुदायिक विकास) → DPAP 1973 → FWP 1977 → TRYSEM 1979.
6. Landslides need gravity, water, weak slopes, and often vegetation loss. Himalaya and north-eastern hills are core belts. Kedarnath (केदारनाथ) **2013** is the cloudburst–landslide–flood chain fact. Roads, mining, and deforestation are human triggers.
7. Tropical cyclones need warm seas around **26–27°C**. The **Bay of Bengal** produces more than the Arabian Sea. The **eye** is calm; the **eyewall** has the strongest winds; **storm surge** (तूफानी महोर्मि) is the coastal flood killer.
8. Mangroves reduce storm-surge damage. Named cyclone memory often uses Fani (**2019**) and Amphan (**2020**).
9. IMD plains heat-wave idea: about **≥40°C** with departure **+4.5°C to +6.4°C** as heat wave; severe when departure is **≥+6.4°C** or temperature **≥47°C**. Peak is May–June in North India. Urban heat island amplifies cities but is not identical to a heat-wave definition.
10. Forest fires need fuel, oxygen, and ignition. Peak season is roughly March–June. Chir pine needles are a Himalayan fuel fact. FSI issues satellite fire alerts.
11. **Piral Lao–Paise Pao** in Uttarakhand (उत्तराखंड) pays for pine-needle collection to cut fire risk and raise income. It is an Uttarakhand flagship, not a default UP scheme name.
12. NDMA Chair = Prime Minister — a recurring institution fact.
13. Eye ≠ strongest winds. Eyewall holds the violence; eye is the calm core.
14. Landslides are not "purely natural" when options ignore slope cutting and deforestation.
15. Heat wave (IMD criteria) ≠ urban heat island (city microclimate effect), though cities worsen heat stress.
16. DPAP year is **1973**, not the older CDP **1952** tag.
17. Cloudburst is extreme local rain in a short time — often the Himalayan flood trigger neighbour to landslides.
18. Storm surge is sea pushed ashore by cyclone winds — distinct from ordinary astronomical tide.
19. Meteorological drought (मौसमी (मौसम विज्ञान) सूखा) is rainfall failure; agricultural drought (कृषि सूखा) is crop-moisture failure; hydrological drought (जल-वैज्ञानिक सूखा) is stream/groundwater failure. Keep the sequence.
20. Sendai **2015** is the global DRR framework fact beside India's NDMA architecture.
21. Forest-fire season and chir-pine fuel matter more in Uttarakhand hills than in deep Gangetic floodplain districts.
22. Cyclone naming and Bay of Bengal dominance are coastal facts; UP's disaster map is flood–drought–heat first.
23. NDRF is the specialised response force; NDMA is the apex policy body chaired by the PM.
24. Bundelkhand drought and Terai/Ganga floods are the two UP geography disaster anchors on this sheet.

📋 Must-score facts — NDMA, drought, cyclone, fire

Institutions / hazards

Item	Lock
NDMA	DM Act **2005**; Chair = **PM**; NDRF response
Sendai	Disaster risk reduction framework
Drought ladder	Meteorological → agricultural → hydrological
DPAP	**1973** (≠ 1952)
Cyclone needs	Warm sea ~26–27°C; BoB > Arabian Sea
Eye / Eyewall	Calm core / strongest winds
Heat wave (plains)	≥40°C + departure rules; severe ≥47°C or +6.4°C
Forest fire season	Roughly Mar–Jun; chir pine fuel
Piral Lao–Paise Pao	UK pine-needle scheme (≠ Delhi)

Topic 24 — Current Environmental Issues

Consolidated — 26 Must-Score Facts

1. **Microplastics** are plastic fragments **smaller than 5 mm**. Primary sources include microbeads; secondary sources come from larger plastic breaking down. They are not "fast biodegradable."
2. Microplastics enter food webs from plankton to fish to humans. India's identified single-use plastic bans tightened around **2022**. UN plastics treaty talks (INC) also run from that period.
3. One **carbon credit** equals **1 tonne of CO_2-equivalent** reduced, removed, or avoided. Kyoto (क्योटो)'s CDM produced CERs; voluntary markets use standards such as Verra and Gold Standard.
4. **Carbon trading** is the market that buys and sells allowances or credits (cap-and-trade). The **EU ETS** is the largest compliance market. India's **CCTS (2023)** under the Energy Conservation Act line is steered with **BEE**.
5. **PAT** creates energy-efficiency certificates (**ESCerts**). PAT is **not** identical to carbon credits under CCTS.
6. **Green Credit Programme (2023)** rewards eco-actions. It is **not** the same instrument as a carbon credit.
7. **Net zero** means anthropogenic emissions balanced by removals. India's target is **2070** (COP26 Glasgow **2021**). China is often locked at **2060**; EU/UK at **2050**. Net zero is not gross zero.
8. India's NDC track includes cutting emission intensity and raising non-fossil capacity share toward the **2030** horizon (including the ~50% non-fossil capacity language in later updates).
9. **Green finance** is capital for climate and environment (पर्यावरण) projects — green bonds, climate funds, GCF. India issued **Sovereign Green Bonds** in FY **2022–23**; SEBI's Green Bond Framework is **2023**.
10. The **100 billion** a year climate-finance pledge is the developed-to-developing finance politics fact. Loss and Damage Fund operationalisation is linked to **COP28 (2023)**.
11. **Blue economy** is sustainable ocean and coastal use — fisheries, ports, tourism, offshore renewables — aligned with **SDG 14**. **Blue carbon** (नीला कार्बन) is carbon stored in mangroves and seagrass.
12. **MISHTI (2023 Budget)** restores mangroves and sits beside Sagarmala and Deep Ocean Mission in the blue–coastal policy neighbourhood.
13. **LiFE** (Lifestyle for Environment (जीवनशैली पर्यावरण)) is a MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) behaviour-change mission. The idea was pitched at **COP26 Glasgow (2021)** and launched in **June 2022**. It is **not** a COP-25 Madrid product.
14. LiFE catalogues pro-planet actions for citizens; it is lifestyle awareness, not a carbon-market statute.
15. Carbon credit (unit) ≠ carbon trading (market) ≠ Green Credit (eco-action incentive) ≠ green finance (capital).
16. India net zero is **2070**, not 2050.
17. Microplastics are **<5 mm**, not 5 cm.
18. Kyoto flexible mechanisms are climate tools; Montreal is ozone — do not swap treaty families in "current issues" stems.
19. Nature-based credits often come from forestry and mangrove (blue carbon) projects.
20. UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम)'s Emissions Gap Report (उत्सर्जन अंतराल रिपोर्ट) tracks the gap between pledges and pathways — neighbour to net-zero politics.
21. SUP item bans are domestic plastic regulation; they do not by themselves equal a global plastics treaty.
22. CCTS is India's carbon market architecture; PAT remains an efficiency-certificate neighbour under BEE.
23. Blue economy is ocean wealth with sustainability; it is not a synonym for inland wetland policy alone.
24. Glasgow **2021** facts both India's 2070 net-zero announcement track and the LiFE idea pitch.
25. Offset means using credits to compensate one's own emissions — related to, but not identical with, "trading" as a market design.
26. Keep MISHTI dated **2023**, not 2019, when mangrove restoration is asked.

📋 Must-score facts — carbon markets, net zero, LiFE

Markets / targets

Item	Lock
Microplastics	<**5 mm**; primary beads / secondary fragments
1 carbon credit	1 tCO_2e reduced/removed/avoided
PAT	ESCerts — ≠ carbon credit under CCTS
Green Credit	**2023** eco-action rewards (≠ carbon credit)
India net zero	**2070** (COP26 Glasgow)
LiFE	Idea COP26 **2021**; launch Jun **2022**
MISHTI	Budget **2023** mangroves
Blue economy	Sustainable ocean use; SDG **14**

Topic 25 — Global Environmental Geography

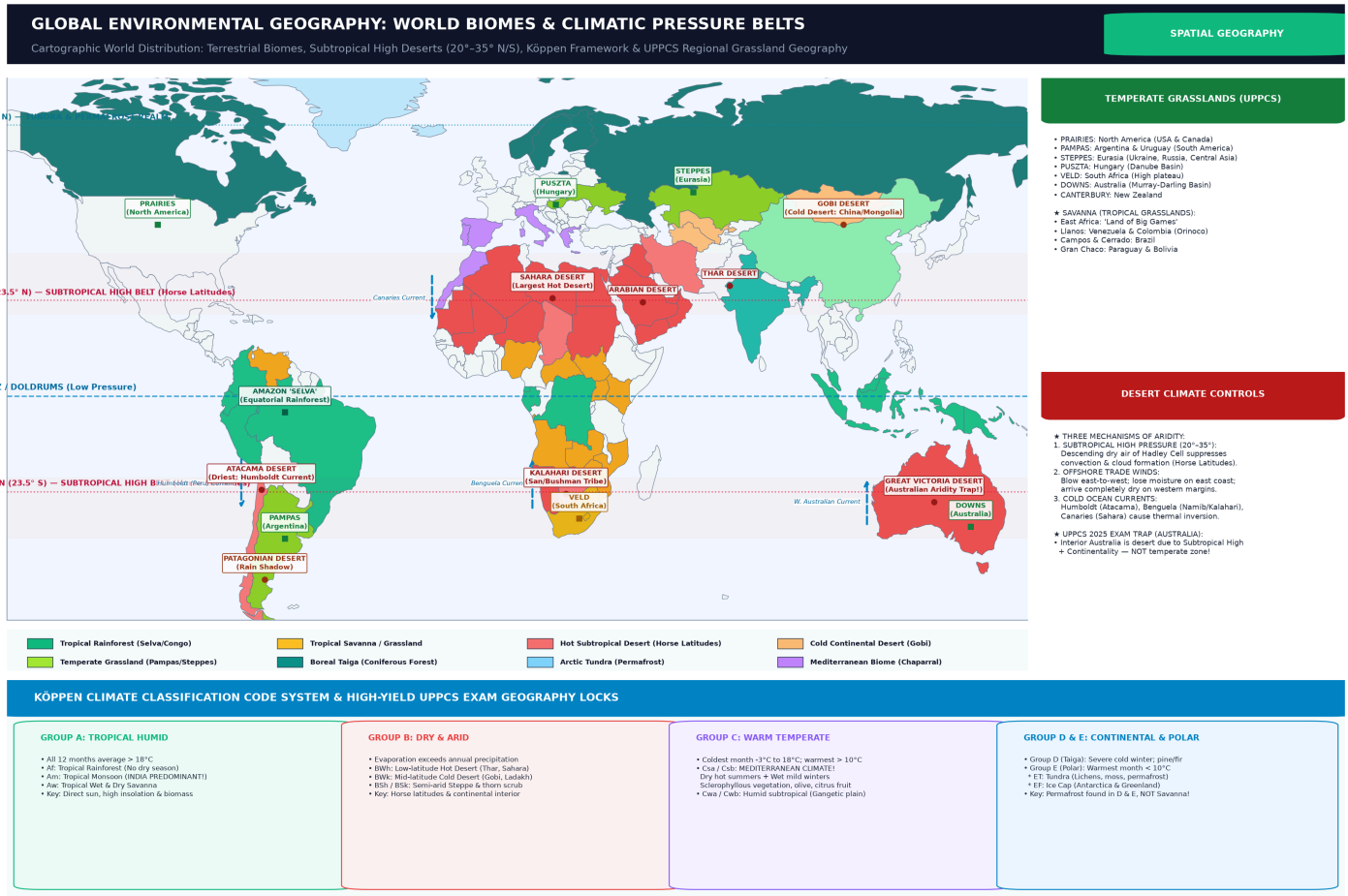
Consolidated — 24 Must-Score Facts

1. Global environmental geography asks **where and why** Earth's climate–biome patterns occur. Climate (temperature and rainfall) shapes biomes, biodiversity, and degradation belts.
2. Major biomes to ratta: tropical rainforest (Amazon **Selva** (सेल्वा)), tropical deciduous (Indian monsoon forests), desert (**<25 cm** rain), savanna (सवाना) (scattered trees with wet–dry seasons), temperate grassland (**Prairie (प्रेयरी), Steppe (स्टेपी), Pampas, Puszta (पुस्ज़टा)**), taiga/boreal, tundra (टुंड्रा), Mediterranean chaparral, and coastal mangrove as a transitional belt.
3. **Köppen** letters: **A** tropical, **B** arid, **C** temperate, **D** cold, **E** polar. India is largely monsoon (**Am** type language); Thar (थार) is hot desert (**BWh**); Gobi is cold desert (**BWk**).
4. **Sahara** is the largest hot desert. **Atacama** is among the driest. **Gobi** spans **Mongolia and China**.
5. Australia's interior desert is explained mainly by **subtropical high pressure and continentality** — not by calling northern Australia "temperate."
6. Brazil facts: **Selva** = Amazon equatorial rainforest name; **Terra Roxa** = fertile dark volcanic soil. Both can be true together in match stems.
7. Floral matching classics: Planting Hearth → **C.O. Sauer**; World Floral Regions → **Adolf Engler**; Cryophyte Steppe → Arctic tundra setting; **Puszta** → **Hungary**.
8. Tribe–region facts: Bushman–Kalahari; Eskimo–northern Canada; Pygmy–Congo. **Masai are East Africa**, not West Africa.
9. **Permafrost** (स्थायी हिम मृदा) appears in **taiga and tundra**. It is **not** a savanna feature.
10. Global biome counts (often **14** in textbook lists) are **not** the same as India's **ten** biogeographic zones (Rodgers and Panwar).
11. Biome is a large climate-defined vegetation–fauna (प्राणीजात) region. Ecosystem is a local biotic–abiotic functional unit. Do not treat them as identical scales.
12. Temperate grassland names: Pampas (Argentina), Prairie (North America), Steppe (Eurasia), Puszta (Hungary).
13. Taiga is boreal conifer forest; tundra is moss–lichen cold treeless belt. Savanna is tropical grassland with trees.
14. Mangrove is coastal transitional vegetation — not a purely inland terrestrial biome.
15. Hotspot (Myers) means high endemism plus severe habitat (वास स्थान) loss (often taught as **≥70%** loss). It is not merely "any rich forest."
16. Gobi is cold desert Asia — not Sahara Africa and not Atacama Chile.
17. Selva ≠ Terra Roxa. Forest name versus soil name.
18. Masai wrong-option trap is West Africa; correct region is East Africa.
19. India's monsoon deciduous belt is not Amazon Selva.
20. Permafrost continuous cover is strongest in tundra; taiga may show patches — still not savanna.
21. Mediterranean biome is winter-rain shrubland (chaparral type), distinct from monsoon India.
22. Desert rainfall fact **<25 cm** is the quick arid filter in many MCQs.
23. Engler floral regions and Sauer planting hearth are cultural–biogeography matchs, not climate-classification letters.
24. Keep "14 biomes vs 10 Indian biogeographic zones" as a numbers trap pair.

📋 Must-score facts — biomes, Köppen, grasslands

Biomes / climate

Item	Lock
Köppen letters	A tropical · B arid · C temperate · D cold · E polar
India / Thar / Gobi	Monsoon Am-type / BWh hot desert / Mongolia–China
Largest hot / driest	Sahara / Atacama
Selva / Terra Roxa	Amazon rainforest name / fertile volcanic soil (Brazil)
Temperate grasslands	Pampas AR · Prairie NA · Steppe Eurasia · Puszta HU
Taiga / Tundra / Savanna	Boreal conifer / moss–lichen treeless / tropical grass+trees
Permafrost	Taiga–tundra (≠ savanna)
Biome ≠ ecosystem	Large climate region / local functional unit
India biogeographic zones	**10** (Rodgers & Panwar) ≠ global biome count



Topic 26 — Water Resources & Water Conservation

Consolidated — 38 Must-Score Facts

1. World water rough shares: oceans about **97.2%**, ice/glaciers about **2%**, groundwater about **0.6%**, rivers about **0.01%**. Descending order is **Oceans > Polar ice > Groundwater > Rivers**.
2. India holds roughly **4%** of world freshwater for about **18%** of world population (जनसंख्या). Per-capita availability around **1400 m^3** marks water stress in many readings.
3. The **water cycle** is driven primarily by **solar radiation** through evaporation from oceans, lakes, and rivers. Humidity is measured with a **hygrometer** or psychrometer.
4. Amount of rainfall depends strongly on **humidity** in the atmosphere — more vapour means greater chance of cloud and rain when temperature falls.
5. Many transplanted seedlings fail because most **root hairs** are lost during transplantation; root hairs absorb water and mineral nutrients.
6. **Freshwater is not a biotic resource.** Biotic resources come from living/organic material (forests, animals, fish); **coal and petroleum** are classed biotic because they form from decayed organic matter.
7. **World Water Day (विश्व जल दिवस) = 22 March** (UN observance from **1993**; UN-Water since **2003**). Theme **2024** = *Leveraging Water for Peace* (also taught as prosperity and peace).
8. **Rajendra Singh** ("Jal Purush" / Waterman of India; **Tarun Bharat Sangh**, Alwar) won **Ramon Magsaysay 2001** and **Stockholm Water Prize 2015**.
9. **IWRM** follows Dublin Principles (**1992**) and treats the **river basin** as the planning unit. National Water Policy **2012** priority language puts drinking first, then ecology (पारिस्थितिकी), agriculture, and industry.
10. **National Water Mission** (NAPCC, from **30 June 2008**) aims at conservation, less wastage, equitable distribution via IWRM. Five goals include public water database + climate impact assessment, citizen/state conservation action, focus on over-exploited areas, **+20% water-use efficiency**, and basin-level IWRM. Urban (नगरीय) wastewater recycling and ocean-water tech for coastal cities fit NWM; **Himalayan–peninsular river linking and free borewell reimbursement do not**.
11. The **Ministry of Jal Shakti (शक्ति)** (**2019**) merged water resources and drinking-water departments.
12. **Rainwater harvesting** captures rain for use or recharge — rooftops, pits, check dams, johads. Tamil Nadu (नाडु)'s mandatory RWH (**2003**) is a state flagship fact.
13. A **watershed** is land draining to a common outlet. **Hariyali** is the GOI watershed project for rural water conservation (≠ Pani Sansad / Jal Kranti as the 2022 key).
14. Desert plants conserve water via hard/waxy leaves, tiny/absent leaves, and thorns instead of leaves.
15. River cleaning spine: **Central Ganga (गंगा) Authority / GAP 1985 → NRCA 1995 → NGRBA 20 Feb 2009 (EPA; PM chair) → Namami Gange 2014** (NMCG). **STP** = domestic; **CETP** = industrial.
16. **Yamuna (यमुना) Action Plan** launched **1993** (YAP-I to ~2003; YAP-II Dec 2004; **YAP-III 7 May 2016** under Namami Gange).
17. **Ganges River Dolphin** (*Platanista gangetica*) = **National Aquatic Animal (2009)**; dams, nets, agrochemicals threaten it — crocodile increase is **not** a standard decline cause in UPSC (संघ लोक सेवा आयोग) framing.
18. Kanpur (कानपुर) is the Ganga tannery pollution fact. **Gomti** is Lucknow (लखनऊ)'s river — do not put Gomti's main story on Kanpur.
19. **NLCP (June 2001)** restored urban/semi-urban lakes; merged with NWCP into **NPCA (Feb 2013)**. NLCP lakes include Bhoj, Sukhna, Pichola, Bhimtal — **Chilika is not NLCP**.
20. Wetlands follow **Ramsar (रामसर) 1971**, Wise Use, World Wetlands Day (विश्व आर्द्रभूमि दिवस) on **2 February** (रामसर), and Wetland Rules **2017**. UP Ramsar sites exist; **Rudrasagar** and **Sultanpur** are **not** UP.
21. **CGWB** prepares the Annual Groundwater Quality Report. **CGWA** (groundwater regulation) is under **EPA 1986** (पर्यावरण संरक्षण 1986). India has the **largest groundwater-irrigated area** in the world. Arsenic drinking-water teaching often points to **Kolkata**.

22. Disinfection: **chlorine**, chloramine, chlorine dioxide, ozone; **UV** kills microbes without changing taste/odour — UV does **not** remove odour or settle turbidity.
23. Water Act **1974** (amended **1988**); Water Cess Act **1977** (enforced **1 Apr 1978**).
24. **Atal Bhujal Yojana (2019)** is a groundwater management scheme tag.
25. **Jal Jeevan Mission** launched in **August 2019** for rural household taps (Har Ghar Jal). Do not date JJM as **2014**.
26. Drinking-water standard fact **IS 10500:2012**: fluoride about **≤1.5 mg/L**, arsenic about **≤0.01 mg/L**, pH about **6.5–8.5**.
27. Heavy sand mining lowers water table and can pollute groundwater — it does **not** decrease river salinity as a standard consequence.
28. **Lisbon Declaration** (UN Ocean Conference) is about **oceans**, not freshwater glaciers.
29. Rivers are **not** second after oceans in global freshwater ranking — ice/glaciers are.
30. CGWB ≠ CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड) on the groundwater quality report; CGWA ≠ CGWB.
31. Namami Gange is **2014**; GAP is **1985**. NGRBA chair is the **Prime Minister**, not rotating CMs.
32. IWRM is basin coordination; watershed is the local drainage unit used in ridge-to-valley works.
33. Aquifer is the underground storage layer; it is not a surface canal.
34. BOD measures organic pollution strength in water (mg/L) — a quality metric, not a volume share.
35. Bundelkhand scarcity, Kanpur tanneries, Gomti–Lucknow, and arsenic plains are the UP colour set.
36. JJM is rural tap connectivity; it is not Namami Gange's river-cleaning mission.
37. STP vs CETP: domestic vs industrial effluent trains — swap them and consent logic breaks.
38. National Water Policy priority order starts with **drinking water**, not industry.

📋 Must-score facts — water shares, IWRM, days, schemes

Shares / policy

Item	Lock
World water order	Oceans ~97.2% > ice ~2% > GW ~0.6% > rivers ~0.01%
India freshwater share	~**4%** water for ~**18%** population
Water cycle driver	Solar radiation / evaporation
World Water Day	**22 March**
Rajendra Singh	Jal Purush; Magsaysay 2001; Stockholm Water Prize 2015
IWRM / Dublin	**1992**; basin as planning unit
NWM	NAPCC; from **30 Jun 2008**
Jal Shakti Ministry	**2019** merger
Freshwater resource type	**Abiotic** (≠ biotic)

GLOBAL WATER PARTITIONING, AQUIFER HYDROGEOLOGY & INDIA'S WATER STRESS

Visual Conceptual Guide: Planetary Freshwater Budget | Falkenmark Stress Index | Unconfined vs Confined Artesian Aquifers

HYDROLOGY MODEL

1. GLOBAL WATER BUDGET: THE CASCADING VOLUME DISTRIBUTION

LEVEL 1: TOTAL EARTH WATER (~1,386 MILLION KM³)

SALINE OCEANS & SEAS: 97.2% (Unusable for drinking / agriculture)

FRESH: 2.8%

LEVEL 2: BREAKDOWN OF THE 2.8% FRESHWATER

GLACIERS & POLAR ICE CAPS: ~68.7% (~2.0% of total Earth water)

GROUNDWATER: ~30.1% (~0.6% of total)

SURFACE: 1.2%

LEVEL 3: BREAKDOWN OF SURFACE FRESHWATER (~0.01% OF TOTAL)

GROUND ICE & PERMAFROST: 69.0%

LAKES: 20.9%

SOIL: 3.8%

RIVERS: ~0.49%!

CRITICAL UPPCS EXAM TRAP — WATER VOLUME DESCENDING ORDER

- CORRECT DESCENDING RANK: Oceans (~97.2%) > Polar Ice & Glaciers (~2.0%) > Groundwater (~0.6%) > Lakes (~0.01%) > Soil Moisture > Atmosphere > Rivers (~0.0001%)
- EXAM TRAP: 'Rivers are second after oceans' is COMPLETELY FALSE! Rivers hold a tiny fraction.
- FRESHWATER RANK: Polar Ice (~68.7%) > Groundwater (~30.1%) > Lakes > Rivers.

FALKENMARK WATER STRESS INDICATOR & INDIA'S VULNERABILITY

> 1,700 m³ / capita / yr	NO WATER STRESS	Adequate water for agriculture & life
1,000 – 1,700 m³	WATER STRESS (INDIA HERE!)	India at ~1,400 m³ (dropped from 5,177 m³ in 1951)
500 – 1,000 m³	CHRONIC WATER SCARCITY	Severe economic & human health stress
< 500 m³ / capita / yr	ABSOLUTE SCARCITY	Insufficient for basic survival & ecosystem collapse

2. AQUIFER HYDROGEOLOGY: UNCONFINED vs CONFINED (ARTESIAN) SYSTEMS

Evaporation and runoff

Land surface rises

VADOSE / UNSATURATED ZONE (Air + Soil Moisture)

WATER TABLE (PHREATIC SURFACE)

UNCONFINED AQUIFER (Permeable Sand & Gravel)

- Direct atmospheric pressure recharge from rain
- Standard dug wells tap this layer (vulnerable to surface contamination)

CONFINING IMPERMEABLE LAYER / AQUITARD (Dense Clay & Shale)

CONFINED AQUIFER (ARTESIAN SYSTEM)

- Water trapped under hydrostatic pressure between two impermeable strata
- Remote recharge zone; protected from direct top contamination

IMPERMEABLE CRYSTALLINE BEDROCK

CGWB GROUNDWATER ASSESSMENT & CRITICAL STATUTORY FRAMEWORK

★ CGWB ASSESSMENT CATEGORIES (STAGE OF GROUNDWATER EXTRACTION):

1. SAFE: Extraction < 70% of annual replenishable recharge.
2. SEMI-CRITICAL: Extraction between 70% and 90%.
3. CRITICAL: Extraction between 90% and 100%.
4. OVER-EXPLOITED: Extraction > 100%! (Punjab, Haryana, W. Rajasthan, NW UP).

★ HIGH-YIELD EXAM DISTINCTIONS & BODIES:

- CGWB (Central Ground Water Board): Scientific survey & assessment body under Ministry of Jal Shakti.
- CGWA (Central Ground Water Authority): REGULATORY & ENFORCEMENT body constituted under Section 3(3) of Environment (Protection) Act, 1986 (NOT under Water Act 1974!).
- India is the LARGEST GROUNDWATER EXTRACTOR in the world (~25% of global groundwater withdrawal), extracting more than China and the USA combined, predominantly for paddy & sugarcane irrigation!

Topic 27 — Renewable & Non-Renewable Energy

Consolidated — 36 Must-Score Facts

1. **Renewable** sources replenish on a human timescale (sun, wind, water, biomass, geothermal). **Non-renewable** sources are finite stocks (coal, oil, gas, uranium). Notes usually treat **nuclear as non-renewable** even though it is low-carbon.
2. In Indian usage, **conventional** energy means coal, oil, gas, **large hydro**, and nuclear. **Non-conventional** means solar, wind, biomass, biogas, tidal, geothermal, ocean, and **small hydro ≤25 MW** — treated as roughly equal to **renewable** in many stems.
3. Non-conventional is treated as roughly equal to renewable in many prelims stems — but large hydro sits on the conventional side of that Indian classification.
4. **Energy crisis** in bank language means the danger of exhaustion of **fossil fuels** (coal, oil, gas) — not merely "short hydro" or malnutrition.
5. UNGA declared **2014–2024** the **Decade of Sustainable Energy for All** (supports SDG 7). Initiative is of the **United Nations**, not World Bank alone.
6. Ultimate source of energy stored in fossil fuels is the **Sun** (ancient photosynthesis).
7. Among coal / oil-gas / nuclear / hydro options, **hydroelectricity** is often the best answer for electricity generation under **sustainable development** (water not depleted by the generation cycle).
8. **Uranium is not a fossil fuel** — nuclear fuel, non-renewable. Coal / petroleum / natural gas are fossils.
9. **Biogas** major constituents are **methane and carbon dioxide**. Butane or carbon monoxide as the major pair is wrong.
10. **Bioethanol** (maize, sugarcane, etc.) blends with petrol. **Biodiesel** (Jatropha, Pongamia, UCO) blends with diesel. Biofuels are eco-friendly and can ease crisis but are often **not cost-effective** vs fossils.
11. Biomass gasification of shells/husk yields **producer gas** (CO, H_2, traces of CH_4 — not CO_2-only); can run IC engines.
12. Solar facts include National Solar Mission **2010**, ISA at **Gurugram**, **500 GW** non-fossil by **2030**, and **PM-KUSUM**.
13. Wind leadership often names Tamil Nadu (नाडु) and Gujarat. **SHP ≤25 MW**. **Tidal** is Moon-driven. **Ocean energy** is a wider set that includes wave, OTEC, and tidal.
14. **OTEC** uses the temperature difference between warm surface water and cold deep water — not the same as tidal barrage power.
15. Energy security: India imports a very large share of crude oil (often taught near **~85%**), while coal is mostly domestic. Strategic Petroleum Reserves sit at **Vizag, Mangalore, and Padur**.
16. **Energy Conservation Act 2001** created **BEE**. **PAT (2012)** trades **ESCerts**. Star labels and UJALA LEDs are efficiency/conservation tools. PAT is **not** CCTS carbon credits.
17. Coal is fossil non-renewable; biomass is renewable organic matter — do not call coal "biomass." Nuclear is **not** a biomass energy (बायोमास ऊर्जा) source.
18. Tidal ≠ wave ≠ OTEC, even though all are ocean-related.
19. Conventional includes large hydro; calling all hydro "non-conventional" is wrong under the Indian split.
20. E20 is the higher ethanol blend ambition beyond E10 — do not freeze policy forever at E10.
21. BEE sits under the EC Act **2001** efficiency architecture.
22. SPR locations are Vizag, Mangalore, Padur — three coastal storage facts.
23. Western UP sugarcane supports bioethanol; bagasse supports cogeneration; UP wind remains limited; rooftop solar is the distributed play.
24. **Hydrogen** is the least-polluting common fuel option (burns to water); called a **future fuel**. National Hydrogen Energy Board set up **3 October 2003**.
25. Fuel cells: pure H_2 → electricity + **heat and water**; can power buildings **and** small devices; output is **DC** (not AC).

26. Microbial fuel cells use living microbes as catalysts; useful in wastewater plants — substrates are mainly **organic**, not "only inorganic."
27. **Bioasphalt** is non-petroleum / renewable-based paving binder — eco-friendlier road surfacing.
28. Green hydrogen can be used in IC engines, blended with natural gas, and in fuel-cell vehicles — all three routes appear in stems.
29. Ocean energy is the umbrella; tidal is one member of that set.
30. ISA is solar diplomacy/IGO; BEE is domestic efficiency regulation — different lanes.
31. PAT certificates measure energy savings trading; carbon markets measure CO_2 units — related climate politics, different instruments.
32. Small hydro cutoff **≤25 MW** is the recurring MNRE number.
33. Nuclear's classification is usually non-renewable/conventional, not "renewable because low carbon."
34. Energy conservation (using less) and energy efficiency (more service per unit energy) are neighbours under BEE/PAT, not synonyms for renewable capacity addition alone.
35. **SATAT** promotes **compressed biogas (CBG)** for transport; oil-marketing companies procure CBG. **GOBAR-Dhan** is the community/dung biogas neighbour scheme — do not invent butane as a major biogas gas.
36. Biogas forms by **anaerobic** digestion; slurry is useful bio-fertiliser after gas capture.

📋 Must-score facts — conventional vs non-conventional

Classifications

Item	Lock
Renewable	Sun, wind, water, biomass, geothermal
Non-renewable	Coal, oil, gas, uranium (nuclear fuel ≠ fossil)
Indian conventional	Coal, oil, gas, **large hydro**, nuclear
Non-conventional	Solar, wind, biomass, biogas, tidal, SHP…
Fossil ultimate source	**Sun** (ancient photosynthesis)
UN Decade Sustainable Energy	**2014–2024** (UNGA; SDG 7)
Biogas majors	CH_4 + CO_2
Bioethanol / Biodiesel	Petrol blend / diesel blend (Jatropha, UCO…)
OTEC vs tidal	Warm–cold water ΔT / Moon-driven sea level
SHP	≤**25 MW**

Topic 28 — Environmental Research & Institutions

Consolidated — 24 Must-Score Facts

1. Headquarters map: **Kolkata** — BSI and ZSI; **Dehradun** (देहरादून) — FSI, FRI, and WII; **New Delhi** — NBPGR; **Chennai** — NBA.
2. **BSI (1890, Kolkata)** surveys **plants** (*Flora* (वनस्पति), plant Red Data Book (रेड डेटा बुक)). **ZSI (1916, Kolkata)** surveys **animals** (*Fauna* (प्राणीजात), animal Red Data Book).
3. **FSI (1981, Dehradun)** maps forest cover (वन आवरण) and publishes **ISFR every two years**. **FRI (1906, Dehradun)** does forestry **research** under the ICFRE world — it does **not** publish ISFR.
4. **WII (1982, Dehradun)** does wildlife research, IFS/officer training, and tiger-census protocols.
5. **NBA** under the Biological Diversity Act sits in **Chennai**, not Delhi. It handles foreign access and ABS at the national level.
6. Biodiversity pyramid (पिरामिड): **NBA (national)** → **SBB (state)** → **BMC (local)**. BMCs prepare the **People's Biodiversity Register (PBR)**.
7. **NBPGR (1978, New Delhi, ICAR)** is the plant gene bank. It is **not** Kolkata BSI and **not** Karnal **NBAGR** (animals).
8. ISFR **2023** class figures often cited: India about **21.76%** forest cover; Uttar Pradesh (उत्तर प्रदेश) about **6.24%**. UP districts with **>20%** forest cover in that reading (रीडिंग) were **Chandauli and Shravasti only** — **Bahraich** was not in that pair.
9. BSI ≠ animals; ZSI ≠ plants; FRI ≠ ISFR; NBA ≠ Delhi; PBR ≠ NBA headquarters function.
10. SBB is the state biodiversity board — not the State Pollution Control Board.
11. NBPGR ≠ Kolkata. Gene bank is Delhi; plant survey is Kolkata.
12. ISFR is biennial, not annual.
13. UPSBB sits in the Lucknow (लखनऊ) state-biodiversity lane; WII supports Terai (तराई) wildlife science links.
14. BMC is local and owns PBR documentation of local biodiversity knowledge.
15. SBB supervises state-level commercial access and BMC work under BDA — domestic track beside NBA's foreign-access focus.
16. FRI research ≠ FSI satellite forest-cover accounting, even though both are Dehradun forestry institutions.
17. WII is wildlife science/training — not the ISFR publisher and not the plant gene bank.
18. Red Data Books from BSI/ZSI are national catalogues; IUCN (आईयूसीएन) Red List remains the global risk ladder (different chapter neighbour).
19. NBA year fact rides on BDA **2002** institutionalisation — headquarters city remains Chennai.
20. Gene-bank NBPGR serves crop genetic resources for all states, including UP, from New Delhi.
21. Forest-cover percentage traps for UP hinge on Chandauli + Shravasti, not a long distractor list of Terai districts.
22. PBR is prepared locally by BMCs; it is not an NBA HQ publication series.
23. Match institution → city before matching function. Wrong city is enough to kill a match list.
24. Dehradun trio: FSI (cover/ISFR), FRI (research), WII (wildlife). Keep the three jobs separate.

📋 Must-score facts — BSI ZSI FSI NBA NBPGR

HQ map

Item	Lock
Kolkata	BSI **1890** plants; ZSI **1916** animals
Dehradun	FSI **1981** ISFR; FRI **1906**; WII **1982**
New Delhi	NBPGR plant gene bank

Item	Lock
Chennai	NBA (≠ Delhi)
Karnal	NBAGR animals
ISFR	Biennial (≠ annual); FSI product
BDA ladder	NBA → SBB → BMC → PBR
SBB ≠ SPCB	Biodiversity board ≠ pollution board

Topic 29 — Indian Environmental Movements

Consolidated — 22 Must-Score Facts

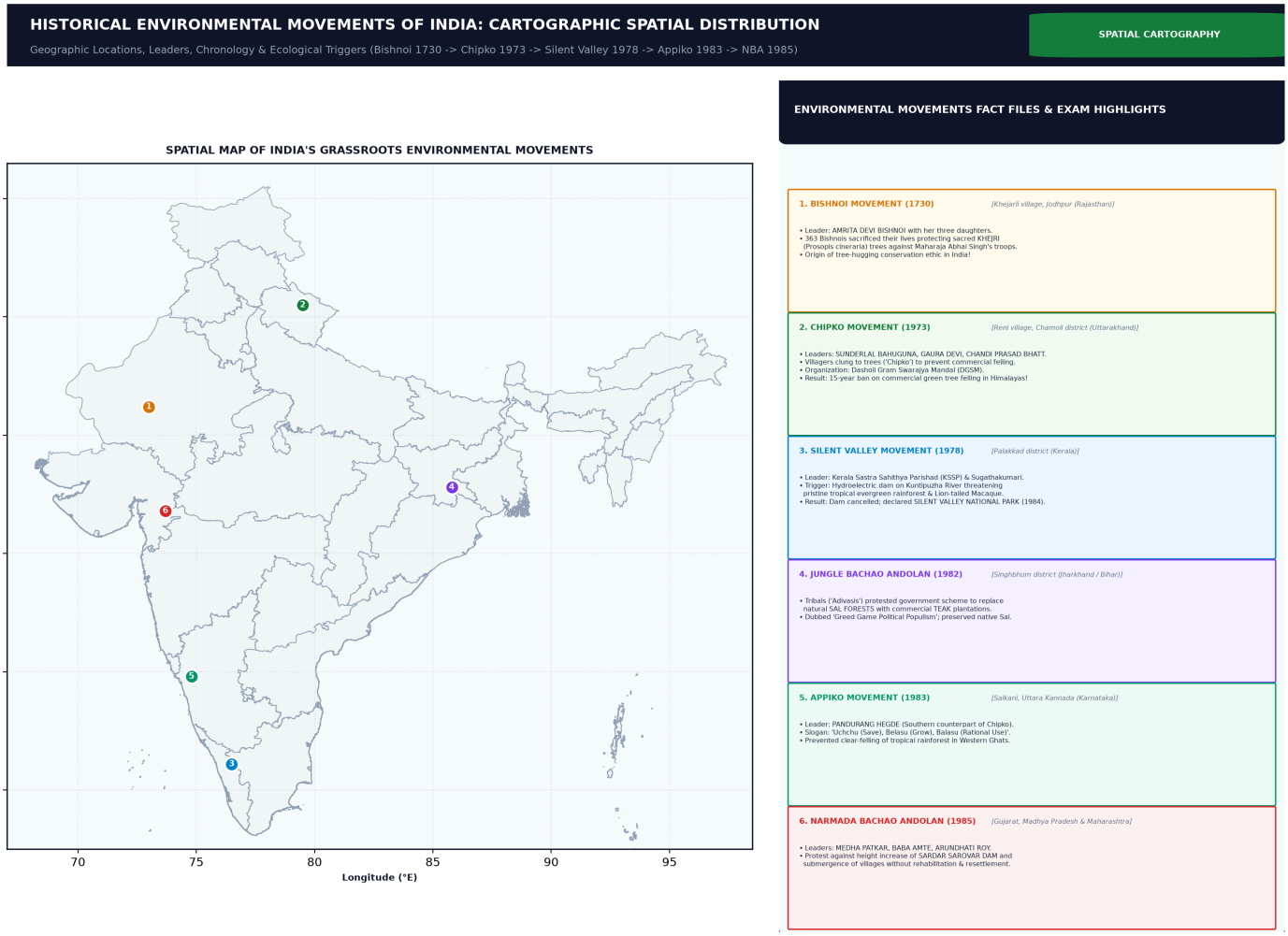
1. **Bishnoi** (बिश्नोई) resistance at **Khejarli, Rajasthan (1730)** is led in memory by **Amrita Devi** (अमृता देवी) and the **363** martyrs who died protecting khejri trees. It is centuries older than Chipko.
2. **Chipko (1973)** began at **Reni, Uttarakhand** (then Uttar Pradesh (उत्तर प्रदेश)). Leaders include **Sunderlal Bahuguna, Chandi Prasad Bhatt (चंडी प्रसाद भट्ट), and Gaura Devi**. Method: embracing trees against logging.
3. **Silent Valley (1970s)** (शांत घाटी) in **Kerala (Palakkad)** opposed a hydro project in rainforest. **KSSP** and scientists were central. It is **not** a Tamil Nadu (नाडु) movement.
4. **Jungle Bachao Andolan (1982)** (जंगल बचाओ) in **Singhbhum, Jharkhand** was tribal (आदिवासी) resistance for forest rights.
5. **Appiko (1983)** in **Karnataka (Sirsi)** was led by **Panduranga Hegde**. It is the southern tree-embrace movement. Appiko is **Karnataka only** — not Uttarakhand, UP, or Kerala.
6. **Narmada Bachao Andolan (from 1985)** spans **Madhya Pradesh, Gujarat, and Maharashtra**, with **Medha Patkar** (मेधा पाटकर) and **Baba Amte** as leader facts. It is anti-large-dam displacement politics.
7. **Save Western Ghats (from about 2011)** is multi-state ecology (पारिस्थितिकी) politics around the **Gadgil** and **Kasturirangan** reports, mining, and ESZ questions.
8. Method families: Chipko, Appiko, and Bishnoi = forest/tree protection. Silent Valley and Narmada Bachao = anti-dam. Save Western Ghats = report/policy ecology protests.
9. Chipko ≠ Appiko. Chipko is Uttarakhand; Appiko is Karnataka.
10. Bishnoi **1730** ≠ Chipko **1973**. Century trap matters.
11. "NBA" as Narmada Bachao Andolan is **not** the National Biodiversity Authority. Same letters, different bodies.
12. Silent Valley is Kerala rainforest–dam (दाम -) conflict, not a Western Ghats mining-only slogan.
13. Jungle Bachao is Jharkhand Singhbhum tribal forest rights — not Chipko's Garhwal (गढ़वाल) story.
14. Narmada Bachao is multi-state Narmada valley politics — not a single-state Kerala story.
15. Save Western Ghats leans on expert committee reports; Chipko leans on village tree hugging.
16. UP link: Chipko's Reni lay in erstwhile UP until Uttarakhand statehood (**2000**). Appiko and Silent Valley are **not** UP movements.
17. Gaura Devi is a Chipko village leadership fact; Panduranga Hegde is Appiko; Medha Patkar is Narmada.
18. Khejarli martyrdom is Rajasthan Bishnoi memory — not Uttarakhand Chipko.
19. Anti-dam movements still differ: Silent Valley saved a Kerala valley rainforest project path; Narmada contested a large multi-state dam-displacement model.
20. Gadgil report is often read as stronger ecological protection; Kasturirangan as a later recalibration — both sit inside Save Western Ghats politics.
21. Tree-hugging method travelled from Chipko north to Appiko south — same method family, different states and years.
22. Match movement → year → state → leader before answering "which state" stems; state alone often decides the key.

📋 Must-score facts — Chipko, Appiko, Narmada, Silent Valley

Movement ↔ place / year

Item	Lock
Bishnoi / Khejarli	**1730** Rajasthan; Amrita Devi; 363 martyrs
Chipko	**1973** Reni, Uttarakhand; Bahuguna / Bhatt / Gaura Devi

Item	Lock
Silent Valley	1970s **Kerala** rainforest–dam; KSSP
Jungle Bachao	**1982** Singhbhum, Jharkhand
Appiko	**1983** Sirsi, **Karnataka**; Panduranga Hegde
Narmada Bachao	From **1985**; MP–GJ–MH; Medha Patkar / Baba Amte
Chipko ≠ Appiko	UK / Karnataka
NBA letters trap	Narmada Bachao ≠ National Biodiversity Authority



Topic 30 — Environmental Literature & Awareness

Consolidated — 28 Must-Score Facts

1. Four (चातुर्याम) pillars on this sheet: **literature** (landmark books/reports), **education** (formal EE), **awareness** (public campaigns), and **communication** (science to citizens).
2. ***Silent Spring* (1962)** by **Rachel Carson** exposed pesticide/DDT harms. It is not Hardin's commons essay.
3. **"Tragedy of the Commons" (1968)** by **Garrett Hardin** is about overuse of shared resources.
4. ***Limits to Growth* (1972)** is the **Club of Rome** systems study. ***Our Common Future* (1987)** is the **Brundtland Report** that popularised sustainable development. Brundtland ≠ Limits to Growth.
5. ***Small Is Beautiful* (1973)** by **E.F. Schumacher** critiques mass-production economics — "economics as if people mattered." It is not Carson's pesticide book.
6. Assertion–reason traps often make Brundtland and Limits to Growth look interchangeable — they are different books/years/authors.
7. Environmental education landmarks include **Tbilisi (1977)**, MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) **NGC/ECO Clubs**, **GSDP (2017)**, and **NEP 2020** ESD language.
8. Environmental awareness hooks include **Article 51A(g)**, **NEAC**, **Van Mahotsav** (July tree planting), and **LiFE** (launched **June 2022** from a COP26 idea).
9. LiFE is **not** a COP-25 product. Idea pitch is Glasgow COP26; launch is June 2022.
10. Environmental communication flagships include CSE's ***Down To Earth***, Gobar Times, and IEC components inside schemes. **KSSP** also sits in science–public communication memory from Kerala.
11. Carson ≠ Hardin. Pesticides book versus commons essay.
12. Brundtland ≠ Limits to Growth. SD definition report versus Club of Rome growth study.
13. Literature shapes thought; education is curriculum and clubs; awareness is campaigns and duties; communication is media/IEC — keep the four jobs distinct.
14. NGC/ECO Clubs are MoEFCC school eco-action programmes — not a Bharat Nirman infrastructure scheme.
15. Agenda 21 is a sustainable-development action plan from Rio; it is not "only environmental education."
16. Article **51A(g)** is the constitutional citizen duty to protect the environment (पर्यावरण) — an awareness/duty fact.
17. Van Mahotsav is the July greening festival fact for schools and public tree planting.
18. NEP 2020 supports education for sustainable development language; it does not replace Tbilisi as the classic EE conference fact.
19. Down To Earth is CSE journalism for environment — a communication pillar example.
20. **Eco-mark (1991)** is the BIS earthen-pot label for certified environment-friendly consumer goods. It is **not** AGMARK and not a FAO mark.
21. **Indira Gandhi Paryavaran Puraskar (1987, MoEF)** honours measurable contribution to environment protection. **Rajiv Gandhi Environment Award** recognises clean technology and industrial pollution reduction.
22. **Tyler Prize** (University of Southern California) is often called the "Nobel of Environment" for environmental science, health, and energy.
23. UNEP's **Global 500 Award** (from **1987**) was later succeeded in spirit by **Champions of the Earth** (from **2005**). Narendra Modi received Champions of the Earth in **2018**.
24. The **Borlaug Award** is for outstanding agricultural research. Norman Borlaug's Nobel was the **Peace** Prize (Green Revolution (हरित क्रांति)), not medicine.
25. UP colour: NEAC district campaigns, Van Mahotsav in schools, NCAP (राष्ट्रीय स्वच्छ वायु कार्यक्रम) city awareness (Kanpur (कानपुर)), and ECO Clubs in UP schools.
26. Silent Spring's year is **1962**; Limits to Growth is **1972**; Brundtland is **1987** — three different decades.
27. Hardin's commons logic is shared-resource overuse; it is not Carson's pesticide narrative.

28. Match pillar → flagship before matching author names: Carson/Hardin/Brundtland/Schumacher for literature; Tbilisi/ECO Club for education; 51A(g)/LiFE for awareness; Down To Earth for communication; awards for recognition traps.

📋 Must-score facts — books, EE, LiFE

Books / campaigns

Item	Lock
Silent Spring	Rachel Carson **1962**; pesticides/DDT
Tragedy of the Commons	Garrett Hardin **1968**
Limits to Growth	Club of Rome **1972**
Our Common Future	Brundtland **1987**
Tbilisi	EE landmark **1977**
LiFE	COP26 idea **2021**; launch Jun **2022**
NGC / ECO Clubs	MoEFCC school programmes
CSE flagship	*Down To Earth* / Gobar Times

Topic 31 — Environmental Days

Consolidated — 28 Must-Score Facts

1. **World Wetlands Day** falls on **2 February** (रामसर) and marks the **Ramsar (रामसर) Convention (1971)** (रामसर कन्वेंशन). It is not 2 March or 3 March.
2. **World Wildlife Day** falls on **3 March** and marks the adoption of **CITES (साइट्स) (1973)** (साइट्स). It is **not** 22 March.
3. **International Day of Forests** falls on **21 March**. It was proclaimed by the UN General Assembly in **2012**.
4. **World Water Day** falls on **22 March**. The classic trap is pairing Wildlife Day with 22 March.
5. **Earth Day** falls on **22 April**. It began in the United States in **1970** as a civic teach-in movement.
6. **International Day for Biological Diversity** falls on **22 May** and commemorates the CBD text adopted (अंगीकृत) at Nairobi in **1992**.
7. **World Environment Day** falls on **5 June**. It marks the opening of the **Stockholm Conference (1972)** (स्टॉकहोम सम्मेलन) and is run by **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम).
8. **World Ozone Day** falls on **16 September** and marks the **Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल) (1987)** (मॉन्ट्रियल प्रोटोकॉल).
9. In calendar order the eight syllabus days run **2 Feb → 3 Mar → 21 Mar → 22 Mar → 22 Apr → 22 May → 5 Jun → 16 Sep**.
10. The March ladder is **Wildlife (3) → Forests (21) → Water (22)**. Never put Wildlife on 22 March.
11. World Environment Day is **not** Earth Day and **not** Wetlands Day. The dates are **5 June**, **22 April**, and **2 February**.
12. India hosted World Environment Day in **2011** (Forests theme) and **2018** (Beat Plastic Pollution).
13. Biological Diversity Day shifted from an older **29 December** link to **22 May** from **2000**; papers use **22 May** only.
14. World Water Day is coordinated under **UN-Water**. Wetlands Day is a **Ramsar** (रामसर) day, not a Water Day synonym.
15. World Wildlife Day was proclaimed by the UN in **2013**, but the date itself is the **1973 CITES** anniversary.
16. Ozone Day belongs to the **Montreal Protocol** (मॉन्ट्रियल प्रोटोकॉल) family. Do not date it as 2 October, 21 April, or 30 January.
17. Earth Day is a civic global day. It is **not** the same UN treaty day as World Environment Day.
18. Biodiversity Day is broader than Wildlife Day. Wildlife Day is fauna (प्राणीजात)–flora (वनस्पति) trade (पण्याध्यक्ष) and crime awareness under the CITES date.
19. Van Mahotsav is a separate **July** plantation campaign in India. It is **not** International Day of Forests.
20. Earth Hour (अर्थ आवर) (last Saturday of March, WWF (डब्ल्यूडब्ल्यूएफ)) is **not** on this syllabus day list and is not Earth Day.
21. Uttar Pradesh (उत्तर प्रदेश) has **10** Ramsar sites — useful Wetlands Day geography, but the day itself stays **2 February**.
22. A correct Ozone–WED pair is **16 September** and **5 June**. A Wildlife–**22 March** pair is always wrong.
23. Stockholm **1972** anchors WED. Ramsar **1971** anchors Wetlands. Montreal **1987** anchors Ozone. CITES **1973** anchors Wildlife.
24. Same-number traps: **22** March is Water, **22** April is Earth, **22** May is Biodiversity — three different months.
25. **World Toilet Day = 19 November** (UN, from 2013). **World No Tobacco Day = 31 May** (not 5 May). **World Habitat (वास स्थान) Day** = first Monday of October (not a fixed 5 October).
26. **Swachh Bharat Mission** launched **2 October 2014** (Gandhi (गांधी) Jayanti). Sierra Leone observes national cleanliness on the last Saturday of every month.
27. Earth Day theme teaching: **2024 = Planet vs. Plastics**. WED **2023 = Beat Plastic Pollution**.
28. **International Tiger Day** falls on **29 July** and marks the **St Petersburg Tiger Summit (2010)**. It is **not** World Wildlife Day (3 March).

📋 Must-score facts — calendar order of key days

Day ↔ date

Item	Lock
Wetlands	**2 February** (Ramsar)
Wildlife	**3 March** (CITES) — ≠ 22 Mar
Forests	**21 March**
Water	**22 March**
Earth	**22 April** (US 1970)
Biological Diversity	**22 May** (CBD)
Environment	**5 June** (Stockholm/UNEP)
Ozone	**16 September** (Montreal)
March ladder	Wildlife 3 → Forests 21 → Water 22
India WED hosts	**2011** Forests; **2018** Beat Plastic

Topic 32 — National Parks & Protected Areas (Advanced)

Consolidated — 34 Must-Score Facts

1. Uttar Pradesh has **one National Park — Dudhwa (दुधवा)** — in **Lakhimpur Kheri** on the Indo-Nepal Terai (तराई).
2. Dudhwa moved sanctuary (**1958**) → National Park (**1977**) → core of **Dudhwa Tiger Reserve (1987)** with Kishanpur and Katarniaghat. Rhino rehabilitation began **1984** (later phases including ~**2018**).
3. **Valmiki** (वाल्मीकि) National Park is in **Bihar**. **Sultanpur** National Park is in **Haryana**. **Corbett** is in **Uttarakhand** (उत्तराखंड). None is a UP national park.
4. Uttar Pradesh is **landlocked**, so it has **zero** marine or coastal national parks.
5. UP has **25 Wildlife Sanctuaries**. **Hastinapur** (हस्तिनापुर) is the largest. **Chandraprabha** is in **Chandauli**. **Katarniaghat** was established in **1975**.
6. **Rudrasagar** Lake is in **Tripura**. **Sultanpur** NP is in **Haryana**. **Samaspur** and **Sarsai Nawar** are in Uttar Pradesh.
7. India's first National Park is **Hailey / Jim Corbett (जिम कॉर्बेट) (1936)** in Uttarakhand. The world's first is **Yellowstone (USA, 1872)** — Old Faithful geyser; Wyoming–Montana–Idaho.
8. India's **largest** National Park is **Hemis** in **Ladakh** (लद्दाख). **Madhya Pradesh** has the most national parks (~**11**). **Andaman (अंडमान) & Nicobar (निकोबार)** has the most wildlife sanctuaries.
9. **Gir (गीर) (Gujarat)** holds the only wild **Asiatic lions**. **Kaziranga (काजीरंगा) (Assam)** is the classic one-horned rhino stronghold.
10. A National Park is the strictest **Wildlife Protection (वन्यजीव सुरक्षा) Act** category (Section **35**). Boundary change needs the **State Legislature**.
11. A Wildlife Sanctuary (Section **18**) is more flexible than a National Park. National Parks are **in-situ** (स्थान पर) conservation.
12. India's first Marine National Park is **Gulf of Kutch (Gujarat)**. Others include **Gulf of Mannar (Tamil Nadu (नाडु))** and the Andaman marine parks.
13. Marine NP facts: **Mahatma Gandhi (महात्मा गांधी) Marine** and **Rani Jhansi (झांसी) Marine (1996)** are in the **Andaman** (अंडमान) group.
14. Natural World Heritage (विश्व धरोहर) overlaps include Kaziranga, Keoladeo (केवलादेव), Manas, Sundarbans (सुंदरबन), Nanda Devi (नंदा देवी)–Valley of Flowers (फूलों की घाटी), Western Ghats (पश्चिमी घाट), Great Himalayan (**2014**), and Khangchendzonga (mixed **2016**).
15. Uttar Pradesh has **no** natural UNESCO World Heritage site of its own.
16. Classic match set: **Dachigam–J&K–Hangul**, **Keoladeo–Rajasthan–birds**, **Kanha (कान्हा)–MP–hard-ground barasingha / Bhoorsingh mascot**, **Periyar (पेरियार)–Kerala–elephant/tiger**.
17. **Pilibhit (2014)** and **Ranipur (2023)** are tiger reserves. They are **not** additional full state national parks of UP.
18. National Chambal (चंबल) Sanctuary spans UP–MP–Rajasthan and is famous for **gharial** and dolphin — **not Haryana**.
19. River–park traps: Corbett = **Ramganga (रामगंगा) + Kosi (कोसी)**; Kaziranga = **Brahmaputra (ब्रह्मपुत्र) / Diphlu**; Silent Valley (साइलेंट वैली) = **Kunthipuzha**.
20. Species–park traps: **Keoladeo Ghana (घन) ≠ lion (सिंह)**; **Eravikulam = Nilgiri (नीलगिरि) Tahr** (not Hoolock); **Hoolock = NE Assam**; **Bhitarkanika = saltwater crocodile**; **Desert NP = Great Indian Bustard** (Jaisalmer–Barmer; human habitation exists).
21. Dudhwa flagship fauna (प्राणीजात) include tiger, reintroduced one-horned rhino, and **barasingha** (swamp deer).
22. Terai Arc pieces with Dudhwa are **Kishanpur** (Lakhimpur Kheri) and **Katarniaghat** (Bahraich).
23. **Rajaji (राजाजी) NP (UK)** = Dehradun (देहरादून)–Haridwar (हरिद्वार)–Pauri (पौड़ी) (Rajaji + Motichur + Chilla merged **1983**); Tiger Reserve **2015**. Named after C. Rajagopalachari.
24. **Pin Valley = Himachal (हिमाचल) (Lahaul–Spiti)**, not J&K. **Van Vihar = Bhopal**. **Salim Ali NP = Srinagar (ex–City Forest)**. **Nagarhole = Rajiv Gandhi NP (Karnataka)**.

25. **Indravati = Bastar (Chhattisgarh)** NP 1981 / TR 1983. **Bandhavgarh (Umaria, MP)** famous for white tiger association. **Betla/Palamu = Jharkhand**.
26. **Keibul Lamjao (Manipur)** on **Loktak** is the only floating NP (**phumdis**; Sangai (सांगाइ) deer). **Valley of Flowers** (फूलों की घाटी) lies wholly in temperate alpine zone.
27. **Govind NP (Uttarkashi (उत्तरकाशी))** includes the source region of the **Tons** (टौंस) (Yamuna (यमुना) tributary). **Great Himalayan NP = Kullu (HP)**.
28. **Wild Ass Sanctuary (Rann of Kachchh, Gujarat)** is India's largest wildlife sanctuary. **Kuno–Palpur (MP)** is the Asiatic lion second-home site.
29. UK west→east sanctuary line: **Kedarnath (केदारनाथ)** → **Nanda Devi** → **Binsar** → **Askot** (Askot/Kedarnath = musk deer country).
30. **Sathyamangalam TR** links Eastern and Western Ghats. **Nagarjunsagar–Srisailam** holds the largest critical tiger habitat (वास स्थान) among common options.
31. Bird/butterfly facts: first butterfly park = **Bannerghatta (Karnataka)**; first State butterfly = **Blue Mormon (Maharashtra)**; Great Indian Hornbill (हॉर्नबिल) = **Western Ghats** (पश्चिमी घाट).
32. World-first versus India-first is a fixed pair: **Yellowstone 1872** versus **Corbett 1936**.
33. Early world peers: **Royal National Park, Australia (1879)**; **Banff NP, Canada (1885)**.
34. Continent icons: **Serengeti (Tanzania)**, **Kruger (South Africa)**, **Kakadu (Australia)**, **Galápagos (Ecuador)**, plus USA icons Yosemite / Everglades / Grand Canyon.

📋 Must-score facts — UP PAs, firsts, marine

UP / India firsts

Item	Lock
UP National Park	**Only Dudhwa** (Lakhimpur Kheri)
Dudhwa path	Sanctuary 1958 → NP **1977** → TR core **1987**
Valmiki / Sultanpur / Corbett	Bihar / Haryana / Uttarakhand (**not** UP NP)
UP marine NP	**Zero** (landlocked)
UP sanctuaries	**25**; Hastinapur largest
India first NP	Hailey/Corbett **1936** (UK)
World first NP	Yellowstone **1872**
Largest India NP	Hemis (Ladakh)
Most NPs / most WLS	MP / A&N
First Marine NP (प्रथम समुद्री उद्यान)	Gulf of Kutch **1980**
Gir / Kaziranga	Only Asiatic lions / one-horned rhino classic

Topic 33 — Biosphere Reserves

Consolidated — 25 Must-Score Facts

1. A Biosphere Reserve is a **UNESCO MAB (1971)** landscape with **core → buffer → transition**. It is **not** a Wildlife Protection (वन्यजीव सुरक्षा) Act category.
2. India has **18** notified national Biosphere Reserves. About **13** sit on the UNESCO World Network.
3. India's **first** Biosphere Reserve is **Nilgiri (नीलगिरि) (1986)**. The **largest** is **Great Rann of Kutch (Gujarat)**. The **smallest** is **Nokrek (Meghalaya (मेघालय))**.
4. **Nokrek** is in **Meghalaya** (मेघालय). Pairing Nokrek with **Manipur** is always wrong.
5. **Manas (Assam)** remains a national Biosphere Reserve but was **removed from the UNESCO World Network in 2011**.
6. **Uttar Pradesh has zero Biosphere Reserves.** Nanda Devi (नंदा देवी) BR is in **Uttarakhand** (उत्तराखंड), not UP.
7. Dudhwa (दुधवा) is a National Park. Valmiki (वाल्मीकि) is a Bihar National Park. Neither is a UP Biosphere Reserve.
8. Match fact: **Simlipal–Odisha**, **Nokrek–Meghalaya**, **Agasthyamalai–Kerala**, **Manas–Assam**.
9. Match fact: **Simlipal–Odisha**, **Pachmarhi–Madhya Pradesh**, **Nanda Devi–Uttarakhand**, **Nokrek–Meghalaya**.
10. **Panna (Madhya Pradesh, 2011)** is India's newest notified Biosphere Reserve.
11. Madhya Pradesh has **three** Biosphere Reserves: Pachmarhi, Achanakmar–Amarkantak (with Chhattisgarh), and Panna.
12. Multi-state classics: **Nilgiri** (नीलगिरि) (Tamil Nadu (नाडु)–Karnataka–Kerala), **Agasthyamalai** (Kerala–Tamil Nadu), **Achanakmar–Amarkantak** (MP–Chhattisgarh).
13. Core allows **no exploitation**. Buffer allows limited research and education use. Transition supports livelihoods.
14. A National Park is the strictest Wildlife Protection Act site. A Biosphere Reserve adds sustainable use and research under MAB.
15. World Network membership is a UNESCO subset of national Biosphere Reserves. National listing alone is not enough for WNBR.
16. Bihar and Rajasthan also have **zero** Biosphere Reserves. Valmiki remains an NP, not a BR.
17. Gulf of Mannar is a **marine** Biosphere Reserve in Tamil Nadu. Sundarbans (सुंदरबन) BR is in West Bengal.
18. Cold Desert BR is in **Himachal (हिमाचल) Pradesh**. Khangchendzonga BR is in **Sikkim**.
19. Seshachalam Hills BR is in **Andhra Pradesh**. Great Nicobar (निकोबार) BR is in the Andaman (अंडमान) and Nicobar Islands.
20. Periodic UNESCO review can delist a site from the World Network — India's example is **Manas 2011**.
21. Yellowstone, Serengeti–Ngorongoro, Galápagos, Lake Baikal, Doñana, and Mount Kenya are classic world Biosphere icons.
22. Do not place Nanda Devi in Uttar Pradesh on a match list. After bifurcation it is an **Uttarakhand** site.
23. Agasthyamalai paired only with Kerala is accepted as correct in recent papers even though the landscape also spans Tamil Nadu.
24. Biosphere Reserves can overlap National Parks and World Heritage (विश्व धरोहर) tags, but the MAB label is still a separate tool.
25. Among common options, **biosphere reserves** are the best strategy for conserving biodiversity **together with traditional human life / cultural diversity**. **Gulf of Mannar** is often cited as India's richest marine biodiversity BR mosaic.

📋 Must-score facts — MAB zones, India list tags

Frame

Item	Lock
BR model	UNESCO MAB **1971**; core → buffer → transition
India count	**18** national; ~**13** UNESCO WNBR
First / largest / smallest	Nilgiri **1986** / Great Rann of Kutch / Nokrek
Nokrek state	**Meghalaya** (≠ Manipur)
Manas UNESCO	Removed from WNBR **2011** (still national BR)
UP BRs	**Zero**; Nanda Devi = Uttarakhand
Newest notified	Panna **2011** (MP)
MP trio	Pachmarhi · Achanakmar–Amarkantak · Panna
Multi-state	Nilgiri TN–KA–KL; Agasthyamalai KL–TN

Topic 34 — Climate Change (Advanced)

Consolidated — 32 Must-Score Facts

1. **Climate change** is a long-term shift in temperature, rainfall, and extremes. **Global warming** is the temperature-rise part only.
2. IPCC AR6 places warming near **1.1°C** above **1850–1900**, with human influence dominant. Past-century rise ≈ **1°C / 1.8°F**.
3. The **Keeling Curve** records atmospheric **CO_2** at **Mauna Loa** since **1958**.
4. The **Emissions Gap Report** (उत्सर्जन अंतराल रिपोर्ट) is an annual **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) product. It is **not** published by the IPCC.
5. Main greenhouse gases are **CO_2, CH_4, N_2O, and F-gases**. **Argon is not a greenhouse gas**. **Joseph Fourier (1820s)** framed the greenhouse-effect idea.
6. WMO-style relative (सापेक्ष) shares often tested: **CO_2 ~64% > CH_4 ~19% > CFCs+others ~11% > N_2O ~6%**. Water vapour dominates the **natural** effect.
7. India ranks near the top in **absolute** (निरपेक्ष) emissions (**China #1**, India ~**3rd**) but stays **low per capita**. **Bhutan** is often carbon-negative.
8. A **carbon sink** absorbs net CO_2. A **carbon budget** is the cumulative CO_2 allowed for a temperature target. **Blue carbon** (नीला कार्बन) = ocean/coastal sinks.
9. A **carbon credit** equals **1 tonne of CO_2 equivalent**. Carbon trading is the market for credits or allowances (Kyoto (क्योटो) origin).
10. **Green Credit (2023)** (ग्रीन क्रेडिट) and **PAT ESCert** are **not** the same thing as a Kyoto-style carbon credit.
11. **Mitigation** cuts emissions and grows sinks. **Adaptation** adjusts society and systems to climate impacts.
12. **Kyoto** (क्योटो) is climate mitigation (जलवायु शमन). **Montreal** is ozone protection. Do not swap the two treaties.
13. India's **NAPCC** launched in **2008** with **eight** missions (**not** nuclear power).
14. India's updated NDC path includes about **45%** emission-intensity cut and about **50%** non-fossil electricity capacity share.
15. India's **net-zero** year fact is **2070**. Net zero = balance emissions with removals, not absolute zero every emission.
16. **LiFE** was floated as an idea at **COP-26 (2021)** and launched in **June 2022**. COP-25 Madrid is the false date trap.
17. India's **Carbon Credit Trading Scheme (2023)** sits with **BEE** under the Power Ministry.
18. The **IPCC** does science assessment under **WMO + UNEP** parents. It does **not** set national targets or write the Emissions Gap Report.
19. The **greenhouse effect** is infrared trapping by greenhouse gases. Without it Earth would be about **−18°C**.
20. The **carbon cycle** moves carbon among atmosphere, oceans, land, and fossil pools. Sequestration is the storing process.
21. **Milankovitch** = eccentricity, obliquity, precession — **not** solar irradiance. **Ice cores** = cryogenic climate archive.
22. **GCF** = Cancun **2010**. **CDM/CERs** = Kyoto. **GHG Protocol** = WRI + WBCSD. **Earth Hour** (अर्थ आवर) = WWF (डब्ल्यूडब्ल्यूएफ).
23. Carbon trading grew from Kyoto-era market tools; Paris-era systems use national NDCs and domestic markets.
24. Do not treat IPCC Assessment Reports as UNFCCC (यूएनएफसीसीसी) treaty text. Science reports and the climate convention are different homes.
25. Adaptation examples include early warning, climate-resilient crops, coastal defence, and heat-action plans — not only emission cuts.
26. **Methane hydrates**, paddy+fertiliser (CH_4+N_2O), and wetland CH_4 are high-yield source traps.
27. **Carbon fertilization** ≠ ocean acidification (महासागरीय अम्लीकरण).
28. **Forest Carbon Partnership Facility** and **BioCarbon Fund ISFL** are **World Bank** programmes — FCPF does **not** fund universities/individuals for research grants.

29. **Climate Action Tracker** = Climate Analytics + New Climate Institute (not IPCC/UNEP wing).
30. **Deep Carbon Observatory** maps deep-Earth carbon (physics, reservoirs, deep energy, deep life).
31. **Common Carbon Metric** (UNEP-linked) measures building-operation carbon footprints.
32. Primary recent warming drivers are fossil fuels, autos, deforestation — **not** solar flares.

📋 Must-score facts — AR6, sinks, NAPCC, NDC

Advanced tags

Item	Lock
AR6 warming	~**1.1°C** above 1850–1900
Keeling Curve	Mauna Loa CO_2 since **1958**
Emissions Gap Report	**UNEP** (≠ IPCC)
Mitigation / Adaptation	Cut emissions+sinks / adjust to impacts
Montreal ≠ Kyoto	Ozone / climate
NAPCC	**2008**; **eight** missions (no nuclear)
NDC class facts	~45% intensity cut; ~50% non-fossil capacity
1 carbon credit	1 tCO_2e
Green Credit / PAT	**2023** / ESCerts — ≠ Kyoto credit

Topic 35 — Atmosphere

Consolidated — 24 Must-Score Facts

1. From bottom to top the layers run **Troposphere → Stratosphere → Mesosphere → Thermosphere → Exosphere**.
2. The **troposphere** (क्षोभमंडल) holds weather. Temperature **falls** with height in this layer.
3. The **stratosphere** (समतापमंडल) holds the **ozone layer**. Temperature **rises** with height here.
4. The **mesosphere** is the **coldest** layer and is where most meteors burn.
5. The **thermosphere** is very hot but thin. Auroras, the ISS path, and much of the ionosphere sit here.
6. The **exosphere** is the outermost fringe and merges into space.
7. Dry air composition is about **N_2 78%**, **O_2 21%**, **argon ~0.93%**, and **CO_2 ~0.04%**.
8. **Nitrogen gas (N_2) is not a greenhouse gas.Nitrous oxide (N_2O)** is a greenhouse gas. Do not confuse the two.
9. **Argon is not a greenhouse gas.** Oxygen and helium are also not greenhouse gases.
10. Earth's air is heated mainly by **long-wave terrestrial radiation (भौमिक विकिरण)** from the warmed surface, not by direct shortwave alone.
11. The Sun sends **shortwave** insolation (सूर्यताप). The surface emits **longwave** radiation that warms the air.
12. Maximum ozone concentration is in the **stratosphere**, not the troposphere.
13. The **homosphere** (सममंडल) is well mixed to about **80 km**. Above that the **heterosphere** (विषममंडल) separates gases by mass.
14. Tropospheric ozone (क्षोभमंडलीय ओजोन) is a pollutant and greenhouse gas. Stratospheric ozone (समतापमंडलीय ओजोन) is the UV shield.
15. Temperature inversion and weather systems live in the troposphere. Commercial jets often cruise near the tropopause.
16. Do not place the ozone layer in the troposphere on a match list.
17. Mesosphere cold and thermosphere hot can both be true because density differs — thin hot air is not the same as surface heat.
18. Exosphere atoms can escape to space. It is not a weather layer.
19. Composition percentages are for **dry air**. Water vapour varies strongly by place and season.
20. Atmospheric studies for climate start from this layer ladder plus greenhouse-gas composition facts.
21. Ionosphere radio reflection is mainly a thermosphere–upper-atmosphere feature, not a troposphere fact.
22. Stratospheric heating with height comes largely from ozone absorbing ultraviolet radiation.
23. Never reverse the order of troposphere and stratosphere in a bottom-to-top ladder.
24. Long-wave terrestrial radiation is the standard answer for how the lower atmosphere is heated after surface warming.

📋 Must-score facts — layers, composition, radiation

Structure

Item	Lock
Order (bottom→top)	Tropo → Strato → Meso → Thermo → Exo
Weather / ozone layer	Troposphere / Stratosphere
Coldest / meteors	Mesosphere
Aurora / ISS	Thermosphere
Dry air	N_2 ~78% · O_2 ~21% · Ar ~0.93% · CO_2 ~0.04%

Item	Lock
N_2 vs N_2O	Not GHG / is GHG
Not GHGs	Ar, N_2, O_2, He
Air heating	Mainly longwave from warmed surface
Homosphere	Well mixed to ~**80 km**

Topic 36 — Ozone Layer

Consolidated — 34 Must-Score Facts

1. **Ozone (O_3)** is triatomic oxygen. **Good ozone** (अच्छा ओजोन) sits in the **stratosphere** (समतापमंडल) (about **15–30/35 km**). Ground-level ozone is a harmful pollutant.
2. About **90%** of atmospheric ozone is in the **stratosphere**; about **10%** is in the **troposphere** (क्षोभमंडल).
3. Stratospheric ozone (समतापमंडलीय ओजोन) forms when **UV** acts on **O_2** (Chapman cycle). Ozone absorbs **ultraviolet**, not IR as its main protective role.
4. Ozone amount is measured in **Dobson Units**. **1 DU ≈ 0.01 mm** of pure ozone at **0°C and 1 atm**. Average column ≈ **300 DU**. An ozone hole is often taken as below **220 DU**.
5. Stratospheric ozone is naturally regulated in part by **nitrogen dioxide (NO_2)**. Column thickness is often highest in **spring** and lowest in the **rainy season** (general seasonal pattern — distinct from the Antarctic spring hole).
6. Ozone depletion is driven by **ODS** such as CFCs through chlorine and bromine catalytic cycles.
7. One chlorine atom can destroy on the order of **100,000** ozone molecules before it is removed.
8. The classic **ozone hole** forms over **Antarctica** in spring (**September–October**) with polar stratospheric clouds, polar front/vortex, and CFCs.
9. Joe **Farman** (फ़रमान), Brian **Gardiner**, and Jonathan **Shanklin** (British Antarctic Survey) reported the Antarctic ozone hole in **Nature (May 1985)**; TOMS mapping was used in that era of monitoring.
10. ODS include **CFCs, Halon-1211/1301, carbon tetrachloride, methyl chloroform, methyl bromide, HCFCs**, and related chlorine/bromine carriers. **HFCs** are **not** classic ODS.
11. Refrigerant trade (पण्याध्यक्ष) name **Mafron** (halocarbon coolants) is common in domestic refrigerators; **ammonia** is used in many large plants. CFCs are **not** primarily "automobile exhaust" gases.
12. CFCs are used in refrigerators/AC, foam blowing, aerosol propellants, and cleaning some electronic components — **not** tubeless-tyre production as a classic CFC use.
13. **HFCs** are high-GWP climate gases under **Kigali**. They are **not** classic ozone-depleting substances.
14. The **Vienna Convention (1985)** (वियना कन्वेंशन) is the framework treaty. The **Montreal Protocol (16 September 1987; force 1 January 1989)** is the binding ODS phase-out.
15. World Ozone Day (विश्व ओजोन दिवस) is **16 September** (Montreal signing; UN observance from **1994**).
16. The **Kigali Amendment (2016; force 1 January 2019)** phases down **HFCs** for climate reasons under the Montreal family.
17. **Montreal = ozone. Kyoto (क्योटो) = climate.** Carbon trading ≠ Montreal.
18. ODP measures ozone-damage potential. GWP measures heat-trapping potential.
19. Stratospheric ozone absorbs harmful UV (**~93–99%** of the most damaging wavelengths). Tropospheric ozone (क्षोभमंडलीय ओजोन) is a pollutant and greenhouse gas.
20. **UV-A (315–400 nm)** largely reaches the surface; **UV-B (280–315 nm)** is mostly blocked by ozone; **UV-C (100–280 nm)** is blocked and does not reach the surface under normal conditions.
21. Extra UV-B raises **skin cancer**, cataracts, crop damage, and plankton stress — not "liver/brain/oral cancer" as the classic stem key.
22. India's Ozone Cell under MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) tracks HCFC phase-out and cooling-sector compliance.
23. Do not place the protective ozone layer in the troposphere.
24. CFCs are both ODS and strong greenhouse gases — "CFC not a GHG" is false.
25. Polar spring timing matters: the deepest Antarctic loss is not midsummer.
26. Vienna does not itself list every control measure; Montreal carries the binding phase-out schedules.
27. **G.W. Kent Moore** (University of Toronto) led the **2005** report of an ozone "halo" / low-ozone feature over the **Tibetan Plateau**.

28. Homosphere (सममंडल) (uniform composition) ≈ troposphere + stratosphere + mesosphere (मध्यमंडल); **ionosphere** is **not** part of the homosphere set in UKPCS-style stems.
29. Troposphere is deepest in the tropics (~**18–20 km**) and shallower near the poles.
30. Dobson Unit is a column measure, not a surface concentration in ppm.
31. A stem pairing Montreal with climate mitigation (जलवायु शमन) is usually the Kyoto/Paris trap.
32. Ozone as a gas near the surface is poisonous to lungs/respiratory systems; as a stratospheric layer it is essential for biotic life via UV shielding.
33. Formation of ozone is slower over polar regions, so depletion effects are strongest there.
34. Acid rain (अम्ल वर्षा) is driven by **SO_2 / NOx**, not "nitric acid alone" as a matched pair with ozone–CFC style keys.

📋 Must-score facts — DU, ODS, Vienna–Montreal–Kigali

Science / treaties

Item	Lock
Good / bad	Stratospheric UV shield / tropospheric pollutant
Share	~90% / ~10%
1 DU	≈0.01 mm pure O_3; hole often <**220 DU**
Antarctic season	Sep–Oct spring
1985 discovery	Farman–Gardiner–Shanklin (BAS)
ODS set	CFCs, Halons, CCl_4, CH_3CCl_3, CH_3Br, HCFCs…
HFCs	High GWP; **not** classic ODS → **Kigali**
Vienna / Montreal / Kigali	**1985** / **1987** (force 1989) / **2016**
Ozone Day	**16 Sep**
Montreal ≠ Kyoto	Ozone / climate

Topic 37 — Greenhouse Gases

Consolidated — 30 Must-Score Facts

1. A greenhouse gas absorbs outgoing infrared radiation and traps heat in the atmosphere.
2. Without the natural greenhouse effect (प्राकृतिक ग्रीनहाउस प्रभाव) Earth would average about **−18°C**. With it the surface averages about **+15°C**. **Joseph Fourier (1820s)** framed the idea.
3. **CO_2** has **GWP = 1**. It dominates long-term forcing from fossil fuels, cement, and deforestation.
4. Pre-industrial CO_2 was about **280 ppm (~0.03%)**. Present levels are above **415–420 ppm**.
5. Rough WMO-style relative (सापेक्ष) shares: **CO_2 ~64%, CH_4 ~19%, N_2O ~6%, CFCs + others ~11%**. Order: **CO_2 > CH_4 > CFCs > N_2O**.
6. **Methane (CH_4)** has GWP about **28–36**, a shorter lifetime near **12 years**, and strong links to paddies, cattle, landfills, termites, and wetlands (**marsh gas**).
7. **Nitrous oxide (N_2O)** is "laughing gas," has very high GWP (about **265**), and is strongly linked to **fertilizer** use. **Rice** emits both CH_4 and N_2O.
8. **Water vapour** is the most abundant natural greenhouse gas (~half of natural effect), but it mainly acts as a **feedback**, not the primary anthropogenic driver.
9. **Tropospheric ozone** (क्षोभमंडलीय ओजोन) is a greenhouse gas and pollutant. **Stratospheric ozone** (समतापमंडलीय ओजोन) is the UV shield.
10. **CFCs** are both greenhouse gases and ozone-depleting substances and fall under Montreal controls.
11. **HFCs** are greenhouse gases only (not classic ODS) and are phased down under **Kigali 2016**.
12. Gases that are **not** greenhouse gases include **argon, N_2, O_2, helium, hydrogen, and propane**.
13. The Kyoto (क्योटो) basket covers **CO_2, CH_4, N_2O, HFCs, PFCs, and SF_6** (later also **NF_3**).
14. **GWP** compares heat-trapping power to CO_2 over a stated horizon, usually **100 years**. Methane's short-horizon GWP is much higher than CO_2 — CH_4 is more potent per molecule.
15. **CO_2e** converts all greenhouse gases into carbon-dioxide equivalent using GWP.
16. Biogas (बायोगैस) is mainly **methane plus CO_2**. Do not treat biogas as CO_2-only.
17. Paddy fields are a classic **methane** source fact in Indian papers.
18. **Direct** GHGs: CO_2, CH_4, N_2O, CFCs, SF_6, NF_3. **Indirect:** NO_x, CO, NMVOCs, SO_2.
19. CFC versus HFC is dual-role ODS+GHG versus climate-only substitute under Kigali.
20. N_2O is not the same as N_2. Molecular nitrogen is inert as a greenhouse gas.
21. Water vapour rises as Earth warms; that feedback amplifies warming but does not replace fossil CO_2 as the driver.
22. F-gases (HFCs, PFCs, SF_6) are small by mass but can be huge by GWP.
23. Cement process emissions are a major industrial CO_2 source beyond fuel combustion.
24. Do not list argon as a greenhouse gas in a multi-statement stem.
25. Methane's shorter lifetime still matters because near-term warming impact is strong.
26. Ozone's greenhouse role is mainly **tropospheric**; do not call stratospheric ozone the main GHG form.
27. **Carbon fertilization** = higher plant growth from elevated CO_2.
28. **Methane hydrates** in Arctic/seafloor can release CH_4 if warmed; CH_4 oxidises to CO_2 in about a decade or two.
29. **China** is the largest CO_2 emitter; India ~3rd absolute (निरपेक्ष); **Bhutan** often carbon-negative.
30. CO_2 is both **useful** (photosynthesis) and **harmful** (enhanced greenhouse effect (मानवजनित संवर्धित प्रभाव)) to life on Earth.

📋 Must-score facts — GWP, shares, Kyoto basket

Gases

Item	Lock
Natural effect temps	~−18°C without; ~+15°C with
CO_2 GWP / levels	**1**; pre-ind ~280 ppm; now >415–420
Shares order	CO_2~64% > CH_4~19% > CFCs+~11% > N_2O~6%
CH_4 GWP / lifetime	~28–36; ~12 years; paddies–cattle–landfills
N_2O	GWP ~**265**; fertilizers; "laughing gas"
Water vapour	Most abundant natural GHG; mainly **feedback**
CFCs / HFCs	GHG+ODS (Montreal) / GHG only (Kigali)
Not GHGs	Ar, N_2, O_2, He, H_2, propane
Kyoto basket	CO_2 CH_4 N_2O HFCs PFCs SF_6 (+NF_3)

Topic 38 — Pollution (Advanced)

Consolidated — 38 Must-Score Facts

1. **Primary** air pollutants are emitted directly (SO_2, PM, CO, NOx, lead). **Secondary** pollutants form in air (O_3, PAN, photochemical smog, acid mist).
2. **Lichens** are pollution indicators and do **not** grow well in badly polluted air.
3. India's AQI uses **eight** pollutants (**PM10, PM2.5, NO_2, CO, O_3, SO_2, NH_3, Pb**) — **not CO_2/CH_4**. Bands: **Good (0–50)** to **Severe (401–500)**. NAQI launched **2014**.
4. **NCAP (2019)** targets non-attainment cities with about a **20–30%** particulate cut goal.
5. **London smog** is SO_2 plus smoke. **Photochemical smog** needs NOx, VOCs, and sunlight and yields **ozone + PAN**; **PAN** is a strong eye irritant.
6. **Fly ash** is a thermal-power primary pollutant usable in bricks/cement; oxides SiO_2/Al_2O_3/CaO **plus toxic metals**. **BS-VI** began **1 Apr 2020**.
7. **SAFAR** (सफर) is an IITM air-quality and weather forecast system. **WAYU** units were installed in **Delhi**.
8. **CO** forms carboxyhaemoglobin (**~200–300×** affinity vs O_2). **Radon** is the top indoor air-pollutant teaching fact.
9. **BOD** measures organic pollution through microbial oxygen demand. High BOD means worse water quality.
10. **COD** is usually greater than or equal to BOD. Clean water often shows DO above about **6 mg/L** and BOD below about **3 mg/L**.
11. **Eutrophication** (सुपोषण) is nutrient (N/P) overload → algal bloom → dissolved-oxygen crash — water, not air.
12. The **Central Ground Water Board (CGWB)** (केंद्रीय भूजल बोर्ड) issues the groundwater quality report. **CPCB** (केंद्रीय प्रदूषण नियंत्रण बोर्ड) is not CGWB.
13. Classic arsenic hotspot teaching points to **Bangladesh** / Gangetic plains. **Knock-knee** = fluoride; **Black Foot** = arsenic; **Minamata** = Hg; **Itai-itai** = Cd.
14. **Damodar** (दामोदर) = biological desert. Ganga (गंगा) BOD peak classically **Kanpur (कानपुर)–Allahabad (इलाहाबाद)**. Leather industry = high chemical pollution.
15. Many pesticides are **not** readily biodegradable — a frequent true statement in papers.
16. Heavy-metal facts include **chromium** around Kanpur plus lead, cadmium, and mercury elsewhere. Mercury sources include **gold mining** and **coal TPPs**; **no known safe exposure level**.
17. Daytime noise standards (dB): Silence **50**, Residential **55**, Commercial **65**, Industrial **75**. **Green muffler** = tree belts for noise.
18. **Moradabad** was flagged among the world's noisiest cities in UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) Frontier reporting.
19. About **80%** of marine pollution comes from land. **MARPOL** governs ship-source pollution. **IMO HQ = London**. Ocean acidification (महासागरीय अम्लीकरण) harms calcareous life and corals.
20. Plastic Waste Management Rules date from **2016**. The SUP ban push is **2022**. EPR is the producer-responsibility tool. Microbeads harm marine ecosystems.
21. India ranks among the world's largest plastic-waste generators (often taught as about **third**).
22. Indoor air pollution is driven by solid cooking fuels, tobacco smoke, poor ventilation, and **radon**.
23. Surface-water pollution includes sewage, industrial effluent, and agricultural runoff into rivers and lakes.
24. Drinking-water quality in India is commonly referenced to **IS 10500** standards.
25. Vehicular emissions: **catalytic converters** (Pt/Pd/Rh); Euro-II diesel S **≤0.05%**; India leapfrogged to **BS-VI (2020)**.
26. Coastal pollution includes oil, plastics, sewage, and industrial discharge along the shoreline belt.
27. Do not call photochemical smog a simple SO_2–smoke fog. Ozone chemistry is the key.
28. SAMEER-type apps publish AQI for public warning; they do not replace NCAP city action plans.
29. **CCAC** focuses on **methane, black carbon, HFCs** — not a G20-only initiative.
30. **Asian Brown Cloud** is especially associated with **South Asia**. Bag filter ≠ cyclone divider.
31. **Bioremediation** cleans via microbes; **Cd/Pb** not readily treated; **Oilzapper** (TERI) for oily waste.

32. **Bhopal MIC (Dec 1984)**; **Chernobyl (1986)** radioactive disaster teaching pair.
33. **SWM Rules 2016** give elaborate landfill/site criteria. **R2 Code** = responsible electronics recycling.
34. **H-CNG** reduces CO/HC/CO_2 vs CNG but does **not eliminate** CO; up to ~**1/5** H_2 by volume; not cheaper than CNG.
35. Geoengineering (भू-इंजीनियरिंग) teaching: **cirrus thinning** and **sulphate aerosol** injection aim at cooling / offsetting warming — not cyclone control.
36. Steel industry emits **CO, CO_2, SOx, and NOx**. **Aflatoxins** = mould toxins on poorly stored grains.
37. Excess **nitrogen and phosphorus** fertilisers drive **eutrophication**, raise nitrate in groundwater, and link to **N_2O** climate forcing — Indo-Gangetic wheat–rice systems are high-input geography.
38. Fertiliser pollution is mostly **nutrient overload**, not heavy-metal tanning chemistry.

📋 Must-score facts — AQI, smog, BOD, disease traps

Air / water

Item	Lock
Primary / Secondary	Direct emit / form in air
AQI 8	PM10 PM2.5 NO_2 CO O_3 SO_2 NH_3 Pb
NCAP / BS-VI	**2019** / **1 Apr 2020**
London / Photochemical	SO_2+smoke / NOx+VOC+sun → O_3+PAN
SAFAR	IITM AQ + weather forecast
BOD / COD / clean water	Microbial O_2 demand / COD≥BOD; DO>~6, BOD<~3 mg/L
Eutrophication	N/P overload → bloom → DO crash (**water**)
CGWB ≠ CPCB	GW quality report / ambient air–surface water
Disease locks	Fluoride knock-knee · As Black Foot · Hg Minamata · Cd Itai-itai
Damodar / Ganga BOD	Biological desert / Kanpur–Allahabad peak class

Topic 39 — Acid Rain

Consolidated — 28 Must-Score Facts

1. **Acid rain** means acidic deposition with **pH below 5.6**.
2. Normal rain (सामान्य वर्षा) is already slightly acidic near **pH 5.6** because CO_2 forms weak carbonic acid. It is **not** pH 7.
3. **Wet deposition** arrives with rain, snow, or fog. **Dry deposition** settles as acidic gases or particles without rain.
4. The main strong acid pathway is **SO_2 → H_2SO_4**. **NOx → HNO_3** is the other major path.
5. **CO_2 → H_2CO_3** sets the weak baseline. Carbon dioxide alone is **not** the main culprit of severe acid rain.
6. On the pH scale, acidic is below **7**, basic is above **7**, and neutral is **7**.
7. Reversed statements that call normal rain alkaline or acid rain alkaline are both false.
8. Lake waters below about **pH 5** can kill fish. Acid soils release toxic **Al^{3+}**.
9. Acid deposition harms conifers and weathers marble and limestone. The Taj marble → **gypsum** story is the classic Indian fact.
10. **Lichens** die in high SO_2 and act as pollution bio-indicators — they indicate pollution but do **not** thrive in polluted air.
11. Prevention tools include **FGD**, catalytic converters, cleaner fuel norms such as **BS-VI**, and regional protocols.
12. The **Gothenburg Protocol (1999)** under **CLRTAP (1979)** targets acidifying air pollutants. **Montreal** targets ozone ODS — different problem.
13. The **Taj Trapezium Zone** covers about **10,400 km^2** around the Taj Mahal (ताज महल), with Supreme Court-driven controls from **1996**.
14. **Mathura (मथुरा) refinery** is a classic SO_2 source in the Taj Trapezium narrative.
15. The **Singrauli** thermal belt and industrial NCR/Kanpur (कानपुर) airsheds are Indian acidifying-emission hotspots.
16. SO_2 is denser than air and can travel long distances before deposition.
17. Acid rain is a **secondary-pollutant** story built from primary SO_2 and NOx emissions.
18. Coal power and some industries are major SO_2 sources. Vehicles and combustion add NOx.
19. Liming lakes is an emergency response, not a substitute for cutting emissions at source.
20. Do not blame acid rain mainly on CO_2 just because normal rain is weakly acidic.
21. Dry deposition can damage materials even on rainless days.
22. Acid mist and acid fog are wet forms even when they are not classic rainfall.
23. Cultural-heritage damage and forest dieback are both valid effect facts alongside aquatic harm.
24. India's focus pairs chemistry (SO_2/NOx) with Taj protection geography.
25. Calling Montreal the acid-rain treaty is a standard distractor.
26. FGD is flue-gas desulphurisation at power-plant chimneys — mitigation hardware, not a treaty name.
27. Secondary pollutants H_2SO_4 and HNO_3 form in the atmosphere from primary SO_2 and NOx.
28. Acid rain is **not** "rain only"; dry deposition counts too.

📋 Must-score facts — pH, Taj Trapezium, protocols

Locks

Item	Lock
Acid rain	pH <**5.6**
Normal rain	~5.6 from CO_2 (≠ alkaline)
Main acids	H_2SO_4 (SO_2) + HNO_3 (NOx)

Item	Lock
Taj marble	→ gypsum ($CaSO_4$) under sulphuric attack
Taj Trapezium	Pollution-control belt for Taj
Gothenburg/CLRTAP	Acidifying air (≠ Montreal ozone)
Prevention	FGD, converters, BS-VI, cleaner fuel

Topic 40 — Desertification

Consolidated — 22 Must-Score Facts

1. **Desertification** is land degradation in **arid, semi-arid, and dry sub-humid (ASDM)** drylands. It is not merely "the desert expanding."
2. Drivers combine climate variation with human pressures such as overgrazing, deforestation, and bad irrigation.
3. **Land degradation** is the broader productivity decline and can occur in any climate zone.
4. India's degraded land share is often taught near **29–30%** (ISRO Atlas **2016**, about **96.4 million ha**). **Rajasthan** (थार) is the worst state hotspot.
5. Drought types run **meteorological** → **hydrological** → **agricultural**. Meteorological drought (मौसमी (मौसम विज्ञान) सूखा) is often linked to rainfall below about **75%** of long-period average.
6. **DPAP** began in **1973**. Drought and desertification reinforce each other in a feedback loop.
7. Bundelkhand's **seven** Uttar Pradesh (उत्तर प्रदेश) districts are a classic UP drought–degradation hotspot.
8. Water erosion stages run **sheet** → **rill** → **gully** → **ravine**. The **Chambal** (चंबल) ravines are the textbook deep badlands.
9. Wind erosion works through **deflation** and **abrasion**, especially in the Thar (थार).
10. **Overgrazing** is the top dryland cause in many coaching facts. Afforestation helps prevent further degradation.
11. The **UNCCD** was adopted (अंगीकृत) in **1994**. **17 June** is Desertification and Drought Day.
12. **Land Degradation Neutrality (LDN)** (भूमि क्षरण तटस्थता) targets **no net loss** of healthy land by **2030** (SDG **15.3**).
13. UNCCD is the desertification convention. **UNFCCC** (यूएनएफसीसीसी) is the climate convention — different treaties.
14. Salinity from poor irrigation is a human-made degradation path even where rainfall is not zero.
15. Accelerated erosion from human misuse is the focus, not only slow geological erosion.
16. Desertification can reverse with sustainable land management; it is not an irreversible death sentence in every plot.
17. Do not define desertification as any bare land in a humid forest belt.
18. Sheet erosion (परत (चादर) अपरदन) removes a thin film; gullies cut channels; ravines are deep, dense badland networks.
19. IMD drought monitoring and DPAP geography often appear beside desertification stems.
20. SDG 15.3 LDN is the policy end-state; UNCCD is the global legal home.
21. Thar wind processes and Chambal water ravines are twin Indian landform facts.
22. Deforestation removes cover and raises runoff and erosion risk in drylands and beyond.

📋 Must-score facts — ASDM, Chambal, UNCCD, LDN

Locks

Item	Lock
Desertification	ASDM dryland degradation (≠ only desert expand)
India share	~29–30%; ~96.4 mha; Rajasthan worst hotspot
Drought types	Met → hydro → agri; met often <75% normal rain (सामान्य वर्षा)
DPAP	**1973**
Bundelkhand UP	Seven districts drought–degradation hotspot
Erosion sequence	Sheet → rill → gully → ravine (Chambal)
UNCCD / Day	**1994** / **17 June**

Item	Lock
LDN	No net loss by **2030** (SDG 15.3)
UNCCD ≠ UNFCCC	Desertification / climate

Topic 41 — Environmental Monitoring

Consolidated — 22 Must-Score Facts

1. The **PSR** model runs **Pressure** → **State** → **Response** for environmental indicators.
2. Air indicators include **PM2.5, PM10, SO_2, NOx, and AQI**. Water indicators include **BOD, COD, DO, pH, and coliform**.
3. Noise is measured in **decibels**. Smoke opacity uses the **Ringelmann** scale **0–5**.
4. **Lichens** are classic bio-indicators for sulphur dioxide stress.
5. **Monitoring** means systematic repeated measurement against standards. It is not the same as an audit.
6. **CPCB** (केंद्रीय प्रदूषण नियंत्रण बोर्ड) runs **NAMP** (manual air), **CAAQMS** (real-time air), and **NWMP** (surface water).
7. **CGWB** monitors groundwater quality and levels and publishes the annual groundwater quality report.
8. **SAFAR** (सफर) is the IITM air-quality and weather forecast system associated with Harsh Vardhan's launch messaging in **July 2018**.
9. **NCAP (राष्ट्रीय स्वच्छ वायु कार्यक्रम) (2019)** expands city monitoring and action. **SAMEER** is an AQI public app.
10. An **environmental audit** verifies compliance with law or clearance conditions — it is more than raw (रॉ) data logging.
11. **Form V** Environmental Statement under EPA Rule **14** goes to the SPCB yearly.
12. **ISO 14001** is the Environmental Management System standard.
13. Consent to Operate links industry permission to both monitoring data and audit compliance.
14. **BOD** is microbial oxygen demand. **COD** is chemical oxygen demand. COD is usually ≥ BOD.
15. NAMP shows long trend with manual stations. CAAQMS gives continuous real-time feeds.
16. CPCB is the air–surface-water regulator home. CGWB is the groundwater home. Do not swap them.
17. Green audits and EC compliance reports sit in the audit family, not in casual one-off sampling.
18. Indicators turn raw numbers into decision signals for policy and public warning.
19. Bio-indicators supplement instruments when living organisms reveal chronic pollution stress.
20. Ringelmann is about visible smoke shade, not a water-quality number.
21. Do not call an audit "just another AQI reading (रीडिंग)." Audit checks legal conformity.
22. Surface-water networks (NWMP) do not replace groundwater networks (CGWB).

📋 Must-score facts — PSR, NAMP, audit, BOD

Tools

Item	Lock
PSR	Pressure → State → Response
Noise / smoke	dB / Ringelmann **0–5**
Lichens	SO_2 bio-indicators
CPCB networks	NAMP · CAAQMS · NWMP
CGWB	Groundwater quality + levels report
SAFAR	IITM forecast (2018 launch messaging)
SAMEER	AQI public app
Form V	EPA Rule 14 Environmental Statement → SPCB yearly
ISO 14001	EMS standard
Monitoring ≠ audit	Repeated measurement / compliance verification

Topic 42 — International Environmental Organizations

Consolidated — 29 Must-Score Facts

1. **UNEP** was created after **Stockholm 1972** (स्टॉकहोम सम्मेलन). Its HQ is **Nairobi**. Flagship products include the **Emissions Gap Report** (उत्सर्जन अंतराल रिपोर्ट) and **GEO**.
2. **UNDP** sits in **New York** and publishes the **Human Development Report / HDI** (from **1990**).
3. **FAO** sits in **Rome** and publishes **SOFO**. It co-hosts Rotterdam work with UNEP.
4. **UNESCO** sits in **Paris** and runs World Heritage (विश्व धरोहर) and the **MAB** Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र) programme.
5. **IUCN** (आईयूसीएन) sits in **Gland** and publishes the **Red List** of extinction risk.
6. **WWF** also sits in **Gland** and publishes the **Living Planet Report / LPI**.
7. **IPCC** sits in **Geneva**. It was created by **WMO + UNEP** in **1988** and shared the **2007** Nobel Peace Prize.
8. Emissions Gap Report = **UNEP**, not IPCC or UNDP. HDR = **UNDP**, not UNEP.
9. IPCC does science assessment. **UNFCCC** (यूएनएफसीसीसी) is the climate treaty. Do not merge the two.
10. IUCN assesses extinction risk. **CITES** (साइट्स) regulates wildlife **trade** (पण्याध्यक्ष). Different jobs.
11. IUCN and WWF share a city but differ: assessment union versus conservation NGO campaigning and Living Planet metrics.
12. UNEP coordinates environment (पर्यावरण). UNDP coordinates development. HQ and report facts separate them.
13. UNESCO heritage–education–MAB is not the same as UNEP environment coordination.
14. The **GEF (1991)** is a multilateral funder. Donors are not "developed countries only" as a rigid fact.
15. IPCC Assessment Reports synthesize climate science. They do not set national NDCs.
16. FAO's food–forest–agriculture mandate explains why SOFO and pesticide PIC work sit there.
17. Red List categories are IUCN risk ranks, not CITES Appendix numbers.
18. Living Planet Index tracks vertebrate abundance trends for WWF messaging.
19. Nairobi versus Geneva versus New York versus Rome versus Paris versus Gland is a high-yield HQ ladder.
20. Do not credit IPCC with the Emissions Gap Report.
21. Do not credit UNEP with HDI.
22. Treaty secretariats hosted or supported by UNEP (Montreal, Basel, Stockholm POPs, Minamata) are still treaty bodies, not UNEP synonyms.
23. WWF is not the Red List author. IUCN is.
24. FAO HQ is Rome, not Paris or Geneva.
25. **WMO** headquarters is **Geneva**. It grew from the International Meteorological Organization (**1873**) and became WMO on **17 March 1951**. It co-created the **IPCC** with UNEP in **1988**.
26. **Greenpeace International** is an environmental NGO headquartered in **Amsterdam**, Netherlands.
27. The **US EPA** was established on **2 December 1970**. Do not confuse it with India's **Environment (Protection) Act, 1986**.
28. **OPCW** implements the **Chemical Weapons Convention** (force **29 April 1997**); HQ is **The Hague**. It won the Nobel Peace Prize in **2013**. It is **not** an EU, NATO, or WHO body.
29. **UN-Habitat** (UN Human Settlements Programme; HQ **Nairobi**, from **1978**) promotes sustainable towns and shelter. Partners are **not** governments-only.

📋 Must-score facts — UNEP UNDP IUCN IPCC HQs

Org ↔ HQ / product

Item	Lock
UNEP	Nairobi; Emissions Gap + GEO; post-Stockholm 1972
UNDP	New York; HDR/HDI
FAO	Rome; SOFO
UNESCO	Paris; WH + MAB
IUCN	Gland; Red List
WWF	Gland; Living Planet
IPCC	Geneva; WMO+UNEP 1988; Nobel 2007
IUCN ≠ CITES	Risk assess / trade regulate
GEF	**1991** multilateral funder

Topic 43 — International Environmental Agreements

Consolidated — 24 Must-Score Facts

1. The climate ladder is **UNFCCC 1992** → **Kyoto 1997** → **Paris 2015**.
2. **Kyoto** (क्योटो) set binding targets mainly for **Annex-I** parties. **Paris** uses universal **NDCs** for all parties.
3. **Montreal** protects the ozone layer. **Kyoto / Paris / UNFCCC** are climate instruments. Never swap Montreal for climate mitigation (जलवायु शमन).
4. **CBD (1992)** is the biodiversity convention. **Cartagena** is the biosafety protocol under CBD.
5. **Ramsar (1971)** covers wetlands. **CITES (1973)** (साइट्स) covers wildlife **trade** (पण्याध्यक्ष). **CMS / Bonn (1979)** covers **migratory** species.
6. **Basel (1989)** covers hazardous **waste**. **Rotterdam (1998)** covers hazardous **chemicals** through the PIC procedure (प्रक्रिया).
7. **Stockholm Convention (2001)** (स्टॉकहोम संधि) covers **POPs** such as DDT and PCBs. It is **not** the Stockholm Conference of **1972**.
8. **Minamata (2013)** covers **mercury**.
9. UNFCCC was opened at **Rio 1992**. Its secretariat sits in **Bonn**. Annual COP meetings drive negotiations.
10. CITES is not the IUCN (आईयूसीएन) Red List. CMS is not CITES. Trade rules and migration treaties are different tools.
11. Ramsar "wise use" allows sustainable human use of wetlands — it is not a total ban slogan.
12. Paris Agreement year fact is **2015**. Kyoto year fact is **1997**. Montreal year fact is **1987**.
13. A stem that says Kyoto is about "saving water" is false. Kyoto is climate.
14. Stockholm **1972** created the UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) moment. Stockholm **2001** is the POPs treaty. Keep the two Stockholms apart.
15. Basel moves on waste shipment and disposal. Rotterdam's PIC is prior informed consent for chemicals trade.
16. CBD's three objectives are conservation, sustainable use, and fair benefit-sharing from genetic resources.
17. CMS protects migratory animals across borders. CITES controls international commercial trade in listed species.
18. Minamata's mercury focus includes artisanal gold mining and product phase-downs in teaching notes.
19. UNFCCC is the parent framework; Kyoto and Paris are protocols/agreements under that climate family.
20. Do not put Ramsar, CITES, or CBD into the climate-mitigation answer slot when Kyoto is present.
21. UP Ramsar geography traps belong beside Ramsar treaty facts but do not change the 1971 identity.
22. Rotterdam is chemicals PIC, not mercury (Minamata) and not POPs listing (Stockholm 2001 (स्टॉकहोम संधि)).
23. Paris NDCs are nationally determined. Kyoto Annex-I targets were top-down binding for listed developed parties.
24. Bonn Convention means **CMS**, not the UNFCCC Bonn secretariat alone.

📋 Must-score facts — UNFCCC Kyoto Paris CBD Basel

Ladders / tags

Item	Lock
Climate	UNFCCC **1992** → Kyoto **1997** → Paris **2015**
Kyoto vs Paris	Annex-I binding / universal NDCs
Montreal	Ozone (**1987**) — not climate
CBD / Cartagena	**1992** / biosafety protocol
Ramsar / CITES / CMS	**1971** / **1973** / Bonn **1979**
Basel / Rotterdam / Stockholm POPs	**1989** / **1998** / **2001**

Item	Lock
Minamata	Mercury **2013**
Two Stockholms	**1972** conference→UNEP / **2001** POPs treaty
UNFCCC secretariat	**Bonn**

Topic 44 — Current Environmental Issues

Consolidated — 24 Must-Score Facts

1. A **carbon footprint** totals greenhouse gases in **tCO_2e**. Scope **1** is direct, Scope **2** is purchased energy, Scope **3** is the wider value chain.
2. India's **per capita** carbon footprint is about **2 tCO_2e** — low versus global and USA averages — even though absolute (निरपेक्ष) emissions are high.
3. An **ecological footprint** (पारिस्थितिक पदचिह्न) measures resource demand in **global hectares (gha)**. It is **not** the same unit as tCO_2e. The **Social Cost of Carbon** is the monetary long-term damage of **one tonne of CO_2**.
4. **Net zero** balances remaining emissions with removals. India's year fact is **2070**. It is not gross zero and not automatically **2050**.
5. NDC-style facts include about **45%** intensity cut, about **50%** non-fossil capacity, and a large forest sink goal.
6. The **Emissions Gap Report** (उत्सर्जन अंतराल रिपोर्ट) is by **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम).
7. A **green economy** (UNEP framing) is low-carbon, resource-efficient, and socially inclusive.
8. A **blue economy** is sustainable ocean use linked to **SDG 14**. **MISHTI** mangrove restoration is from the **2023** Budget, **not 2019**.
9. A **circular economy** closes material loops through 3R/9R and EPR. It sits inside the broader green-economy idea and links to **SDG 12**.
10. **Green hydrogen** uses electrolysis powered by renewables. India's mission (January **2023**) targets about **5 MMT** by **2030**.
11. **Grey hydrogen** reforms natural gas. **Blue hydrogen** adds CCS (सुरक्षा कैबिनेट समिति) to grey pathways. Do not call grey "green."
12. **Microplastics** are plastic pieces smaller than **5 mm**. They are primary (made small) or secondary (fragmented) and are not fast biodegradable.
13. The **SUP ban** push of **2022** targets single-use plastics that feed microplastic pathways.
14. **Climate refugees** are mostly **internally** displaced. They are **not** a separate legal category under the **1951** Refugee Convention.
15. **Nature-based Solutions** protect, manage, or restore ecosystems (IUCN (आईयूसीएन) framing). They are not geoengineering (भू-इंजीनियरिंग) (cirrus thinning / sulphate aerosols are geoengineering ideas for cooling, not NbS).
16. **LiFE** was an idea at **COP-26 (2021)** and launched in **June 2022** under MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) with a long action list. **COP-25 Madrid** is the false trap.
17. Carbon footprint ≠ ecological footprint. Units are **tCO_2e** versus **gha**.
18. Green economy ≠ circular economy as identical synonyms. Circular is the material-loop design inside a wider green agenda.
19. Net zero allows balancing removals. Gross zero means nearly eliminating emissions without relying on offsets.
20. MISHTI is a mangrove initiative. Pairing it with 2019 is wrong.
21. Microplastic size fact is **<5 mm**, not 5 cm.
22. NbS examples include mangroves, wetlands, and urban (नगरीय) forests — ecosystem (पारिस्थितिकी तंत्र) actions, not aerosol injection.
23. LiFE is lifestyle and demand-side behaviour change, not a substitute for the entire NDC package.
24. Blue economy is oceans and coasts. Green hydrogen is energy. Do not merge the two labels.

📋 Must-score facts — footprints, net zero, hydrogen, plastics

Current locks

Item	Lock
Carbon footprint	tCO_2e; Scope 1/2/3
Ecological footprint	gha demand (≠ tCO_2e)
India net zero	**2070**
Emissions Gap	**UNEP**
Green / Blue / Circular	Low-carbon inclusive / ocean SDG14 / 3R–EPR SDG12
MISHTI	Budget **2023** (≠ 2019)
Green H_2 mission	Jan **2023**; ~5 MMT by 2030
Grey / Blue / Green H_2	Gas reform / +CCS / RE electrolysis
Microplastics	<5 mm; SUP ban push **2022**
Climate refugees	Mostly IDPs; not 1951 Refugee Convention category

Uttarakhand — 01 — Natural Resources & Climate Contribution

Consolidated — 20 Must-Score Facts

1. Uttarakhand is a **Himalayan state** whose forests, glaciers, rivers, and biodiversity contribute to India's climate and water security.
2. **Chipko** (चिपको) began in **1973** at **Reni (Chamoli (चमोली))**. It is the classic UK people's movement for forest protection.
3. Dehradun institutions: **FSI** (ISFR), **FRI** (forestry research), **WII** (wildlife science/training). Do not swap their jobs.
4. **NAPCC** includes the **National Mission for Sustaining the Himalayan Ecosystem (पारिस्थितिकी तंत्र)**.
5. UK rivers (Ganga (गंगा)–Yamuna (यमुना) headwaters, Bhagirathi (भागीरथी), Alaknanda (अलकनंदा), and tributaries) feed the Gangetic water system.
6. Forests and alpine meadows act as **carbon sinks** and regulate local hydrology through interception and infiltration.
7. Glacier and snowmelt contribution to Himalayan rivers is a climate-vulnerability fact for UKPCS.
8. **Weathering** breaks rocks into sediments, clays, and soils at the Earth's surface. Erosion then transports the loosened material.
9. National facts that UKPCS also asks: **NEERI = Nagpur 1958**; **WED = 5 June**; atmosphere **N_2 ~78%**; **EPA 1986** (पर्यावरण संरक्षण 1986); **Tansley 1935** coined ecosystem; ecosystem is an **open** system.
10. **Fukuoka** pioneered natural farming — a UKPCS-asked literature/agriculture fact.
11. Sustainable development, Agenda 21, Brundtland, and SDGs follow the same national sustainable-development spine.
12. UK biodiversity sits in the **Himalaya** (हिमालय) biogeographic zone with high endemism and altitudinal vegetation belts.
13. Landslides and flash floods rise where deforestation, road cutting, and extreme rainfall combine — disaster–environment (पर्यावरण) link.
14. Eco-tourism and hydropower both use natural capital (प्राकृतिक पूंजी); unmanaged expansion can degrade slopes and rivers.
15. **Article 48A / 51A(g)** and the right to a healthy environment under **Article 21** (अनुच्छेद 21) apply nationwide, including UK.
16. Protected-area names and district facts belong in the biodiversity UK chapter; this card keeps resource–climate focus.
17. Remote sensing and GIS support SD road-mapping and hazard monitoring — UKPCS General Science/Ecology (पारिस्थितिकी) mentions these tools.
18. Natural capital of UK includes forests, water, minerals, and biodiversity. Roads are **not** natural capital.
19. Over-urbanisation and unplanned industrialisation stress ecology and biodiversity in hill towns as well as plains.
20. Shared national ecology stems (sustainable development, institutions, agreements, days) also appear in UKPCS — learn them as national facts with UK colour where needed.

Uttarakhand — 02 — Biodiversity & Protected Areas

Consolidated — 22 Must-Score Facts

1. Uttarakhand sits in the **Himalaya** biogeographic zone with high endemism and steep altitudinal habitat belts from Tarai–Bhabar (भाबर) to alpine meadows.
2. Forest and biodiversity richness is explained primarily by **topography and climate** packing many climatic zones into a short horizontal distance (A/R both true; R explains A).
3. Habitats run Tarai–Bhabar forests → mid-hill forests → alpine meadows and cold-desert edges — not a single plains biome.
4. **Corbett National Park (1936)** anchors Tarai tiger conservation and Project Tiger (प्रोजेक्ट टाइगर) memory.
5. **Rajaji National Park (1983)** covers Shiwalik (शिवालिक)–dun (दून) elephant corridors between Dehradun (देहरादून) and Haridwar (हरिद्वार).
6. **Gangotri National Park** dates to **1989**. **Govind National Park** dates to **1990** — do **not** date Govind as 1989.
7. **Nanda Devi** and **Valley of Flowers** (फूलों की घाटी) form one UNESCO World Heritage (विश्व धरोहर) property.
8. **Nanda Devi Biosphere Reserve** holds **snow leopard**, Himalayan **musk deer**, and **blue sheep (bharal)**.
9. Biosphere teaching uses **core–buffer–transition** zonation — stricter core, more flexible outer rings — not "just another National Park label."
10. Elephant and tiger corridor logic links **Corbett–Rajaji–Terai** (तराई) landscapes with neighbouring UP Terai reserves.
11. Sanctuary–district pairs: **Askot–Pithoragarh**; **Binsar–Almora**; **Govind–Uttarkashi**; **Nandhaur–Nainital**.
12. Wildlife Sanctuaries allow regulated human uses that a National Park usually forbids — keep the WPA hierarchy from national Protected Areas notes.
13. **Asan Conservation Reserve** (संरक्षण आरक्षित क्षेत्र) is UK wetland PA colour and also a **Ramsar** (रामसर) site in national wetland teaching.
14. National Park ≠ Wildlife Sanctuary ≠ Biosphere Reserve — different legal and management ideas under WPA versus MAB.
15. Chipko (चिपको), FSI–FRI–WII, and weathering facts stay on Topic 1 (Natural Resources); this card keeps biodiversity–PA focus.
16. Endemism is high relative (सापेक्ष) to many plains states because altitude stacks many climates in a short distance.
17. High Garhwal (गढ़वाल) catchments are protected by Gangotri and Govind parks — snow-leopard and musk-deer colour sits here.
18. Match-list sanctuary stems are decided by district keys, not by alphabetical order of sanctuary names.
19. Corbett year **1936**, Rajaji **1983**, Gangotri **1989**, Govind **1990** is the core NP chronology spine for UKPCS.
20. Calling Nanda Devi fauna only "tiger and elephant" is wrong — the BR set is snow leopard / musk deer / blue sheep.
21. Treating Govind NP as **1989** is the planted chronology trap against Gangotri's correct **1989**.
22. Assertion that UK is biodiversity-rich without topography–climate explanation fails the classic A/R pair.

Uttarakhand — 03 — Climate Vulnerability & Governance

Consolidated — 18 Must-Score Facts

1. Uttarakhand forests and soils act as **carbon sinks** supporting India's forest-sink narrative.
2. Snow and glacier systems make the state a **water tower** for the Gangetic plains.
3. The **National Mission for Sustaining the Himalayan Ecosystem (पारिस्थितिकी तंत्र)** under **NAPCC** is the named Himalayan climate mission.
4. International climate commitments (Paris, Net Zero **2070**, Panchamrit) are India's; UK's role is ecosystem services and vulnerability reduction — not a separate UNFCCC track.
5. Fragile slopes, glacier-fed rivers, and extreme rainfall make Uttarakhand **highly vulnerable** to climate change.
6. Documented climate effects include **water stress** (drying springs) and pressure on **biodiversity** / habitats — both statements true together.
7. Landslides, cloudbursts, flash floods, and **GLOF** risk sit on the same physical vulnerability picture.
8. **State Council for Climate Change (SCCC)** was established in **2011** — not at statehood **2000** and not **2009/2013**.
9. The **Uttarakhand Action Plan on Climate Change (UAPCC)** was prepared through the **forest and environment** set-up, **not** the Revenue Department.
10. Adaptation tools include spring rejuvenation, disaster-sensitive land use, afforestation, and early-warning systems.
11. Institutional facts (SCCC / UAPCC) sit **on top of** physical vulnerability — they do not replace it.
12. Chipko (चिपको) and Dehradun FSI–FRI–WII stay on Topic 1; parks and sanctuary–district pairs stay on Topic 2.
13. Panchamrit / COP venues belong to national climate and agreements chapters; UKPCS still expects the Himalayan Mission name under NAPCC.
14. "Highly vulnerable" without water-stress / biodiversity effects is incomplete for the classic two-statement stem.
15. UAPCC ≠ Revenue Department is the planted governance trap.
16. SCCC year fact is **2011** — do not invent 2009 or 2013.
17. UK contributes via India's NDCs and ecosystem services — it does not run a separate UNFCCC party track.
18. Cloudburst / GLOF seasons refresh vulnerability colour without changing the SCCC **2011** or UAPCC department keys.